# Fruit Market · Decision register

*Generated by `npm run records` from `docs/workflow/records/`. Do not edit by hand. Last recorded change: 2026-10-27 15:37 (Asia/Kuala_Lumpur).*

Approval status and delivery status are separate. Only the owner's explicit words set "approved".

## Pending approvals

| ID | What | Status |
|---|---|---|
| DEC-001 | Adopt the personal workflow guide 1.2 for Fruit Market | Proposed |
| DEC-004 | PDF exports with pandoc and the existing Playwright browser | Proposed |
| DEC-005 | Budget interpretation: RM500 per month across all games | Proposed |
| DEC-014 | Use the Concept and Theme Companion for creative work | Proposed |
| DEC-020 | UI and UX polish in three phases | Awaiting approval |
| PLAN-003 | Package C: online, capacity and release evidence (outline only) | Proposed |

## Current decisions

| ID | Decision | Approval | Delivery | Reports |
|---|---|---|---|---|
| DEC-001 | Adopt the personal workflow guide 1.2 for Fruit Market | Proposed | In progress | REP-001 |
| DEC-002 | Package A scope = PLAN-001 | Approved (owner, 2026-10-26) | Verified | REP-001, REP-002 |
| DEC-003 | Live deploys only through the manual Deploy workflow after verify | Approved (owner, 2026-10-26) | Verified | REP-001, REP-002 |
| DEC-004 | PDF exports with pandoc and the existing Playwright browser | Proposed | Implemented | REP-001 |
| DEC-005 | Budget interpretation: RM500 per month across all games | Proposed | Not started | REP-001 |
| DEC-006 | Preview = separate staging Worker | Approved (owner, 2026-10-26) | Verified | REP-002, REP-003 |
| DEC-007 | Package B scope = PLAN-002 | Approved (owner, 2026-10-26) | Verified | REP-003, REP-004 |
| DEC-008 | Canonical record format: JSON records plus Markdown narrative reports | Approved (owner, 2026-10-26) | Verified | REP-003, REP-004 |
| DEC-009 | Records check runs in verify and CI | Approved (owner, 2026-10-26) | Verified | REP-003, REP-004 |
| DEC-010 | Reconnect fix scope = PLAN-004 (evidence first, then fix) | Approved (owner, 2026-10-26) | Verified | REP-005, REP-006 |

| ID | Decision | Approval | Delivery | Reports |
|---|---|---|---|---|
| DEC-011 | Stress test runs on demand, not on every push | Approved (owner, 2026-10-26) | Verified | REP-005, REP-006 |
| DEC-012 | 4-player clarity scope = PLAN-005 (screen text only) | Approved (owner, 2026-10-26) | Verified | REP-007, REP-008 |
| DEC-013 | Publish version 1.1.1 to players | Approved (owner, 2026-10-26) | Released | REP-009, REP-013 |
| DEC-014 | Use the Concept and Theme Companion for creative work | Proposed | In progress | REP-010 |
| DEC-015 | Adopt three design pillars | Approved (owner, 2026-10-26) | Implemented | REP-010, REP-011, REP-013 |
| DEC-016 | First creative test: reveal-comprehension playtest | Approved (owner, 2026-10-26) | Not started | REP-010, REP-011, REP-013 |
| DEC-017 | Setting: a market where the whole world shops | Approved (owner, 2026-10-26) | Implemented | REP-011 |
| DEC-018 | Rename Panda Lim while keeping the cast coherent | Approved (owner, 2026-10-26) | Verified | REP-011, REP-012 |
| DEC-019 | Rename scope = PLAN-006, shipped in 1.1.2 | Approved (owner, 2026-10-26) | Verified | REP-011, REP-012 |
| DEC-020 | UI and UX polish in three phases | Awaiting approval | Not started | REP-014, REP-015 |
| DEC-021 | Art direction: B, toy-box tabletop | Approved (owner, 2026-10-26) | In progress | REP-015, REP-016 |
| DEC-022 | Look and tone: cunning edition of direction B | Approved (owner, 2026-10-27) | Implemented | REP-017 |
| DEC-023 | Look: cosy pixel edition of direction B | Approved (owner, 2026-10-27) | Implemented | REP-018, REP-020 |
| DEC-024 | Layout: one shared market table (board game, not menus) | Approved (owner, 2026-10-27) | Implemented | REP-019, REP-020 |
| DEC-025 | Market board: vertical decay tracker and a delivery lorry | Approved (owner, 2026-10-27) | Implemented | REP-021 |

## Work packages

| ID | Package | Approval | Delivery | Release |
|---|---|---|---|---|
| PLAN-001 | Package A: baseline, one verification path, preview and owner brief | Approved (owner, 2026-10-26) | Released | 1.1.1 |
| PLAN-002 | Package B: canonical records, generated views and checks | Approved (owner, 2026-10-26) | Released | 1.1.1 |
| PLAN-003 | Package C: online, capacity and release evidence (outline only) | Proposed | Not started | Unreleased |
| PLAN-004 | Reconnect fix: the game screen sometimes doesn't return after rejoining an online room | Approved (owner, 2026-10-26) | Released | 1.1.1 |

| ID | Package | Approval | Delivery | Release |
|---|---|---|---|---|
| PLAN-005 | 4-player clarity: make it obvious that up to 4 people can play | Approved (owner, 2026-10-26) | Released | 1.1.1 |
| PLAN-006 | Rename the panda rival (cosmetic, within the animal-trader cast) | Approved (owner, 2026-10-26) | Verified | Unreleased |
| PLAN-007 | UI polish phase 1: fix what looks broken | Superseded | Not started | Unreleased |
| PLAN-008 | UI polish: art-direction style test and layout mockups (design only) | Approved (owner, 2026-10-26) | Implemented | Unreleased |
| PLAN-009 | Cosy table build 1: seated rivals and the new market board | Approved (owner, 2026-10-27) | Verified | Unreleased |
| PLAN-010 | Cosy table build 2: the delivery lorry (outline) | Approved (owner, 2026-10-27) | Verified | Unreleased |
| PLAN-011 | Cosy table build 3: pricing, reveal and evening on the table (outline) | Approved (owner, 2026-10-27) | Verified | Unreleased |
| PLAN-012 | Market rebalance and cosy home screen: tighter town, fair Handcart, better apples, optional tweaks | Approved (owner, 2026-10-27) | Verified | Unreleased |
| PLAN-013 | Full-screen pricing and event sheets; clearer rival status while pricing | Approved (owner, 2026-10-27) | Verified | Unreleased |
| PLAN-014 | Tabletop edition rules: printed supply and demand, posted prices, market pulse, fruit payouts | Approved (owner, 2026-10-27) | Implemented | Unreleased |
| PLAN-015 | Readable price duel: five-beat day, prototypes A/B/C and the Market Gossip card | Approved (owner, 2026-10-27) | Implemented | Unreleased |

## Decision details

### DEC-001 · Adopt the personal workflow guide 1.2 for Fruit Market

- **Question:** Should Fruit Market be run with the workflow guide (one approved package at a time, evidence-based checks, decision records)?
- **Choice:** Yes; one package at a time, starting with Package A.
- **Rationale:** The game is complete but releases were unreviewed and records lived in chat history.
- **Player impact:** None directly; fewer broken releases.
- **Technical impact:** Adds checks, records and a release gate.
- **Data and security:** None.
- **Cost:** RM0
- **Owner:** Murshidi
- **Approval:** Proposed
- **Delivery:** In progress
- **Review trigger:** If the process costs more time than it saves after three packages.
- **Reports:** REP-001

### DEC-002 · Package A scope = PLAN-001

- **Question:** What should the first setup package contain?
- **Choice:** One verify command, preview before publishing, short records and an owner brief (PLAN-001).
- **Rationale:** Smallest package that stops unreviewed releases and gives the owner a clear way to try builds.
- **Player impact:** None until a release; protects players from silent breakage.
- **Technical impact:** package.json scripts, CI and deploy workflows, new docs.
- **Data and security:** None.
- **Cost:** RM0
- **Owner:** Murshidi
- **Approval:** Approved: “Approve PLAN-001 only, including its listed tests and documentation” (2026-10-26)
- **Delivery:** Verified (PLAN-001)
- **Review trigger:** Acceptance A4 (owner preview playtest) still pending.
- **Reports:** REP-001, REP-002

### DEC-003 · Live deploys only through the manual Deploy workflow after verify

- **Question:** When should the live game change?
- **Choice:** Only when the owner runs Deploy LIVE game and types PUBLISH; it runs verify first and a smoke test after. Never on push.
- **Rationale:** Makes 'a preview is not a release' true for this repository.
- **Player impact:** Updates arrive only after the owner approves them.
- **Technical impact:** deploy-web.yml is manual only.
- **Data and security:** None.
- **Cost:** RM0; one extra tap per release.
- **Owner:** Murshidi
- **Approval:** Approved: “Approve PLAN-001 only, including its listed tests and documentation” (2026-10-26)
- **Delivery:** Verified (PLAN-001)
- **Review trigger:** If urgent fixes are delayed because nobody pressed Deploy.
- **Reports:** REP-001, REP-002

### DEC-004 · PDF exports with pandoc and the existing Playwright browser

- **Question:** Which tool builds the decision PDFs?
- **Choice:** pandoc (Markdown to HTML) plus the project's existing Playwright Chromium (HTML to PDF). No installs.
- **Rationale:** Both are already available; no new dependency.
- **Player impact:** None.
- **Technical impact:** PDF export runs locally, not in CI.
- **Data and security:** None.
- **Cost:** RM0
- **Owner:** Murshidi
- **Approval:** Proposed
- **Delivery:** Implemented
- **Review trigger:** If pandoc is unavailable on the machine that exports.
- **Reports:** REP-001

### DEC-005 · Budget interpretation: RM500 per month across all games

- **Question:** How should the RM500 monthly ceiling be applied?
- **Choice:** Treat it as portfolio-wide. Fruit Market's current spend is unknown (hypothesis: Cloudflare Free, or Workers Paid at about USD 5).
- **Rationale:** The workflow guide's conservative default.
- **Player impact:** None.
- **Technical impact:** Paid services need a quote and approval.
- **Data and security:** None.
- **Cost:** Unknown; to be recorded.
- **Owner:** Murshidi
- **Approval:** Proposed
- **Delivery:** Not started
- **Review trigger:** When the owner states the Cloudflare plan and other paid accounts.
- **Reports:** REP-001

### DEC-006 · Preview = separate staging Worker

- **Question:** How should changes be previewed, given that Cloudflare preview and version URLs are not generated for Workers with Durable Objects?
- **Choice:** A separate Worker, fruit-market-staging, with its own rooms, leaderboard and stats, updated by CI after checks pass on main.
- **Rationale:** Closest free option to a per-change preview URL; keeps the live data untouched.
- **Player impact:** None; players never see the preview.
- **Technical impact:** A second Worker and a preview job in CI.
- **Data and security:** Separate storage; preview link is unlisted, not private.
- **Cost:** No new paid service; tiny usage within the existing plan.
- **Owner:** Murshidi
- **Approval:** Approved: “Confirm DEC-006, and plan PLAN-002” (2026-10-26)
- **Delivery:** Verified (PLAN-001)
- **Review trigger:** If Cloudflare adds preview URLs for Durable Object Workers.
- **Reports:** REP-002, REP-003

### DEC-007 · Package B scope = PLAN-002

- **Question:** What should the records package contain?
- **Choice:** Canonical JSON records, generated views (register, changelog, state block, PDFs), a records check in verify, release tagging on Deploy, and migration of existing records.
- **Rationale:** Records were kept by hand in several places with nothing checking that they agreed.
- **Player impact:** None directly; releases become traceable.
- **Technical impact:** New records, a generator and checker script, and a deploy step.
- **Data and security:** None.
- **Cost:** RM0
- **Owner:** Murshidi
- **Approval:** Approved: “Approve PLAN-002 only, including its listed tests and documentation” (2026-10-26)
- **Delivery:** Verified (PLAN-002)

- **Review trigger:** If keeping records costs more than it saves.
- **Reports:** REP-003, REP-004

### DEC-008 · Canonical record format: JSON records plus Markdown narrative reports

- **Question:** Where is the single source of truth for decisions and packages?
- **Choice:** JSON files in docs/workflow/records; hand-written REP reports stay as narrative snapshots; Markdown, changelog and PDF views are generated.
- **Rationale:** Node reads JSON without dependencies and the check can be strict. YAML needs a parser; plain Markdown can't be validated; a database is ruled out by the guide.
- **Player impact:** None.
- **Technical impact:** Claude edits the JSON; the owner reads the generated views.
- **Data and security:** None.
- **Cost:** RM0
- **Owner:** Murshidi
- **Approval:** Approved: “Approve PLAN-002 only, including its listed tests and documentation” (2026-10-26)
- **Delivery:** Verified (PLAN-002)
- **Review trigger:** If hand edits to JSON become a frequent source of errors.
- **Reports:** REP-003, REP-004

### DEC-009 · Records check runs in verify and CI

- **Question:** Should record disagreements fail the build?
- **Choice:** Yes: npm run verify runs records:check, so CI fails on broken records.
- **Rationale:** Once branch protection is on, broken records can't be merged.
- **Player impact:** None.
- **Technical impact:** verify is about 1 second slower.
- **Data and security:** None.
- **Cost:** RM0
- **Owner:** Murshidi
- **Approval:** Approved: “Approve PLAN-002 only, including its listed tests and documentation” (2026-10-26)
- **Delivery:** Verified (PLAN-002)
- **Review trigger:** If the check blocks urgent fixes for record reasons.
- **Reports:** REP-003, REP-004

### DEC-010 · Reconnect fix scope = PLAN-004 (evidence first, then fix)

- **Question:** How should the intermittent rejoin failure be fixed?
- **Choice:** Collect evidence and reproduce it with a stress test before changing game code; fix only the measured root cause, with a regression test.
- **Rationale:** Reading the code didn't identify a certain cause, and two failures in five runs are too rare to debug by guessing. The workflow guide says to reproduce and isolate before fixing.
- **Player impact:** Fewer stuck screens after a reload or dropped connection, once the fix is published.

- **Technical impact:** New stress test and diagnostics; a small fix in the online client or room server.
- **Data and security:** Diagnostics stay in CI artifacts for 7 days; they contain test players only.
- **Cost:** RM0 (GitHub Actions minutes are free for public repositories)
- **Owner:** Murshidi
- **Approval:** Approved: “Approve PLAN-004 only, including its listed tests and documentation” (2026-10-26)
- **Delivery:** Verified (PLAN-004)
- **Review trigger:** If not reproduced after 300 cycles: stop and report.
- **Reports:** REP-005, REP-006

### DEC-011 · Stress test runs on demand, not on every push

- **Question:** Should the long reconnect stress test run in every CI run?
- **Choice:** No: a manual CI workflow plus local runs. Every-push CI keeps the existing tests (about 15 minutes).
- **Rationale:** Hundreds of cycles would make every CI run far slower; the stress test is for investigation and before releases.
- **Player impact:** None.
- **Technical impact:** One extra manual workflow.
- **Data and security:** None.
- **Cost:** RM0
- **Owner:** Murshidi
- **Approval:** Approved: “Approve PLAN-004 only, including its listed tests and documentation” (2026-10-26)
- **Delivery:** Verified (PLAN-004)
- **Review trigger:** If reconnect bugs slip through between releases.
- **Reports:** REP-005, REP-006

### DEC-012 · 4-player clarity scope = PLAN-005 (screen text only)

- **Question:** How should the game make 4-player play discoverable?
- **Choice:** Clearer screen text and open-seat rows in the lobby, short hints on the home, quick-match and new-game screens, plus an automated 4-person test. No rule or server changes.
- **Rationale:** 4-player play already works; only discoverability is missing. Text changes are low risk and fast to verify.
- **Player impact:** Groups of 3 or 4 can find how to play together.
- **Technical impact:** src/ui/Online.tsx, src/ui/App.tsx, styles; one new browser test.
- **Data and security:** None.
- **Cost:** RM0
- **Owner:** Murshidi
- **Approval:** Approved: “Approve PLAN-005 only, including its listed tests and documentation” (2026-10-26)
- **Delivery:** Verified (PLAN-005)
- **Review trigger:** If playtesters still miss how to invite more than one friend.
- **Reports:** REP-007, REP-008

### DEC-013 · Publish version 1.1.1 to players

- **Question:** Should the 1.1.1 candidate (reconnect fix, 4-player clarity, release process) replace 1.1.0 on the live game?
- **Choice:** Recommended after the owner's 15-minute preview playtest passes. Publication happens only when the owner runs Deploy LIVE game and types PUBLISH.
- **Rationale:** Fixes a bug players can hit (stuck on Connecting after a reload or rejoin) and makes group play discoverable. The game-code change is small (4 files, about 40 lines), the server code is unchanged, and all automated checks pass.
- **Player impact:** Fewer stuck online screens; clearer 2 to 4 player options.
- **Technical impact:** Live Worker updated to the candidate commit and tagged v1.1.1.
- **Data and security:** No data or storage format change; rolling back is safe.
- **Cost:** RM0
- **Owner:** Murshidi
- **Approval:** Approved: “yes to 1 and 2 (1 = release 1.1.1: approve DEC-013, then the owner presses Deploy LIVE game)” (2026-10-26)
- **Delivery:** Released
- **Review trigger:** If the preview playtest finds a problem, or the post-deploy smoke test fails.
- **Reports:** REP-009, REP-013

### DEC-014 · Use the Concept and Theme Companion for creative work

- **Question:** Should Fruit Market's creative decisions follow the Game Concept and Theme Companion (edition 1.0) alongside the development workflow?
- **Choice:** Yes, with one shared record system (same reports, register and PDFs); creative approval never implies implementation approval.
- **Rationale:** Keeps creative and development decisions in one history and avoids a parallel register.
- **Player impact:** None directly.
- **Technical impact:** None; documentation process only.
- **Data and security:** None.
- **Cost:** RM0
- **Owner:** Murshidi
- **Approval:** Proposed
- **Delivery:** In progress
- **Review trigger:** If creative work starts needing its own package type.
- **Reports:** REP-010

### DEC-015 · Adopt three design pillars

- **Question:** Which design pillars guide Fruit Market's creative choices?
- **Choice:** 1) Fresh fruit, fast money. 2) Outprice rivals in secret. 3) A cheerful town market. Each with inclusion and exclusion rules (REP-010 section 3).
- **Rationale:** Derived from the actual rules and the strongest theme-to-mechanic matches found in the review.
- **Player impact:** Future features and content are judged against these.
- **Technical impact:** None now.
- **Data and security:** None.
- **Cost:** RM0
- **Owner:** Murshidi

- **Approval:** Approved: “yes to 1 and 2 (2 = approve DEC-015 and DEC-016)” (2026-10-26)
- **Delivery:** Implemented
- **Review trigger:** If playtests show players value something the pillars exclude.
- **Reports:** REP-010, REP-011, REP-013

### DEC-016 · First creative test: reveal-comprehension playtest

- **Question:** What is the smallest useful test of the riskiest creative hypothesis?
- **Choice:** 3 to 5 first-time players, Quick game vs 1 computer shop, no coaching; continue if at least 3 of 4 can explain a lost sale by day 3 (provisional, owner-set). No code.
- **Rationale:** Pillar 2 depends on players understanding why they won or lost a sale; no player evidence exists yet.
- **Player impact:** Testers play the existing game.
- **Technical impact:** None.
- **Data and security:** Notes only; no personal data kept.
- **Cost:** RM0
- **Owner:** Murshidi
- **Approval:** Approved: “yes to 1 and 2 (2 = approve DEC-015 and DEC-016)” (2026-10-26)
- **Delivery:** Not started
- **Review trigger:** After the first playtest round.
- **Reports:** REP-010, REP-011, REP-013

### DEC-017 · Setting: a market where the whole world shops

- **Question:** Where is Fruit Market's market?
- **Choice:** Option C: a deliberate 'the whole world shops here' mix. A sunny harbour market; customers from every culture and fantasy role; seaside palette; plain market dollars.
- **Rationale:** Keeps the existing art and cast, turns the mix into an intentional identity, and costs nothing now.
- **Player impact:** No visible change now; guides future content.
- **Technical impact:** None now; recorded in the creative bible.
- **Data and security:** None.
- **Cost:** RM0
- **Owner:** Murshidi
- **Approval:** Approved: “1. Where is this market? C: a deliberate “the whole world shops here” mix.” (2026-10-26)
- **Delivery:** Implemented
- **Review trigger:** If playtesters find the setting confusing.
- **Reports:** REP-011

### DEC-018 · Rename Panda Lim while keeping the cast coherent

- **Question:** Should the 'Panda Lim' rival be renamed?
- **Choice:** Yes: an animal nickname that fits beside Foxy Rosa, Big Benny, Uncle Toad, Rocky and Pinky; picture and lines unchanged. Cast rules recorded in the creative bible.
- **Rationale:** Avoids pairing a surname with a national symbol while preserving the animal-trader theme.
- **Player impact:** New games show the new name.

- **Technical impact:** One data line plus tests (PLAN-006).
- **Data and security:** Saved games keep the old name until they end.
- **Cost:** RM0
- **Owner:** Murshidi
- **Approval:** Approved: “Rename Panda Lim? - sure, but dont run away from the general theme coherence” (2026-10-26)
- **Delivery:** Verified (PLAN-006)
- **Review trigger:** If the new name tests poorly with players.
- **Reports:** REP-011, REP-012

### DEC-019 · Rename scope = PLAN-006, shipped in 1.1.2

- **Question:** How and when should the rename be built?
- **Choice:** As PLAN-006 (display name, reserved names, compatibility tests), released as 1.1.2 after 1.1.1 is live.
- **Rationale:** Keeps the 1.1.1 candidate unchanged; the rename is small and safe but still needs its own checks.
- **Player impact:** New name from 1.1.2.
- **Technical impact:** Small data change plus tests.
- **Data and security:** None.
- **Cost:** RM0
- **Owner:** Murshidi
- **Approval:** Approved: “Approve PLAN-006 only, including its listed tests and documentation, name: Patches” (2026-10-26)
- **Delivery:** Verified (PLAN-006)
- **Review trigger:** If 1.1.1 is delayed for long.
- **Reports:** REP-011, REP-012

### DEC-020 · UI and UX polish in three phases

- **Question:** How should the game's look and feel be made to feel finished?
- **Choice:** Phase 1 fixes 9 visible defects (PLAN-007, ships in 1.1.2); phase 2 calms the table layout and phase 3 adds a design system, motion and sound, both designed as phone-viewable mockups first (PLAN-008) and built only after the owner picks; playtest before and after.
- **Rationale:** The review found concrete causes (defects, clutter, no focus, inconsistent system). Fixing defects is cheap and safe; layout changes need the owner's eye and player evidence.
- **Player impact:** A cleaner, clearer game in steps.
- **Technical impact:** Presentation code only; no rules, saves or server.
- **Data and security:** None.
- **Cost:** RM0
- **Owner:** Murshidi
- **Approval:** Awaiting approval
- **Delivery:** Not started
- **Review trigger:** If the baseline playtest shows a different priority.
- **Reports:** REP-014, REP-015

### DEC-021 · Art direction: B, toy-box tabletop

- **Question:** Which art direction should the polish follow, given the owner dislikes the current art style and points to It Takes Two?
- **Choice:** Direction B: a smooth 2D toy-like tabletop style (rounded shapes, wood, felt and clay materials, warm light, soft shadows, motion on every action), drawn by Claude as vector art; a paid illustrator stays a later, separately priced option. The A/B/C style test is dropped.
- **Rationale:** B addresses clutter, feel and art style, keeps phones fast and costs nothing to test; C is too large and risky now; A does not answer the art-style concern.
- **Player impact:** A warmer, toy-like look with more motion, if chosen and built.
- **Technical impact:** Redraw of fruits, about 30 characters, rivals, about 40 icons and the UI (only after approval of a build plan).
- **Data and security:** None.
- **Cost:** RM0 for the test; any illustrator needs a quote and approval
- **Owner:** Murshidi
- **Approval:** Approved: “go with plan b” (2026-10-26)
- **Delivery:** In progress (PLAN-008)
- **Review trigger:** After the style test, or if the owner wants a paid illustrator.
- **Reports:** REP-015, REP-016

### DEC-022 · Look and tone: cunning edition of direction B

- **Question:** How should direction B reflect the game's undercutting mechanic?
- **Choice:** Keep the toy-box tabletop style and add a cheeky, scheming layer: plum and gold for secret moments (pricing, reveal, evening), wax-sealed face-down price boards, 'Psst' whisper bubbles that restate public facts, sly rival faces (Foxy Rosa winks), a spotlight reveal with an 'Outfoxed by $1!' headline, and the tagline 'Undercut. Outfox. Sell it fresh.'
- **Rationale:** The owner asked for the look to match the undercut mechanic. Cheeky rather than mean keeps pillar 3 (a cheerful town market) and pillar 2 (outprice rivals in secret) aligned; the market stays sunny by day.
- **Player impact:** Secret phases feel secret and the reveal feels like a showdown, if built.
- **Technical impact:** Same redraw as DEC-021 plus a plum secret-phase theme; only after a build plan is approved.
- **Data and security:** None. Whispers only restate information the game already shows; no hidden prices or choices.
- **Cost:** RM0
- **Owner:** Murshidi
- **Approval:** Approved: “yes but align more with the cunning aesthetic given the undercut mechanism of the game” (2026-10-27)
- **Delivery:** Implemented (PLAN-008)
- **Review trigger:** If playtesters read the rivals as mean, or the plum screens feel too dark.
- **Reports:** REP-017

### DEC-023 · Look: cosy pixel edition of direction B

- **Question:** The cunning edition lost the cosy feel and the toy-box art moved too far from pixel art. What should the look be?

- **Choice:** Go back to the first toy-box mockups (warm cream, teal and wood; the owner's own Evening layout edit) and use the game's own pixel characters, fruit and icons with its pixel heading font, inside the smooth, polished frame. Keep a light sly touch in warm colours: dotted 'Psst' whispers from Foxy Rosa, a sly-eyed Foxy Rosa and Patches, sealed prices, 'Outfoxed by $1!'. Drop the dark plum screens, curtain and spotlight.
- **Rationale:** Owner feedback: the first toy-box version felt cosy; the cunning edition did not; balance the art back toward pixel art. Reusing the game's sprites also means far less redraw work than a full vector cast.
- **Player impact:** Familiar pixel cast and fruit, cleaner screens, cosy warm colours, if built.
- **Technical impact:** Much smaller art job than DEC-021: reuse and tidy the existing sprites; the UI frame is the main build.
- **Data and security:** None. Whispers repeat only public information.
- **Cost:** RM0
- **Owner:** Murshidi
- **Approval:** Approved: “going back to your mockup, i like it now, but tweak the layout so that the fruit market is more visible” (2026-10-27)
- **Delivery:** Implemented (PLAN-008)
- **Review trigger:** Owner review of the cosy pixel mockups; if confirmed, DEC-022's plum layer is superseded.
- **Reports:** REP-018, REP-020

### DEC-024 · Layout: one shared market table (board game, not menus)

- **Question:** The cosy pixel mockups still felt like menus. How should the game screens be laid out?
- **Choice:** Every game screen is the same market table, like the current game: customers queue along the street at the top, Foxy Rosa's and Patches' stalls flank the market in the middle, your stall sits at the bottom. Pricing seals boards on each stall; the reveal flips the boards on the stalls themselves, the winner's stall glows, coins fly to it and the customer walks over; at night the stalls light lanterns and the market is covered. Keeps DEC-023's pixel art, warm colours and smooth buttons.
- **Rationale:** Owner: "revamp the current UI/UX to match the cosiness of the previous UI/UX board layout" and "the one currently feels very menu based, instead of board game". Seeing everyone at one table is what made the current game feel cosy.
- **Player impact:** The game feels like a board game you sit at, with less clutter than today.
- **Technical impact:** Reuses the current game's table structure; the build restyles it rather than replacing it.
- **Data and security:** None. Rival boards stay sealed until the bell; whispers repeat only public information.
- **Cost:** RM0
- **Owner:** Murshidi
- **Approval:** Approved: “going back to your mockup, i like it now, but tweak the layout so that the fruit market is more visible” (2026-10-27)
- **Delivery:** Implemented (PLAN-008)
- **Review trigger:** Owner review of the table mockups.
- **Reports:** REP-019, REP-020

### DEC-025 · Market board: vertical decay tracker and a delivery lorry

- **Question:** How should the market look and how should fruit arrive each day?
- **Choice:** The market is a vertical decay tracker: one column per fruit, left to right (blueberry, watermelon, apple, mango); new and ripening fruit at the top, spoiling fruit at the bottom; each slot shows its price, and empty slots stay visible. Your stall uses the same tracker in small strips. At the start of buying, a lorry drives in along the street, customers shuffle aside, and crates drop onto the board; it drives off when buying ends, before selling.
- **Rationale:** Owner's explicit instruction. The tracker is the game's existing decay rule made visible, and the lorry turns the daily delivery into a moment.
- **Player impact:** Easier to see what is fresh and what is about to spoil; a lively start to each day.
- **Technical impact:** Visual only: the tracks follow src/engine/data.ts exactly; no rule or balance change.
- **Data and security:** None.
- **Cost:** RM0
- **Owner:** Murshidi
- **Approval:** Approved: “change the board layout to a vertical layout arranging the fruits from left to right instead, revert back to the decay style tracker for the game board, at the start of the buying turn, animate a lorry offloading the fruit on to the board, the customers are to make space accordingly, the lorry would leave before the selling turn” (2026-10-27)
- **Delivery:** Implemented (PLAN-008)
- **Review trigger:** If the vertical board is too tall on small phones, or the lorry slows down play (keep it skippable).
- **Reports:** REP-021

## History (oldest first)

- 2026-10-26 13:15 · REP-001: baseline audit; PLAN-001 proposed; DEC-001 to DEC-005 proposed.
- 2026-10-26 13:20 · Owner: "Approve PLAN-001 only, including its listed tests and documentation". DEC-002 and DEC-003 approved; PLAN-001 approved.
- 2026-10-26 13:25 · PLAN-001 implemented on main (2e4b240). Push started CI and Android only; no live deploy.
- 2026-10-26 13:42 · REP-002: PLAN-001 verified except A4 (owner playtest). DEC-006 recorded, awaiting confirmation.
- 2026-10-26 13:46 · Owner: "Confirm DEC-006, and plan PLAN-002". DEC-006 approved. REP-003: PLAN-002 proposed; DEC-007 to DEC-009 awaiting approval.
- 2026-10-26 13:48 · Owner: "Approve PLAN-002 only, including its listed tests and documentation". PLAN-002, DEC-007, DEC-008 and DEC-009 approved.
- 2026-10-26 13:50 · Correction: the report headers of REP-002 (13:45) and REP-003 (13:50) were rounded estimates; the commit times 13:42 and 13:46 are used above. The reports are left as written.
- 2026-10-26 13:52 · PLAN-002 implemented on workflow/plan-002: records migrated; B1, B2, B3, B5 and B7 PASS; B4 and B6 pending.
- 2026-10-26 14:09 · PLAN-002 verified: B1 to B7 PASS (CI run 40276699547 on a7fa577). DEC-007, DEC-008 and DEC-009 delivered.

- 2026-10-26 14:09 · Known issue found: CI run 40275680793 (docs-only commit 63c3682) failed because the online test's page reload did not show the game within 20 s. Same symptom as the quick-match failure in CI run 1. Pre-existing and intermittent (2 of 5 CI runs); not part of PLAN-002. Proposed as the next package in REP-004.
- 2026-10-26 14:16 · Owner: "PLAN the reconnect fix". REP-005: PLAN-004 proposed; DEC-010 and DEC-011 awaiting approval.
- 2026-10-26 14:18 · Owner: "Approve PLAN-004 only, including its listed tests and documentation". PLAN-004, DEC-010 and DEC-011 approved. (Time approximate: between the REP-005 commit at 14:17 and the first work commit at 14:27.)
- 2026-10-26 14:53 · PLAN-004: root cause reproduced (lost first update after rejoining; 20 of 20 hidden-tab reloads stuck before the fix) and fixed in the client (9d9e911).
- 2026-10-26 15:01 · PLAN-004: R1 to R5 PASS (165 hidden-tab cycles after the fix, 0 failed). R6 waiting for 3 CI runs.
- 2026-10-26 15:34 · REP-006: PLAN-004 verified (R1 to R6 PASS). The fix is on the preview; the live game is unchanged until a release is prepared and deployed.
- 2026-10-26 18:34 · Owner asked how to play with 4 people and requested a plan. Check: 4-person online room works locally (4 joined and played; a 5th refused). REP-007: PLAN-005 proposed; DEC-012 awaiting approval.
- 2026-10-26 18:36 · Owner: "Approve PLAN-005 only, including its listed tests and documentation". PLAN-005 and DEC-012 approved.
- 2026-10-26 18:44 · PLAN-005 implemented: C1 to C5 PASS; C6 waiting for CI.
- 2026-10-26 19:01 · REP-008: PLAN-005 verified (C1 to C6 PASS). On the preview; goes live with the next release.
- 2026-10-26 19:02 · Correction: the REP-008 header time (19:05) was an estimate; the report was completed at 19:01. The report is left as written.
- 2026-10-26 20:26 · Owner: "PREPARE RELEASE". Version set to 1.1.1 (package and Android versionName); release candidate entry added; DEC-013 (publish 1.1.1) awaiting approval. REP-009.
- 2026-10-26 20:27 · CI run 13 (f472186, records-only commit) was cancelled: the browser download took 16 minutes, so the job hit its 20-minute limit. Not a code failure. Limit raised to 35 minutes in the 1.1.1 release commit.
- 2026-10-26 21:47 · Owner asked to apply the Game Concept and Theme Companion. REP-010: theme review (no changes made); DEC-014 proposed; DEC-015 (pillars) and DEC-016 (first playtest) awaiting approval. Open questions: setting identity, renaming Panda Lim, availability of 3 to 5 testers.
- 2026-10-26 22:14 · Owner answers to REP-010: setting C approved (DEC-017); rename Panda Lim approved, keeping the cast coherent (DEC-018); 3 to 5 first-time testers are available. REP-011: creative bible written; PLAN-006 (rename) and DEC-019 awaiting approval. DEC-015 and DEC-016 still await explicit approval.
- 2026-10-26 22:17 · Owner: "Approve PLAN-006 only, including its listed tests and documentation, name: Patches". PLAN-006 and DEC-019 approved; new name Patches.
- 2026-10-26 22:18 · PLAN-006 implemented on workflow/plan-006: N1 to N4 PASS (5 new tests); N5 waiting for CI.
- 2026-10-26 22:48 · REP-012: PLAN-006 verified (N1 to N5 PASS) on workflow/plan-006. Waits off main until 1.1.1 is live; then ships as 1.1.2.
- 2026-10-26 22:56 · Owner: "yes to 1 and 2". DEC-013 (publish 1.1.1) approved; the owner runs Deploy. The owner's own preview playtest result was not reported (recorded as NOT RUN).

DEC-015 (pillars) and DEC-016 (first playtest) approved. Recorded on workflow/plan-006 so main stays on the 1.1.1 candidate.

- 2026-10-26 22:57 · Correction: the REP-013 header time (22:58) was an estimate; the report was written at 22:56. The report is left as written.
- 2026-10-26 23:06 · Owner: the game doesn't feel polished; propose changes on the whole UI and UX. REP-014: review of all main screens at 360×640 and 390×844; DEC-020, PLAN-007 (phase 1 fixes) and PLAN-008 (mockups) awaiting approval. PDF export now inlines report images.
- 2026-10-26 23:18 · Owner answers to REP-014: most bothered by clutter, lack of animation and feel, and the art style itself; polished reference: It Takes Two. REP-015: art-direction options; PLAN-008 revised to a style test plus mockups (still awaiting approval); DEC-021 awaiting approval; PLAN-007 suggested to shrink to 3 fixes. PLAN-007 and PLAN-008 not yet approved.
- 2026-10-26 23:50 · Owner: "go with plan b". DEC-021 approved: art direction B (toy-box tabletop). Interpreted also as approval to proceed with PLAN-008 in B only (design-only mockups; the 3-way style test dropped); owner told they can stop it. PLAN-007 and DEC-020 still awaiting a decision.
- 2026-10-26 23:57 · PLAN-008 mockups published (https://claude.ai/artifact/YQUpnHeNNWkAUqFhKRz9VB): style sample plus home, buying, pricing, sales reveal and evening in direction B. REP-016. Waiting for the owner's review.
- 2026-10-27 00:12 · Owner on the REP-016 mockups: "yes but align more with the cunning aesthetic given the undercut mechanism of the game". DEC-022 approved (cunning edition of direction B). Mockups revised and republished at the same link; owner's Evening layout edit kept. REP-017. Waiting for the owner's review of the revision.
- 2026-10-27 00:17 · Owner: "its still not enough, maybe balance the artstyle back to pixel art, the ui/ux here looks more polished but lacks the cosiness of how the previous board was set", then "no i meant the version before you tweaked here, that felt cosy". Mockups rebuilt from the pre-cunning version (owner's Evening edit kept) with the game's pixel sprites and font and a light, warm sly touch; dark plum screens dropped. DEC-023 awaiting the owner's confirmation. REP-018.
- 2026-10-27 00:25 · Owner: "revamp the current UI/UX to match the cosiness of the previous UI/UX board layout" and "the one currently feels very menu based, instead of board game". Buying, pricing, reveal and evening mockups rebuilt around one shared market table (customers on the street, rival stalls either side of the market, your stall below), keeping the pixel art and warm frame. DEC-024 awaiting confirmation. REP-019.
- 2026-10-27 01:21 · Owner: "going back to your mockup, i like it now, but tweak the layout so that the fruit market is more visible (now it feels to small/ narrow)". DEC-023 (cosy pixel look) and DEC-024 (one shared table) approved. Rival stalls slimmed and the market widened with bigger crates and fruit; republished (version 13). REP-020. DEC-022's dark plum layer is superseded by DEC-023. Next: propose the 1.2.0 build plan.
- 2026-10-27 01:28 · Owner asked for a vertical decay-tracker market (fruits left to right), the decay tracker on your stall, and a delivery lorry that arrives at the start of buying (customers make space) and leaves before selling. DEC-025 approved by that instruction. Buying mockup rebuilt (version 14). REP-021.
- 2026-10-27 01:31 · Owner: "revert back the tracker on player's stall". DEC-025 narrowed: the vertical decay tracker is for the market only; your stall goes back to fruit chips showing nights left. Buying mockup republished (version 15).

- 2026-10-27 01:32 · Owner supplied the delivery lorry art ("use this as the lorry"). Saved as docs/workflow/reports/img/owner-lorry.png; the Buying mockup uses a cut-out of it (background removed, resized). Version 16. Before the build, confirm the picture's origin and that we may use it in the game.
- 2026-10-27 01:35 · Owner: the lorry picture was made with Gemini; asked to trim it so no ground shows below. Ground and background removed (wheels kept); cut-out saved as docs/workflow/reports/img/owner-lorry-cutout.png; Buying mockup republished (version 17). Note for the build: AI-made art may not be protected by copyright, so others could reuse it; check Gemini's terms and Google Play's AI-content rules before publishing it in the game.
- 2026-10-27 01:36 · Owner: "rescale the customers, now it looks disproportionate to the lorry". Street customers shrunk from 54px to 36px on buying, pricing and reveal so they match the lorry (68px tall). Version 18.
- 2026-10-27 01:38 · Owner: "keep the demand supply indicator from the evening's build inside the buying phase". The current game's want / in-shops strip (same meaning and colours as src/ui/table.tsx pulse) added to the Buying mockup above the stalls; it updates as you buy. The street notes it repeated were removed. Version 19.
- 2026-10-27 01:43 · Owner: move the other players off the left and right so the board is the main focus. Rivals now sit in a slim row of seats across the top of the table (portrait, cash, what they hold; sealed board when pricing, flipping board at the reveal, lantern at night). The market board takes the full width; the want / in-shops chips now sit on top of each fruit column. Applied to buying, pricing, reveal and evening. Version 20.
- 2026-10-27 01:46 · Owner: tap a rival's name in the buying phase to expand their stall tracker and see freshness. Added to the Buying mockup: tapping Foxy Rosa or Patches opens a drop-down decay tracker of their stall (same columns and colours as the market, counts per slot, a one-line summary); tap again or × to close. Shows only what the current game already shows about rival stock (no prices or choices). Version 21.
- 2026-10-27 01:48 · Owner: make the expanded rival tracker show freshness the same way as your stall. The rival drop-down now uses the same fruit chips as your stall (fruit at its freshness, plus nights left) instead of the grid. Version 22.
- 2026-10-27 01:48 · Owner: "sort the fruit from top to bottom aligning at the top". Market columns are now top-aligned: each starts with its newest slot and runs down to not-so-fresh; shorter tracks (blueberry) end early. Version 23.
- 2026-10-27 01:52 · Owner: remove the freshness bar beside the board and just colour-code freshness. Side bar removed; every market slot, empty ones included, is tinted by its freshness (unripe pale green, fresh green, okay gold, not so orange-brown) with a colour key under the board; your stall and rival chips use the same colours. Version 24.
- 2026-10-27 01:58 · Owner: "ok now do this changes", read as: build the mockup look into the game. Under the workflow this needs an approved plan, so PLAN-009 (table and board), PLAN-010 (lorry, outline) and PLAN-011 (pricing, reveal, evening, outline) were proposed; awaiting approval. No game code changed. REP-022.
- 2026-10-27 01:59 · Owner: "yes" to REP-022. Read as approval of PLAN-009 (started) and of folding PLAN-007's text fixes into PLAN-011. The lorry art choice (Gemini picture or pixel lorry) is still open; it only affects PLAN-010.
- 2026-10-27 02:19 · PLAN-009 built and verified on branch workflow/plan-009 (b2b495a): verify and verify:full PASS, new board tests PASS, 9-size screenshot check PASS. Not merged: main holds the 1.1.1 candidate until the owner deploys it. REP-023.

- 2026-10-27 02:20 · Owner: "use the gemini lorry, approve PLAN-011". PLAN-011 approved and started (pricing, reveal and evening on the table). PLAN-010 will use the owner's Gemini lorry picture; owner accepts that AI-made art may not be protected by copyright.
- 2026-10-27 02:51 · PLAN-011 built and verified on workflow/plan-009 (4408cd8): verify and verify:full PASS (leak check 0), screenshots at 3 sizes. PLAN-007 closed as superseded; its text fixes shipped in PLAN-011. Customers walking to the winner was not built (noted in scope). REP-024. Remaining for 1.2.0: PLAN-010 (lorry). Main still waits for the 1.1.1 deploy.
- 2026-10-27 03:26 · Owner: "deploy". Deploy LIVE game run 15 (40385840230) on main 83de99b succeeded, including its live smoke test; tagged v1.1.1. Release 1.1.1 marked published. Then workflow/plan-009 (PLAN-006 Patches rename, PLAN-009, PLAN-011, creative records) fast-forwarded into main so the preview updates; these will ship together as the next release (planned 1.2.0, with PLAN-010).
- 2026-10-27 03:39 · Owner asked for a full audit of how the game is won, lost, progressed and paced, without code changes. REP-025: rules read line by line plus computer-only simulations (balance at 300 games per test and 4 read-only probes). Main findings: league difficulty cliff at stages 5-6, 4-player economy squeezed, snowballing, Juice Press has no measurable value, trait spread 29-42%, hidden seat tie-break, final-day night events do nothing. No decision taken; proposals need plans and approval.
- 2026-10-27 07:39 · Owner: "approve" (reply to the request to approve PLAN-010). PLAN-010 approved and started: delivery lorry with the owner's Gemini art. The league fix from REP-025 is not yet planned.
- 2026-10-27 07:54 · PLAN-010 built and verified (bc2e760): lorry with the owner's Gemini art; verify and verify:full PASS. Merged to main for the preview. All three cosy-table packages (PLAN-009, 010, 011) plus the Patches rename are now on main, ready to become release 1.2.0 when the owner asks.
- 2026-10-27 09:11 · Owner asked for a balance redesign (fruit identity, strategy depth, pacing) with analysis and a patch plan only. REP-027: 813 lab test cells (384,200 computer games) on a separate engine copy, validated on held-out seeds. Proposed PLAN-012 (tighter town, Handcart no longer grows the market, 4-player demand +15%, apple cost 3/2/1, RULES_VERSION 3). Production rules unchanged; awaiting approval.
- 2026-10-27 09:12 · Record note: the PLAN-010 verification (history line of 07:54) is written up in REP-026.
- 2026-10-27 09:58 · Owner: "yes approvrd together with the optional changes and the new layout from the mockup" (reply to the PLAN-012 approval request). PLAN-012 approved and started, widened to the six optional items of REP-027 section 6b and the Home screen from the cosy pixel mockup (the only mockup screen not yet built; interpretation stated to the owner). Optional items still needing values are set by lab test first.
- 2026-10-27 10:47 · PLAN-012 built and verified on workflow/plan-012 (REP-028): rules version 3 with the market rebalance, trait and mango changes, tomorrow's lorry, Juice Press juice tip, retuned Market League and the cosy home screen. verify and verify:full PASS; production equals the lab package on 450 games. Not changed after lab tests failed: Long-mode tuning and Cool Room rework. Skill gap (Hard beats Normal) 60%, below the 65% aim, flagged to the owner. Playtest NOT RUN. Merged to main for the preview; the live game is unchanged.
- 2026-10-27 12:03 · Owner: "edit the UI/UX so that 1. secret price and events pops up to fill the screen (instead of just the bottom half of the screen 2. rework the rivals indicator during

secret price, now its not clear still". Explicit instruction, recorded as approval of PLAN-013 (UI only, no rule change); started on workflow/plan-013.

- 2026-10-27 12:21 · PLAN-013 built and verified (REP-029): full-screen pricing and event sheets, 'Your rivals' strip (sealed / deciding / next in line, wanted fruit held), matching seat badges. verify and verify:full PASS (price leaks 0). Merged to main for the preview; live game unchanged.
- 2026-10-27 13:00 · Owner asked to re-analyse fruit pricing, decay, events, traits, upgrades and supply/demand against three guidelines (tabletop-playable, real market feel, fruit playstyle matching payout). REP-030: rules version 3 only partly meets them. Lab sweeps 12-15 (600 held-out games per test): printed numbers reproduce the formula flow; posted prices plus a market pulse plus blueberry/apple price trims give a 12-point plan spread and payouts ordered mango > blueberry > watermelon > apple. Proposed PLAN-014 (awaiting approval). No game code changed.
- 2026-10-27 13:23 · Owner: "approve all the idea, work out the mechanics, and summarise to me game back highlighting the concept and game mechanics as i want to benchmark against similar genre board games and traditional popular board games". PLAN-014 approved including the supporting ideas (queue, Market Days, crates, event redesign). First deliverable: finalised mechanics and a concept summary for benchmarking; code build follows.
- 2026-10-27 13:46 · REP-031: final tabletop mechanics (rules version 4) and a concept and mechanics summary for benchmarking. Lab sweeps 16-18: fruit-specific events (Blueberry Glut, Apple Blight), Good Harvest scaled by players, no Inspector on the final day; Market Days kept as an optional rule (most even plans but computers can't plan ahead). Open: Shop Bell, Juice Press and Cool Room retune; Long games favour ripening. Also fixed the coach tip covering the Seal button (the cause of the stuck CI run).
- 2026-10-27 14:36 · Owner shared a benchmark review of REP-031 and asked for tweaks plus a favour card that reveals the face-down demand. REP-032: five-beat day (open stalls, bell, rush), prototypes A/B/C, Market Gossip replacing Free Cash. Lab sweep 19: cash catch-up has no exploit (upgrader 33%); uniform freshness improves balance (26-38%); core without traits/favours/upgrades lets ripening dominate (60%). PLAN-015 proposed; PLAN-014 build held until the owner decides.
- 2026-10-27 14:51 · Owner: "approve plan 15". PLAN-015 approved and started on workflow/plan-015; it is built together with the approved PLAN-014 items as rules version 4.
- 2026-10-27 15:37 · PLAN-015 with the PLAN-014 items built as rules version 4 (REP-033): open stalls, the bell, Market Days and Last Call, four specialities, Market Gossip with server-side privacy, day ledger, playtest set-up options, rules text. verify and verify:full PASS (leak check 0). Balance: Hard beats Normal 56%, comebacks 31%, traits 30-38%; Cold Room and Juice Bar weak for computers; little undercutting among computers. Playtest NOT RUN. Merged to main for the preview; live game unchanged.

# REP-001 · Baseline audit and PLAN-001 (Package A)

| | |
|---|---|
| Game | Fruit Market ( `murshidihashim/Fruit-Market` ) |
| Report date | 2026-10-26 13:15 (Asia/Kuala_Lumpur) |
| Build audited | `main @ c915e49` (CI run 2 green, deployed) |
| Prepared by | Claude (studio lead), from the "Personal workflow for Murshidi", edition 1.2 |
| Type | Planning only. **No game implementation changes were made.** |
| Approval status | PLAN-001 **awaiting approval** |

## 1 · Baseline: where the game stands

**Stage:** between *MVP / private alpha* and *browser beta*. The game is complete and publicly reachable, but no unfamiliar players have been observed yet, so player understanding and fun are unproven.

### Working and must be preserved

- Rules engine: deterministic, seeded, a single entry point ( `applyAction` ), rules version 2. 63 unit tests, including 200 simulated games.
- Saves: versioned envelope ( `v2` plus rules version); old saves still load.
- Online play is server-authoritative (Cloudflare Worker with Durable Objects):
  - secret prices are redacted; the leak check reads every network message and found 0 leaks
  - reconnect, a 75 s grace period then computer takeover, and quick match
  - per-IP rate limits, a name policy, and a replay-verified daily leaderboard
- Android build (Capacitor) is produced by CI on every push.
- Docs: `RULES.md` , `ONLINE.md` , `PUBLISHING.md` , `docs/ANALYTICS.md` , the studio audit `docs/studio/01–10` (backlog = `10` ).

### Gaps against the workflow guide

| # | Finding | Type | Severity |
|---|---|---|---|
| G1 | Every push to `main` deploys straight to the public URL. The guide requires a separate publication approval. | Process defect | High |
| G2 | `main` has **no branch protection** (checked via GitHub: `protected: false` ). CI results don't block anything. | Measured | High |
| G3 | No single verification command. There are separate `lint` , `typecheck` , `test` , `build` , `test:e2e` scripts. | Measured | Medium |
| G4 | No `CLAUDE.md` , current-state record, game brief, or owner brief. Each session relies on long chat history. | Measured | Medium |
| G5 | No decision records or PDF register before this report. | Measured | Low |
| G6 | Real-device testing (Android Chrome, iPhone Safari): never done. | NOT RUN | High for release |

| # | Finding | Type | Severity |
|---|---|---|---|
| G7 | No backup or restore for server data (leaderboards, stats). Room crash recovery not tested. | NOT RUN | Medium |
| G8 | Monthly hosting cost unknown (Cloudflare plan not recorded). | Unknown | Medium |
| G9 | The game-screen open time is noisy (217–489 ms locally). Quick match failed once in CI and passed on the rerun. | Measured, pre-existing | Low |

## 2 · Readiness baseline (evidence as of today)

| Gate | Check | Result | Evidence and limitation |
|---|---|---|---|
| Build ready | Clean install, lint, typecheck, unit tests, build | PASS | CI run 40266399588 on `c915e49` |
| Play ready | Automated solo game to results | PASS | `scripts/smoke.cjs` , local and CI |
| Play ready | Owner completes a match in a preview | NOT RUN | Needs you |
| Online ready | Two independent clients finish a match; results agree | PASS | `e2e-online.cjs` , local and CI run 2 |
| Online ready | Private state: no secret prices in any message | PASS | 40 frames checked, 0 leaks |
| Online ready | Illegal actions, wrong phase, votes, takeover | PASS | `tests/room.test.ts` (16 tests) |
| Online ready | Reconnect, abandonment, quick match | PASS | e2e; one CI flake, then a pass |
| Online ready | Server crash recovery | NOT RUN | Room state is persisted per action; never tested |
| Release ready | Real Android Chrome and iPhone Safari | NOT RUN | Emulation only |
| Release ready | Backup and restore, rollback drill | NOT RUN | — |
| Release ready | Live smoke test after deploy | NOT RUN | The sandbox proxy blocks the live URL; the deploy job itself succeeds |
| Release ready | Publication approval gate | FAIL | See G1 |

Release-ready is **not claimed**.

## 3 · PLAN-001 · Package A: baseline, one verification path, preview and owner brief

**Player problem:** updates reach players without your approval and without one clear pass/fail check, so a bad change could ship silently. You also have no short guide for trying a build yourself.

**Included scope**

1. **One verification command.** `npm run verify` runs lint, typecheck, unit tests and build, and stops at the first failure (about 1 minute). `npm run verify:full` adds the browser suite (about 15 minutes). CI calls `verify:full`. Existing scripts are reused; nothing new is installed.
2. **Preview before publication.**
   - Pushes to `main` run checks only.
   - Production deploys happen only when you trigger *Deploy* (a manual GitHub button), which first runs `verify`.
   - Each change also gets a private Cloudflare **preview URL** that doesn't touch the live game. I'll confirm in Cloudflare's current docs that this is free on your plan; if it isn't, the local preview steps are the fallback.
3. **Branch protection instructions.** A 5-step guide for you to turn it on in GitHub settings (I can't change repository settings). It requires the CI check before merging.
4. **Short records** (reusing the studio docs, not duplicating them):
   - `CLAUDE.md`, about 90 lines: rules, actual commands, preservation constraints, pointers
   - `docs/workflow/STATE.md`, about 500 words
   - `docs/workflow/GAME-BRIEF.md`: audience hypothesis, core loop, session length, exclusions
5. **Owner brief.** `docs/workflow/OWNER-BRIEF.md`: how to open a preview, a 10-minute playtest checklist, and what each check means in plain English.

**Excluded:** any change to rules, balance, art, saves, the server protocol or hosting; paid services; publication; Package B (record automation) and Package C (devices, backups, capacity).

**Roles:** producer (scope), DevOps and release (deploy gate, CI), QA (verify command), technical lead (`CLAUDE.md`). Self-review only; low risk.

**Likely files:** `package.json` (scripts), `.github/workflows/{ci,deploy-web}.yml`, new `CLAUDE.md`, new `docs/workflow/{STATE,GAME-BRIEF,OWNER-BRIEF}.md`, `README.md` (pointer).

**Acceptance (observable)**

- A1. `npm run verify` exits 0 on the current code. A deliberately broken test makes it exit non-zero (shown, then reverted).
- A2. Pushing to `main` does **not** change the live site. Running *Deploy* does, and only after `verify` passes.
- A3. A preview URL (or documented local preview) lets you play the build before you approve publication.
- A4. You can follow `OWNER-BRIEF.md` to open a preview and finish a match, without asking me anything.
- A5. `CLAUDE.md` stays under 120 lines; `STATE.md` stays at or under about 600 words.

**Checks:** `verify` (PASS and FAIL demonstration), one CI run, one manual deploy dry run (after your separate publication approval, or skipped and marked NOT RUN), and the line and word counts.

**Cost and data:** RM0 extra. No player data is touched.

**Risks:** a preview URL feature may need a Cloudflare setting (fallback: local preview). If you forget to press *Deploy*, fixes wait (that's the point).

**Rollback:** revert the package commit; the deploy workflow returns to deploying on every push.

## 4 · Packages B and C (outline only; detailed later)

- **PLAN-002 · Package B, records and PDFs:**
  - one canonical record per package (Markdown with an ID header)
  - a deterministic local script that builds the owner brief, changelog, report PDFs and cumulative register, using pandoc and the project's existing Playwright browser (no new dependency)
  - export failures are labelled PDF EXPORT BLOCKED
- **PLAN-003 · Package C, online, capacity and release evidence:**
  - real-device checklist runs (you plus your phones)
  - Durable Object data export and restore drill
  - a room crash-recovery test
  - a small load test within an approved budget, with written workload assumptions (peak concurrent players, 2–4 per room, about 15 min matches)
  - cost record with a dated Cloudflare quote
  - live smoke test from an unrestricted machine (CI)

## 5 · Decisions

| ID | Question | Recommendation | Status | Delivery |
|---|---|---|---|---|
| DEC-001 | Adopt the workflow guide 1.2 for Fruit Market? | Yes; one package at a time | Proposed | Not started |
| DEC-002 | What goes in Package A? | PLAN-001 scope above | Awaiting approval | Not started |
| DEC-003 | When does production deploy? | Only on your manual *Deploy* after `verify`; never on push | Awaiting approval | Not started |
| DEC-004 | PDF tooling | pandoc plus the existing Playwright browser; no installs | Proposed | Used for this report |
| DEC-005 | Budget interpretation | RM500/month portfolio-wide ceiling; current Fruit Market spend unknown (hypothesis: Cloudflare Free, or Workers Paid at about USD 5) | Proposed (assumption) | Not started |

Rationale for DEC-003: it is the only way to make "a preview is not a release" true for this repo. Trade-off: one extra tap per release.

## 6 · Questions for you (max 3)

1. **Devices:** which real phones can you test on (Android model, iPhone model)? This decides Package C.
2. **Hosting cost:** is your Cloudflare account on the Free plan or Workers Paid? Any other paid accounts yet (domain, Google Play developer account)?
3. **Approve PLAN-001** as written (including DEC-003)? Or change it?

## 7 · Sources

- Repository inspection, `main @ c915e49`, 2026-10-26.
- GitHub branch API (`main protected: false`), 2026-10-26.
- CI run 40266399588 (green), deploy run 40266399518 (success), Android run 40266399555 (success).
- Workflow guide: "Personal workflow for Murshidi", consolidated edition 1.2, 26 Oct 2026.
- Preview-URL and plan-pricing facts to be confirmed against current Cloudflare docs during PLAN-001 (not yet verified).

# REP-002 · PLAN-001 (Package A) implemented and verified

| | |
|---|---|
| Game | Fruit Market |
| Report date | 2026-10-26 13:45 (Asia/Kuala_Lumpur) |
| Package | PLAN-001, approved by the owner 2026-10-26: "Approve PLAN-001 only, including its listed tests and documentation" |
| Build | `main @ 2e4b240` ; records on `workflow/plan-001` |
| Live game | **Unchanged** (still `c915e49` ). No publication approval was given and none was performed. |
| Delivery status | Implemented; verified except A4, which needs the owner |

## What changed

### Added

- `npm run verify` : lint, types, unit tests, build; stops at the first failure.
- `npm run verify:full` : `verify` plus browser tests.
- A **preview** copy of the game, updated automatically after CI passes on `main` , then smoke-tested: https://fruit-market-staging.murshidi-hashim.workers.dev. It is a separate Worker with its own rooms, leaderboard and stats.
- `scripts/smoke-live.mjs` : after a deploy, checks that the page loads and the server runs the expected rules version.
- `CLAUDE.md` (72 lines), `docs/workflow/STATE.md` , `GAME-BRIEF.md` , `OWNER-BRIEF.md` (preview, playtest checklist, publishing, branch-protection steps).

### Changed

- The live deploy now runs only through GitHub Actions → *Deploy LIVE game* → type `PUBLISH` . It runs `verify` first and a live smoke test after.
- CI runs `verify:full` . README and ONLINE.md describe the new path.
- Lint now covers the docs export script (it found 12 errors in it, which `verify` caught; now fixed).

**Removed:** automatic live deploys on every push to `main` .

**Player impact:** none yet. Players only benefit once you publish, because broken changes can no longer reach them silently.

## Acceptance results

| # | Criterion | Result | Evidence |
|---|---|---|---|
| A1 | `verify` exits 0 on current code; non-zero with a broken | PASS | Local: exit 0 (63 tests, build), then exit 1 with one test deliberately broken ("1 failed, 62 passed"), then exit 0 |

| # | Criterion | Result | Evidence |
|---|---|---|---|
| | test | | once restored |
| A2 | Push to `main` does not change the live site | PASS | Push of `2e4b240` started only *CI* (run 40274184197) and *Android build*; no live deploy run |
| A2 | *Deploy* runs only after `verify` passes | PASS (by design) / NOT RUN (live) | `verify` is the step before deploy; not run because publication isn't approved |
| A3 | A preview lets you play before publishing | PASS | Preview job: "Deployed fruit-market-staging", then "SMOKE OK … rules version 2" (version `adf6c304`) |
| A4 | Owner can open the preview and finish a match using the brief | NOT RUN | Needs you (OWNER-BRIEF §1–2) |
| A5 | `CLAUDE.md` under 120 lines; STATE about 500–600 words | PASS | 72 lines; STATE about 350 words |
| — | Full CI (`verify:full`) on `2e4b240` | PASS | Checks job green in 14 min: lint, types, 63 tests, build, solo game, performance budget, online match with a leaver (0 leaks), quick match |
| — | Smoke script fails on a dead address | PASS | Local: "SMOKE FAILED … fetch failed", exit 1 |
| — | Preview online rooms (Durable Objects) in a real browser | NOT RUN | The smoke test checks the page and API only. Covered by the optional online step in the playtest checklist |

## Deviation from the plan (needs your confirmation)

PLAN-001 said "a private Cloudflare preview URL; if it isn't free, local preview steps". Cloudflare's docs state that preview and version URLs **are not generated for Workers that implement a Durable Object**, and Fruit Market's server uses Durable Objects for rooms. So I used the next closest option: a second, separate Worker (`fruit-market-staging`). This is recorded as **DEC-006**.

- Cost: no new paid service. It runs within the Cloudflare plan you already use; usage is tiny. Your plan type is still unknown (DEC-005).
- "Private" means unlisted: anyone with the link can open it.
- To undo: delete the `fruit-market-staging` Worker in Cloudflare and remove the `preview` job.

## Risks and costs

- Expected extra cost: RM0 (no new paid service). Not yet confirmed against your plan type.
- Until branch protection is on, a direct push can still skip CI, but it can no longer reach players without *Deploy*.
- If CI is red, the preview isn't updated; the live game is unaffected.

## Decisions

| ID | Decision | Approval status | Delivery |
|---|---|---|---|
| DEC-002 | Package A scope = PLAN-001 | Approved (owner, 2026-10-26) | Verified (A4 pending) |
| DEC-003 | Live deploys only via manual *Deploy* after `verify` | Approved (owner, 2026-10-26, as part of PLAN-001) | Verified |
| DEC-006 | Preview = separate staging Worker, because per-version URLs don't support Durable Objects | Awaiting confirmation | Implemented |

## Sources

- Cloudflare docs, Version URLs and Preview URLs (`developers.cloudflare.com/workers/versions-and-deployments/version-urls/` and `/preview-urls/`): not generated for Workers that implement a Durable Object. Retrieved 2026-10-26 via search.
- Wrangler config reference vs the Durable Objects environments page disagree on per-environment migrations; resolved empirically: the staging deploy with top-level migrations succeeded.
- CI run 40274184197 (checks and preview jobs green).

# REP-003 · DEC-006 confirmed; PLAN-002 (Package B) proposed

| | |
|---|---|
| Game | Fruit Market |
| Report date | 2026-10-26 13:50 (Asia/Kuala_Lumpur) |
| Build | `main @ 63c3682` (live game still `c915e49`) |
| Type | Planning only. **No game implementation changes were made.** |
| Approval status | DEC-006 **approved** (owner, 2026-10-26: "Confirm DEC-006"). PLAN-002 **awaiting approval** |

## 1 · DEC-006 confirmed

The preview stays a separate staging Worker (`fruit-market-staging`). Approval status: approved; delivery: verified (REP-002, A3).

## 2 · Problem Package B solves

Records exist, but they're kept by hand:

- Each decision's status is typed in two or three places (the register, every report that mentions it, STATE). Nothing checks that they agree. REP-002 already needed a manual word-count correction.
- Nothing stops an approval being recorded without your words as evidence, or history being edited.
- There's no player-facing changelog and no version tags, so "what changed in this release" can't be answered from the repo.
- The PDF export works (built in REP-001) but isn't part of any check.

Players never see this directly. It matters because it keeps approvals honest and makes releases traceable when there's money or player data involved later.

## 3 · PLAN-002 · Package B: canonical records, generated views, checks

**Included scope**

1. **One canonical source per record** (JSON, no new tools):
   - `docs/workflow/records/decisions.json`: every DEC with question, choice, rationale, impacts, owner, approval evidence (your exact words and date), delivery status, links, review trigger, supersession. A `history` list is only ever appended to.
   - `docs/workflow/records/PLAN-###.json`: one per package with scope, exclusions, acceptance, approval evidence, changes (Added / Changed / Fixed / Removed), checks, build, release state.
   - Reports (`reports/REP-###.md`) stay as hand-written narrative snapshots and are never rewritten.
2. **Generated views**, built by `npm run records` with a deterministic script:

    - `DECISIONS.md`: current decisions and pending approvals first, then the full history.
    - `CHANGELOG.md`: player-facing, grouped by release, with "unreleased" vs "published" state. It is seeded with one summary entry for everything before the workflow, taken from git history.
    - The "Pending approvals" block inside `STATE.md` (between markers; the rest stays hand-written).
    - PDFs: each report, plus the cumulative register (current decisions, then pending approvals, then all reports in order).
3. **Checks**, via `npm run records:check`, added to `npm run verify` so CI runs it. It fails when:
    - an ID is duplicated or skipped, or a report mentions a DEC, PLAN or REP that doesn't exist
    - a decision is "approved" without owner evidence, or a delivery status isn't one of the allowed values
    - a history entry was changed or removed compared with the last commit (append-only)
    - a generated view is out of date with its source
4. **Release tagging**: when you publish, the deploy records the release in the changelog and tags the commit (`v1.1.1` and so on), so every live build can be traced. This only runs when you press *Deploy*.
5. **Migration**: move DEC-001 to DEC-006, PLAN-001 and REP-001 to REP-003 into the new records without losing anything.

**Excluded**

- Game code, rules, server and hosting.
- Package C (devices, backups, capacity).
- A cross-game studio register (NOT APPLICABLE until there's a second game).
- Installing anything. PDF building stays local because it uses pandoc. If pandoc is missing, the script prints PDF EXPORT BLOCKED and keeps the last good PDFs; it isn't part of CI.

**Roles:** producer (record design), technical lead (scripts), QA (failure cases), DevOps (tagging). Self-review only; low risk.

**Likely files:** new `docs/workflow/records/*.json`, new `scripts/records.mjs`, new `CHANGELOG.md`, `docs/workflow/export-pdf.cjs` (reads the records), `package.json` (scripts), `.github/workflows/deploy-web.yml` (tag and changelog step), `CLAUDE.md` and `OWNER-BRIEF.md` (one line each).

**Acceptance (observable)**

- B1. `npm run records` rebuilds all views. Running it twice in a row changes nothing (no git diff in the Markdown outputs).
- B2. `records:check` fails, with a plain-English message, for each of these deliberate breaks (shown, then reverted):
    - a duplicate ID
    - an unknown DEC referenced in a report
    - "approved" without evidence
    - an edited history line
    - a stale `DECISIONS.md`

- B3. With pandoc made unavailable, the export prints PDF EXPORT BLOCKED, exits non-zero and leaves the previous PDFs untouched.
- B4. The cumulative PDF opens with current decisions and pending approvals, followed by REP-001 to REP-003 in order.
- B5. After migration, every decision, status and approval quote from REP-001 to REP-003 appears in the records.
- B6. `npm run verify` (and CI) runs `records:check` and stays green on the migrated records.
- B7. Tagging is tested on a throwaway branch: a dry run shows the tag and changelog entry it would create. The real tag happens only at your next publication.

**Checks:** B1–B7, plus `verify`, plus one CI run.

**Cost and data:** RM0. No player data. `npm run verify` gets about 1 second slower.

**Risks:**

- Record-keeping overhead: JSON is less friendly to edit by hand. Mitigation: Claude edits the records; you only read the generated Markdown and PDFs.
- The append-only check compares with the last commit, so a deliberate correction needs a new "correction" history entry rather than an edit. That is intended.

**Rollback:** revert the package commits. The hand-written `DECISIONS.md` and reports remain valid.

## 4 · Decisions

| ID | Question | Recommendation | Status | Delivery |
|---|---|---|---|---|
| DEC-006 | Preview = separate staging Worker | Keep | Approved (owner, 2026-10-26) | Verified |
| DEC-007 | Package B scope = PLAN-002 | As above | Awaiting approval | Not started |
| DEC-008 | Canonical record format | JSON records plus Markdown narrative reports; Markdown, changelog and PDF views generated | Awaiting approval (part of PLAN-002) | Not started |
| DEC-009 | Records check in `verify` and CI | Yes, so disagreements block merges once branch protection is on | Awaiting approval (part of PLAN-002) | Not started |

Options considered for DEC-008:

- Keep hand-written Markdown only: free, but can't be validated reliably.
- YAML front matter: friendlier, but needs a parser dependency.
- A database: ruled out by the workflow guide.
- **JSON (recommended):** Node reads it without dependencies, and the check can be strict.

## 5 · Still open from earlier (not blocking PLAN-002)

- Which real phones you can test on (needed for Package C).
- Your Cloudflare plan (Free or Workers Paid) and any other paid accounts (DEC-005).
- Preview playtest (REP-002, A4) and branch protection (owner brief, section 5).

# REP-004 · PLAN-002 (Package B) implemented and verified

| | |
|---|---|
| Game | Fruit Market |
| Report date | 2026-10-26 14:10 (Asia/Kuala_Lumpur) |
| Package | PLAN-002, approved by the owner 2026-10-26: "Approve PLAN-002 only, including its listed tests and documentation" |
| Build | `main @ a7fa577` (records), plus this report |
| Live game | **Unchanged** (still `c915e49`, version 1.1.0). No publication. |
| Delivery status | Verified (B1–B7 PASS) |

## What changed

### Added

- `docs/workflow/records/`, the single source for every record:
  - `decisions.json`: all decisions, plus a history that can only grow
  - `PLAN-001.json` to `PLAN-003.json`
  - `releases.json`
- `npm run records`: rebuilds `DECISIONS.md`, `CHANGELOG.md`, the pending-approvals block in `STATE.md`, and every PDF.
- `npm run records:check` (now the first step of `verify`, so CI runs it). It rejects:
  - broken or duplicate IDs
  - references to things that don't exist
  - an approval without your words and a date
  - an edited history line or a rewritten approval
  - out-of-date views
- `CHANGELOG.md`, in plain player language. It is seeded with version 1.1.0 (what's live now).
- Release step in *Deploy LIVE game*: it refuses to run without release notes for the new version, and after a successful deploy it tags the commit (for example `v1.1.1`).

### Changed

- The PDF export is all-or-nothing. On failure it prints PDF EXPORT BLOCKED and keeps the previous PDFs.
- CI fetches the full history so the records check can compare with the previous commit.

### Removed

- The hand-edited `DECISIONS.md` (it is now generated).

**Player impact:** none directly. The release process is now traceable: every live build will have a version tag and changelog notes.

## Acceptance results

| # | Criterion | Result | Evidence |
|---|---|---|---|
| B1 | Rebuilding twice changes nothing | PASS | Checksums of all three generated files identical after a second run |
| B2 | Check fails on 5 deliberate breaks | PASS | Duplicate DEC-008; REP-003 citing a nonexistent DEC; DEC-001 "approved" with no evidence; history line 1 edited; stale register. Each exits 1 with a plain message. All reverted |
| B3 | No pandoc → PDF EXPORT BLOCKED, old PDFs kept | PASS | Exit 1; all 4 PDFs byte-identical; no temporary files left |
| B4 | Cumulative PDF order | PASS | Pending approvals, then current decisions, then REP-001, REP-002, REP-003 |
| B5 | Migration lost nothing | PASS | Every row of the old register matches. DEC-004 "In use" maps to the allowed value "implemented"; DEC-007 to DEC-009 changed only because of today's approval (in history) |
| B6 | `verify` and CI run the check, green | PASS | CI run 40276699547: "RECORDS OK … history checked against 63c3682", full suite green, preview updated |
| B7 | Tagging dry run | PASS | Throwaway branch: refuses while 1.1.0 is published; with a 1.1.1 candidate, "would tag v1.1.1" plus the notes. Branch deleted, no tag created |

## Known issue found (not part of this package)

**The game screen sometimes doesn't appear after rejoining an online room.** The automated tests caught it in 2 of the last 5 CI runs, with no game-code change between them:

- Run 1: after quick match, the second player's game didn't show within 15 s.
- Run 4 (a docs-only commit): after reloading the page mid-match, the game didn't come back within 20 s.

Every other run passed, including locally (more than 8 runs). Real players could hit this when their phone reloads the page or their connection drops.

This is pre-existing and intermittent; the cause isn't known yet. I recommend it as the next package, before Package C: reproduce it by repeating the reconnect test many times, find the cause, fix it, and prove the fix with repeated runs.

## Risks and costs

- RM0. `verify` is about 1 second slower.
- The first live release under the new process needs "PREPARE RELEASE" first, because Deploy refuses without release notes. That is intended.
- Until the reconnect issue is fixed, CI can occasionally fail for reasons unrelated to the change being tested. Read red runs before assuming the latest change caused them.

## Decisions

| ID | Decision | Approval | Delivery |
|---|---|---|---|
| DEC-007 | Package B scope = PLAN-002 | Approved (owner, 2026-10-26) | Verified |
| DEC-008 | JSON records plus narrative reports; generated views | Approved (owner, 2026-10-26) | Verified |
| DEC-009 | Records check in `verify` and CI | Approved (owner, 2026-10-26) | Verified |

Still pending your decision: DEC-001 (adopt the workflow), DEC-004 (PDF tooling), DEC-005 (budget assumption) and PLAN-003 (outline). See the register's "Pending approvals".

# REP-005 · PLAN-004 proposed: fix the rejoin problem in online play

| | |
|---|---|
| Game | Fruit Market |
| Report date | 2026-10-26 14:20 (Asia/Kuala_Lumpur) |
| Build | `main @ 935a287` (live game still 1.1.0, `c915e49`) |
| Type | Planning only. **No game implementation changes were made.** Read-only code inspection only. |
| Approval status | PLAN-004 **awaiting approval** (DEC-010, DEC-011) |

## 1 · The problem

Sometimes, after a player rejoins an online room, the game screen doesn't come back:

| CI run | Situation | Waited | Result |
|---|---|---|---|
| Run 1 (`da2eea9`) | Second player joins via Quick match | 15 s | Game screen never appeared |
| Run 4 (`63c3682`, docs only) | Player A reloads the page mid-match | 20 s | Game screen never appeared |
| Runs 2, 3 and 5, plus 8+ local runs | Same steps | — | Passed (usually about 1 s) |

There was no game-code change between the failing and passing runs, so the problem is timing-dependent. Real players can trigger it: phones reload pages when memory is low, and mobile connections drop. The tests saved **no evidence** when it happened: the local server's log was discarded, and the online test takes no screenshot on failure.

## 2 · What the code shows (read-only)

How rejoining works today:

1. The phone keeps a seat token per room.
2. On reload, the online screen sees `?room=CODE` plus the token and reconnects.
3. The server's `hello` matches the token to the seat and sends the room state.
4. The game screen appears once the state says the match has started.

Hypotheses, none confirmed yet:

| # | Hypothesis | Where | Fits run 1 | Fits run 4 | Evidence so far |
|---|---|---|---|---|---|
| H1 | In a quick-match lobby, a seat is deleted when its connection closes. If the waiting player's connection | `server/room.ts closed()` | Yes | No | Plausible from code |

| # | Hypothesis | Where | Fits run 1 | Fits run 4 | Evidence so far |
|---|---|---|---|---|---|
| | blips just as the stranger joins, the match never starts | | | | |
| H2 | The old connection's "closed" event reaches the server *after* the new `hello`, wrongly marking the seat offline or freeing it | `server/room.ts` `closed()` / `hello()` | Yes | Yes | Plausible; order depends on server timing |
| H3 | The server answers the rejoin with a non-fatal error and the screen stays on "Connecting…" with no retry | `server/room.ts`, `src/ui/Online.tsx` | Maybe | Yes | No error is shown or logged in the test |
| H4 | The client drops or ignores the first room state after reconnecting (version check or listener timing) | `src/net/client.ts`, `useGameController.ts` | Maybe | Yes | Code looks right, but it isn't tested under slow timing |
| H5 | Only the test environment: an overloaded CI machine is slower than the timeout | Test scripts | Yes | Yes | Normal time is about 1 s, so 15–20 s would be extreme. Not assumed without data |

Ruled out: a seat being lost when the server sleeps. Seats are stored on each connection in a way that survives the server sleeping (`serializeAttachment`).

## 3 · PLAN-004 · Reconnect fix (evidence first)

**Player problem:** a player who reloads or loses connection can be stuck on a waiting screen instead of returning to their match.

**Included scope (small steps)**

1. **Evidence.** When the online or quick-match test fails, it saves:
    - both players' screens, page text and browser console
    - the room messages each player received
    - the local server log, kept as a CI artifact for 7 days
2. **Reproduce.** A new stress test, `npm run test:reconnect`, repeats three kinds of cycle and times how long the game screen takes to return:
    - reload mid-match
    - a connection drop (the browser goes offline, then online)
    - quick-match pairing

    It runs normally and with the CPU slowed 4× (like a cheap phone). It can be run locally and from a manual CI workflow (DEC-011). I'll run it in batches (target 300 cycles) and record the failure rate.
3. **Isolate.** Compare failing and passing cycles to confirm or reject H1–H5.

4. **Fix the measured cause** with the smallest change, plus a regression test that fails before the fix and passes after.
    - If the cause is in seat or token handling (who owns a seat), a fresh, separate review checks the change, because that's where security matters.
    - If the cause is only test timing (H5), the fix goes in the tests, but only with stress-test data showing real rejoins work.
5. **Prove.** The stress test and CI must stay green repeatedly (acceptance below).

**Excluded**

- Rules, balance, `RULES_VERSION` and saves.
- Message-format changes that old app versions can't understand (if one is needed, I stop and replan).
- Publishing (that needs "PREPARE RELEASE" and your Deploy).
- Package C.
- Making tests "pass" by just waiting longer or retrying.

**Roles:** network/backend engineer and QA (lead), technical lead, plus security review if seat or token logic changes.

**Likely files:**

- `scripts/e2e-online.cjs`, `scripts/e2e-quick.cjs`, `scripts/e2e.mjs` (diagnostics, server log)
- new `scripts/e2e-reconnect.cjs`
- new `.github/workflows/reconnect-stress.yml` (manual)
- `package.json`
- then, depending on the cause: `server/room.ts` or `src/net/client.ts` or `src/ui/Online.tsx`, plus a test in `tests/room.test.ts`

**Acceptance (observable)**

- R1. `npm run test:reconnect` prints time-to-game for every cycle, the failures, and the 95th percentile.
- R2. The root cause is named with evidence from a failing cycle (a log or message excerpt). If it isn't reproduced after at least 300 cycles, I report **NOT REPRODUCED** with the conditions tried, and stop rather than guess.
- R3. If reproduced: the failure rate before the fix is recorded, and after the fix at least 100 slowed-down cycles have 0 failures.
- R4. A deliberately forced failure in the online test produces the diagnostics files and the server log artifact.
- R5. A regression test fails on the old code and passes on the fixed code.
- R6. `verify:full` is green on 3 consecutive CI runs; the leak check stays at 0; all unit tests stay green; no rules or save changes.

**Checks:** R1–R6, `verify`, the manual stress workflow, and CI.

**Cost and data:** RM0. GitHub Actions is free for public repositories. Diagnostics contain only test players and expire after 7 days.

**Risks**

- It may not reproduce outside CI. Mitigation: the slowed-down mode and the CI workflow, and a hard stop at 300 cycles with a report.
- A fix in seat handling could affect reconnect security. Mitigation: a regression test, the existing room tests, and a separate review.
- Time: likely 1–2 sessions. I'll report at the end of step 3 if the cause turns out larger than a small fix.

**Rollback:** revert the package commits. Players are unaffected until a release is published.

**How you'll see the result:** once the fix is on the preview, reload the page mid-match on your phone, or switch to airplane mode for 5 seconds and back. The match should return within a few seconds.

## 4 · Decisions

| ID | Question | Recommendation | Status |
|---|---|---|---|
| DEC-010 | How to fix the rejoin problem | Evidence and reproduction first, then fix only the measured cause (PLAN-004) | Awaiting approval |
| DEC-011 | Run the stress test on every push? | No: on demand (manual workflow and local runs) | Awaiting approval |

## 5 · Still open

- Pending decisions from earlier: DEC-001, DEC-004, DEC-005, and PLAN-003 (outline only).
- Your preview playtest, branch protection, your phones and your Cloudflare plan.

# REP-006 · PLAN-004 done: the online "stuck on Connecting..." bug is found and fixed

| | |
|---|---|
| Game | Fruit Market |
| Report date | 2026-10-26 15:25 (Asia/Kuala_Lumpur) |
| Package | PLAN-004, approved by the owner 2026-10-26: "Approve PLAN-004 only, including its listed tests and documentation" |
| Build | `main @ d7e5d24` (fix in `9d9e911`) |
| Live game | **Unchanged** (1.1.0). The fix is on the preview; players get it only after "PREPARE RELEASE" and your Deploy |
| Delivery status | Verified (R1–R6 PASS) |

## The bug, in plain words

After a player reloaded the page or rejoined an online match, the game sometimes stayed on **"Connecting to room ..."**, even though the match was there and the server had already sent it.

**Why:** the screen started listening for room updates a moment *after* it first appeared. The connection was already open by then. If the server answered in that gap, the answer was missed, and the screen waited for a next message that might never come (for example, when everyone was waiting on that player).

The gap is tiny on a fast computer, so it usually worked. On a slow phone, or when the app is in the background (browsers pause drawing for hidden tabs), the gap is long and the bug happens often. The game screen had the same weakness for updates arriving just as it opened.

**Fix:** two small changes in the app (`src/ui/Online.tsx`, `src/ui/game/useGameController.ts`). Right after starting to listen, each screen checks what has already arrived.

- No server change, no message-format change, no rules or save change.
- Old and new app versions stay compatible.

## Evidence

| Check | Result |
|---|---|
| Normal timing, old code (local) | 270 cycles (reloads, network drops, quick matches; normal and 4× CPU slowdown), 0 failed. The bug is too rare to see this way |
| Background-tab mode, **old code** | **20 of 20 reloads failed**; every one stuck on "Connecting to room" with the full game already received (`room v2 status=playing`) |
| Background-tab mode, **fixed code** (local) | 65 cycles, 0 failed |

| Check | Result |
|---|---|
| Background tab plus 4× slowdown, **fixed code** (local) | 100 cycles, 0 failed (reload p95 0.59 s, drop p95 0.63 s, quick match p95 1.3 s) |
| Same on a CI machine (where the original failures happened) | 110 cycles, 0 failed (Reconnect stress run 40283316566) |

## Acceptance results

| # | Criterion | Result | Evidence |
|---|---|---|---|
| R1 | Stress test with per-cycle timings | PASS | `npm run test:reconnect` ; modes: slowdown, background tab, preview target |
| R2 | Root cause named with evidence | PASS | Lost first update after (re)joining; see the old-code evidence above |
| R3 | Before rate recorded; at least 100 slowed cycles with 0 failures after | PASS | 20/20 failed before; 100/100 passed after (plus 175 more) |
| R4 | Forced failure leaves evidence | PASS | Page text, room messages, console, both screenshots and the server log were saved |
| R5 | Regression test fails before, passes after | PASS | New suite step "rejoin from a background tab" |
| R6 | 3 consecutive green CI runs; leak check 0; unit tests green; no rule or save changes | PASS | CI runs 7 (push), 8 and 9 (manual) on `d7e5d24` : all green, each with leaks 0 and 63 unit tests; no rules or save files changed |

## Side observations (no action taken)

- **The local test server never finishes closing a socket when the page closes it.** Cloudflare's docs say production replies automatically for this project's settings, and the real app never waits on a socket it closed itself, so no player impact was found. If it ever matters, the documented safe fix is one line in the server's close handler.
- **Playwright's offline mode doesn't break open connections**, so the stress test cuts them with a small local relay instead. A "silent" dead connection (no signal at all) isn't simulated yet. That would be part of Package C.

## What changed

### Added

- `npm run test:reconnect` and a manual *Reconnect stress* workflow (local or preview, optional slowdown and background-tab mode).
- Failure evidence for all online browser tests.
- The regression step in the browser suite.

### Fixed (player-facing, on the preview, not yet live)

- After reloading or rejoining an online match, the game could stay stuck on "Connecting to room...", especially on slower phones or when the app was in the background.

## Risks and costs

- RM0.
- The browser suite is about 1 minute longer (the regression step).
- The fix affects all online screens. Mitigation: the full suite, including the secret-price leak check, stayed green, and 545 stress cycles ran in total.
- Remaining NOT RUN: a real phone in a real background tab. Your preview playtest covers that (see below).

## How to try it (preview)

1. Open https://fruit-market-staging.murshidi-hashim.workers.dev on two phones (or one phone and a computer).
2. Start an online game (Play a friend → Create room → open the link on the other device).
3. During the match, on one phone: switch to another app for 10 seconds, come back, then pull down to reload.
4. Expected: the match comes back within a few seconds both times, never stuck on "Connecting to room...".

# REP-007 · PLAN-005 proposed: make 4-player games easy to find

| | |
|---|---|
| Game | Fruit Market |
| Report date | 2026-10-26 18:36 (Asia/Kuala_Lumpur) |
| Build | `main @ cc27848` (live game still 1.1.0) |
| Type | Planning only. **No game implementation changes were made.** One read-only test was run locally. |
| Approval status | PLAN-005 **awaiting approval** (DEC-012) |

## 1 · What you noticed

You couldn't see an option for 4 human players. 4-player games are supported, but the screens don't say so.

**Online room (Play a friend)**, as it looks today on a 360 px phone:

- The heading says **"Invite a friend"** (singular).
- There is one empty row: **"Waiting for a friend to join"**, which disappears once one person joins.
- The count "1/4" is in small grey text next to "Players".

**Same phone (New game):** 4 people means choosing **3** rival shops and switching each from Computer to Friend. Nothing on the screen explains that.

**Quick match** is always 2 strangers, by design. Nothing points groups to Play a friend instead.

## 2 · Check that 4 players really work (read-only, local server)

- 4 browsers joined one room by the same link. The lobby showed "Players 4/4". All 4 pressed ready, the host started, and **all 4 reached the game** (each saw 3 rivals).
- A **5th** person was turned away: "Couldn't join · This room is full."
- Limitation: tested on one machine, not 4 real phones. The automated test below makes this a permanent check.

## 3 · PLAN-005 · 4-player clarity

**Player problem:** groups of 3 or 4 think the game is for 2 and give up, or play with computers instead.

**Included scope (screen text and layout only)**

| Screen | Today | Proposed |
|---|---|---|
| Online lobby, invite box | "Invite a friend" | **"Invite friends (up to 3)"** |
| Online lobby, player list | "Players 1/4" (small, grey), one row "Waiting for a friend to join" | **"Players 1 of 4"** in normal text, plus one row per free seat: "Open seat · send the link" |
| Online lobby, when full | (nothing) | "Room full · start when everyone is ready" |
| Home, under "Play a friend" | (nothing) | Small line: "2 to 4 players online" |
| Quick match screen | (nothing) | "Quick match is always 2 players. For 3 or 4, use Play a friend." |
| New game, under "Rival shops" | (nothing) | "Up to 4 shops. Pick Friend to pass this phone around." plus a live total, for example "4 shops: you, 2 friends, 1 computer" |

Computer shops added in the lobby take seats, so the open-seat rows and the count include them. For example: 2 people + 1 computer = "Players 3 of 4", one open seat.

**Also included:** a new automated browser test where 4 people join one room and all reach the game, and a 5th is refused. It runs in CI on every change.

**Excluded**

- Rules, balance, the 4-player limit and quick-match size.
- Any server or message change.
- Renaming the buttons people tap to get there ("Play a friend", "New game"), so tests and habits keep working.
- Publishing.

**Roles:** UX designer (wording), gameplay engineer, QA. Self-review; low risk.

**Likely files:** `src/ui/Online.tsx` (lobby), `src/ui/App.tsx` (home, new game), `src/ui/styles.css`, new `scripts/e2e-four.cjs`, `scripts/e2e.mjs`.

**Acceptance (observable)**

- C1. The lobby shows "Players N of 4" and one open-seat row per free seat. Both update when people join or leave and when computer shops are added.
- C2. With 4 people there are no open seats and the lobby says the room is full.
- C3. The new browser test passes: 4 people join one room, all see the game, and a 5th is refused.
- C4. The new lines fit on a 360 px wide phone without overflow (screenshots of each screen in the report).
- C5. Store screenshots change only where intended (home and lobby text), checked with the visual-diff tool.
- C6. `verify:full` is green in CI; no rules, server or save changes.

**Cost and data:** RM0. No data changes.

**Risks:** longer text could wrap awkwardly on small phones; C4 checks this. Wording is a matter of taste, so you can change any line before or after.

**Rollback:** revert the package commit.

**After it's built:** open the preview, tap Play a friend → Create room, and check that the lobby clearly invites up to 3 friends.

## 4 · Decision

| ID | Question | Recommendation | Status |
|---|---|---|---|
| DEC-012 | How to make 4-player play easy to find | Screen text and open-seat rows plus a 4-person test (PLAN-005); no rule or server change | Awaiting approval |

## 5 · Note on order

The reconnect fix (PLAN-004) is on the preview and not live yet. PLAN-005 can be built first and both can go out together in 1.1.1, which saves you one Deploy.

# REP-008 · PLAN-005 done: 4-player games are easy to find

| | |
|---|---|
| Game | Fruit Market |
| Report date | 2026-10-26 19:05 (Asia/Kuala_Lumpur) |
| Package | PLAN-005, approved by the owner 2026-10-26: "Approve PLAN-005 only, including its listed tests and documentation" |
| Build | `main @ 04b72b5` |
| Live game | **Unchanged** (1.1.0). On the preview; goes live with the next release |
| Delivery status | Verified (C1–C6 PASS) |

## What changed (screen text and layout only)

| Screen | Before | After |
|---|---|---|
| Online lobby, invite box | "Invite a friend" | "Invite friends (up to 3)". Fewer if computer shops are added, for example "(up to 2)" |
| Online lobby, player list | "Players 1/4" in small grey text, one row "Waiting for a friend to join" | "Players 1 of 4", plus one "Open seat · send the link" row per free seat |
| Online lobby, when full | Nothing | "Room full · start when everyone is ready" |
| Online screen | "Play a stranger..." / "send to a friend" | Quick match: "Always 2 players; for 3 or 4, start a private game below." Private game: "send to up to 3 friends... 2 to 4 players." |
| Home | "Play a friend" | Same button, with a small line "2 to 4 players online" |
| New game | Nothing explained 4 players | "Up to 4 shops. Pick Friend to pass this phone around." plus a live total, for example "4 shops: you, 2 friends, 1 computer" |

**Wording note:** the quick-match hint says "start a private game below" instead of the planned "use Play a friend", because both options are on the same screen. The meaning is the same.

No rules, server or save changes. The buttons you tap keep their names.

## Acceptance results

| # | Criterion | Result | Evidence |
|---|---|---|---|
| C1 | Lobby count and open seats update with joins and computer shops | PASS | New test: 1 of 4 with 3 open; 2 people + 1 computer = "3 of 4" with 1 open and "(up to 2)" |

| # | Criterion | Result | Evidence |
|---|---|---|---|
| C2 | Full room shows no open seats and "Room full" | PASS | New test: "Players 4 of 4", 0 open, "Room full" |
| C3 | 4 people play; a 5th is refused | PASS | New test: 5th sees "This room is full"; all 4 reach the game, each seeing 3 rivals |
| C4 | Fits a 360 px phone | PASS | Home, online and new-game screens: page exactly 360 px wide, no sideways scrolling |
| C5 | Store screenshots change only where intended | PASS | Home 7.3% different (the button's new second line moves the rows below it). All 6 in-game screens 0.00% |
| C6 | Full CI green; no rule, server or save changes | PASS | CI run 11 on `04b72b5` green, including the new four-player step and the preview update; diff touches only screen code, styles, tests and docs |

**Process note:** my first `verify` run failed because I had updated a record without regenerating the views. The records check caught it, as designed (PLAN-002), and the run passed after `npm run records`.

## Risks and costs

- RM0.
- The store screenshot of the home screen (`store/screenshots`) should be retaken before the next Play listing update. It still shows the old button.

## How to see it

On the preview (https://fruit-market-staging.murshidi-hashim.workers.dev):

1. The home screen shows "2 to 4 players online" under Play a friend.
2. Tap Play a friend → Create room. The lobby says "Invite friends (up to 3)" and shows three open seats.
3. Tap New game → 3 rival shops → switch two to Friend. It reads "4 shops: you, 2 friends, 1 computer".

## Next

Both fixes (reconnect and 4-player clarity) are on the preview. After your preview playtest, say "PREPARE RELEASE" to send them to players as 1.1.1.

# REP-009 · Release 1.1.1 prepared: readiness and how to publish

| | |
|---|---|
| Game | Fruit Market |
| Report date | 2026-10-26 20:30 (Asia/Kuala_Lumpur) |
| Request | Owner: "PREPARE RELEASE" |
| Candidate | Version **1.1.1**: the `main` commit that adds this report (the Deploy button tags exactly that commit `v1.1.1`) |
| Live now | 1.1.0 ( `c915e49` ) |
| Approval status | DEC-013 (publish 1.1.1) **awaiting your approval**. Nothing has been published |

## 1 · What players get (changelog for 1.1.1)

- **Fixed:** after reloading the page or rejoining an online match, the game could stay stuck on "Connecting to room...", especially on slower phones or when the app was in the background.
- **Clearer group play:** the online lobby shows how many of the 4 seats are taken and which are still open, and the menus explain that 2 to 4 people can play together, online or by passing one phone.
- **Behind the scenes:** updates now reach players only after a preview and your approval, with automatic checks before and after.

Included packages: PLAN-001, PLAN-002, PLAN-004 and PLAN-005.

## 2 · Size of the change since 1.1.0

- **Game code:** 4 files, about 40 lines ( `src/ui/Online.tsx` , `src/ui/App.tsx` , `src/ui/game/useGameController.ts` , `src/ui/styles.css` ).
- **Server code:** unchanged. **Server settings for the live game:** unchanged (only a separate preview section was added).
- **Rules, balance, rules version (2) and saves:** unchanged. Old and new app versions can play together.
- **Version number:** 1.1.0 → 1.1.1 (shown in Settings; also the Android version name).
- **CI setting:** job time limit raised from 20 to 35 minutes (see section 3). No effect on players.

## 3 · Readiness gates

| Gate | Check | Result | Evidence and limitation |
|---|---|---|---|
| Build ready | Clean install, records, lint, types, 63 unit tests, build | PASS | Latest green CI: run 12 on `f319548` (same game code as the candidate). Run 13 on `f472186` was **cancelled**: its browser download took 16 minutes (usually about 20 seconds), so it hit the job's 20-minute limit. The limit is now 35 minutes. The candidate's own CI run must be green before you press Deploy |

| Gate | Check | Result | Evidence and limitation |
|---|---|---|---|
| Play ready | Computer-played solo game; speed budget | PASS | Browser suite in CI |
| Play ready | **You finish a match on the preview** | **NOT RUN** | Needs you, about 10–15 minutes (owner brief, section 2) |
| Online ready | Two players, secret prices never leak, a player leaves | PASS | Leak check 0 on every CI run |
| Online ready | Rejoin and reconnect | PASS | 545 stress cycles after the fix, 0 failures; regression test in CI |
| Online ready | 4 players in one room; a 5th refused | PASS | New test in CI |
| Online ready | Server crash recovery | NOT RUN | Package C. Server code is unchanged since 1.1.0, so this release adds no new risk here |
| Release ready | Real phones (Android Chrome, iPhone Safari) | NOT RUN | Emulated phones only; Package C |
| Release ready | Data backup and restore | NOT RUN | No data format change in this release; Package C |
| Release ready | Live smoke test after deploy | Automatic | The Deploy workflow checks the live site and its rules version, then tags v1.1.1 |
| Release ready | Rollback path | Ready | See section 5 |

**Release-ready is not claimed**, because required checks are NOT RUN. The workflow guide lets you choose between:

- **A (recommended):** do the 15-minute preview playtest first. If it passes, publish. Real phones, backups and crash recovery stay as documented gaps for Package C. That's reasonable here because 1.1.0 is already public and this release changes no server code or data.
- **B:** publish now and explicitly accept the playtest as NOT RUN (a "scoped exception"). Not recommended: it skips the only human check.

## 4 · How to publish (after you decide)

1. Make sure CI is green for the candidate. I'll confirm in my reply.
2. GitHub → **Actions** → **Deploy LIVE game** → **Run workflow** → type `PUBLISH` → **Run workflow**.
3. It runs the checks again, confirms the release notes for 1.1.1 exist and that `v1.1.1` isn't already used, publishes, smoke-tests the live site, and tags the release.
4. Tell me when it's green. I'll mark 1.1.1 as published (date, commit, tag) and check the live site's version.

If anything else lands on `main` before you press Deploy, ask me to re-check, because the Deploy publishes whatever `main` holds.

## 5 · If something goes wrong after publishing

- **The Deploy run fails:** the old version keeps running. Nothing to undo. Tell me.
- **A problem is found after a successful deploy:**
  - Fastest: Cloudflare dashboard → Workers → `fruit-market` → Deployments → roll back to the previous deployment (takes effect in seconds; no data changes in this release, so it's safe).
  - Standard: I prepare 1.1.2 with the change reverted; you press Deploy (about 30 minutes including checks).

## 6 · Costs and data

RM0 change. No new services. No player data or storage format changes.

## 7 · Decision

| ID | Question | Recommendation | Status |
|---|---|---|---|
| DEC-013 | Publish 1.1.1 to players? | Yes, after your preview playtest passes (option A) | Awaiting approval |

To approve, reply for example: **"Playtest passed. Approve DEC-013, publish 1.1.1."** Then press Deploy. To choose option B, say so explicitly, for example "Approve DEC-013 with the playtest NOT RUN exception".

# REP-010 · Theme review: does Fruit Market's theme support how it plays?

| | |
|---|---|
| Game | Fruit Market |
| Report date | 2026-10-26 21:47 (Asia/Kuala_Lumpur) |
| Request | Apply the "Game Concept and Theme Companion" (edition 1.0) to this game |
| Build reviewed | `main @ 83de99b` (release 1.1.1 candidate); content in `src/engine/data.ts` and `RULES.md` |
| Type | Creative review only. **No game, rule or content changes were made.** |
| Approval status | DEC-014 proposed; DEC-015 and DEC-016 **awaiting your choice** |

The direction is already chosen and the game is live, so this follows the companion's path for an existing game: check what works, review coherence, propose the smallest useful test. No new concepts.

## 1 · Creative brief (what is confirmed, what is assumed)

| Item | Status | Value |
|---|---|---|
| Player fantasy | Confirmed (game brief) | Run a fruit stall; outsell rival shops before your stock goes bad |
| Core interaction | Confirmed (rules) | Buy from the farmers, set secret prices, reveal, age overnight |
| Players and session | Confirmed | 2–4 shops; Quick 8 / Standard 10 / Long 14 days; 10–20 min (assumption, unmeasured) |
| Platform | Confirmed | Phone browser and Android; touch first |
| Audience | **Assumption** | Casual players 12+, families sharing a phone, friends online |
| Desired feeling | **Assumption** | Light, cheeky tension: "will my price win?" plus "sell it before it rots" |
| Setting and identity | **Open** | See section 4 |

## 2 · Does the theme show up in what players do?

| Repeated action | How the theme expresses it | Fit |
|---|---|---|
| Buying at the farmers market; poorest buys first | A morning market where small shops get first pick | **Strong**. The catch-up rule feels like market goodwill |
| Fruit ages each night; older fruit sells for less; spoiled fruit is binned | The freshness track and "Not so" fruit | **Strong**. This *is* the fantasy; every choice is a race against rot |
| Secret prices, all revealed at once; cheapest sells first | "Showdown" reveal, customer by customer | **Mixed**. Exciting and fair, but real stalls post prices openly. The reveal needs a stall-side reason (presentation idea in section 5) |

| Repeated action | How the theme expresses it | Fit |
|---|---|---|
| Customers with orders and price caps | 24 named characters with quirks | **Strong** for orders and caps. The quirks are cosmetic (they never change play), which is fine but should stay labelled cosmetic |
| Events (Heatwave, Health Inspector, Lorry Breakdown…) | Each bends the market in a believable way | **Strong**. Heatwave aging fruit and the Inspector fining old fruit are the best theme-mechanic matches |
| Upgrades (Handcart, Cool Room, Juice Press, Signboard…) | Things a stall owner would actually buy | **Strong** |
| Favour cards | Rewards for serving a whole order yourself | **Mixed**. "Free Cash" and "Good Riddance" read as abstract game cards rather than market favours |
| Rival shops (animal traders) | Six characters with reaction lines | **Mixed**. Fun, but personality and play style don't match: lines are cosmetic, and how cleverly they price comes from difficulty, not character |

Overall the core loop expresses the fantasy well. The weak spots are presentation: why prices are secret, a few abstract card names, and cast identity. None needs a rule change.

## 3 · Proposed design pillars (for your approval: DEC-015)

| Pillar | Include when… | Exclude when… |
|---|---|---|
| **1. Fresh fruit, fast money:** every decision is a race against spoilage | It makes freshness visible or raises the cost of waiting | It lets fruit sit forever at no cost, or hides freshness |
| **2. Outprice rivals in secret:** commit a price blind, then see the reveal | It keeps prices hidden until the reveal and makes the reveal explain *why* | It shows rivals' prices early, or adds open haggling or chat |
| **3. A cheerful town market:** playful characters, short sessions, readable on a phone | Jokes target behaviour (a vampire who hates garlic); games stay under about 20 min | Mean or stereotype-based jokes, grind, pay-to-win, long text |

## 4 · Coherence issues found (review before expanding)

1. **Mixed setting identity** *(owner decision, see question 1)*:
    - Customer names are Malaysian and multicultural (Aisyah, Mei Ling, Kumar, Wei Jie), and one rival is "Uncle Toad".
    - The palette is Mediterranean (terracotta, harbour teal, olive).
    - Prices are in **$**, and the fruits are blueberry, watermelon, apple and mango (two are imports in Malaysia).

    It reads as "anywhere town". That's fine, but it isn't a choice anyone made deliberately.

2. **Possible ethnic association:** "Panda Lim" pairs a Chinese surname with a panda. Most other names pair freely with professions or fantasy roles (Astronaut Aisyah, Ninja Farid), so this one stands out. Recommend renaming it (cosmetic only). "Kumar the Genie" is milder; worth a second look.

3. **Why are prices secret?** The best moment in the game has no in-world reason. That's a presentation fix, not a rule change (section 5).

4. **Repeated jokes:** each rival has 2–3 lines per situation, and a 10-day game has many reveals, so lines probably repeat. Hypothesis; check in the playtest.

5. **Icon style:** some emoji still sit beside the pixel art (🔒 Prices locked, 📊, 🙋/🤖, 🕵). Already in backlog B-33.

## 5 · Representative day (actual rules)

*Day 4 of 10, Standard game. You and Foxy Rosa (computer). You have $14; she has $19.*

1. **Morning:** the event is 🔥 **Heatwave**: watermelons sell for $4 more, but all fruit ages an extra step tonight. Tomorrow's event, shown early: 🕵 Health Inspector.
2. **Buy:** you're poorer, so you **buy first**. Fresh watermelon costs $3. You take 3 and leave the cheap blueberries: in a heatwave they'd rot by tomorrow. *Choice: profit now vs. stock that survives tonight.*
3. **Customers:** Cowboy John wants 2 watermelons and pays up to $6 + 4 =  * *10** for fresh ones.
4. **Secret price:** you offer 2 at **$9**; Rosa secretly offers 2 at $8.
5. **Showdown:** prices flip; the cheapest sells first. Rosa's $8 fills John's order, and you sell nothing. Rosa quips "A dollar cheaper. Sorry, not sorry!" *Feedback:* your offer is tagged "$1 too high".
6. **Evening:** you build the ❄ **Cool Room** ($7). Your fruit now ages only every other night, and heatwaves can't touch it. *Choice: spend on protection or keep cash for tomorrow.*
7. **Night:** heatwave aging hits Rosa's shelf. Your Cool Room saves your 3 watermelons for tomorrow's customers, and tomorrow's Inspector will fine Rosa for her old fruit.

**Theme-only idea for pillar 2 (not approved, no rule change):** present secret pricing as *chalkboards*. Each stall chalks its price face-down, the market bell rings, and the boards flip. Same rule, with a market reason.

## 6 · Mini creative bible (draft: samples only, per the companion)

- **World and tone:** a sunny town market where anyone might shop: cowboys, astronauts, a vampire after sunset. Light, cheeky, never mean. Stakes: a good day's takings, never ruin.
- **Sample character:** *Grandpa Tan* (cosmetic). Silhouette: cardigan, cane, paper bag. Desire: a fair price. Flaw: remembers when apples cost ten cents. Sample reaction when you sell at full price: "Ten dollars? In my day..."
- **Sample event:** *Health Inspector*. The clipboard arrives at night; every Not-so fruit costs $2. This teaches "sell old fruit first" without a tutorial line.
- **Key screen:** the reveal. Face-down boards flip highest to lowest, the winner's coins fly to their stall, and the losing offer is tagged "$1 too high". It works without sound and without colour alone: text plus movement.
- **Do:** jokes about what characters *do*; short lines; a mix of names and roles across cultures. **Avoid:** pairing a person's ethnicity with an animal or cultural cliché; jokes longer than one line; new lore that doesn't change a decision.

## 7 · Riskiest creative hypothesis and the smallest test (DEC-016)

**Hypothesis:** first-time players understand *why* they won or lost a sale at the reveal. That is pillar 2, and it's what makes the game feel fair rather than random.

**Format:** a playtest with the existing game (preview or live 1.1.1). **No code needed.** It can't tell us retention, market demand or online-server behaviour.

**Playtest card**

| | |
|---|---|
| Testers | 3–5 people who have never played; no coaching |
| Setup | Their own phone. New game → Quick (8 days) → 1 computer shop (Normal) |
| Tasks | 1) Start a game. 2) Buy fruit. 3) Set your prices. 4) After the first sales, say why you sold or didn't. 5) Finish the game |
| Watch for | Time to first price; whether they mention "cheapest sells first" or freshness; whether they read the event card; where they hesitate |
| Ask after | "What were you trying to do?" "Why did you lose that customer?" "Would you play again?" (note if they start another game unprompted) |
| Provisional criteria (you set them; these are suggestions, not benchmarks) | **Continue** if at least 3 of 4 can explain a lost sale by day 3. **Revise** (reveal presentation, section 5) if 2 or fewer can |
| Limits | Small sample; friends are kind; not proof of fun or retention |

I'll record what happens (observations separate from interpretations) in the next report.

## 8 · Questions for you (max 3)

1. **Where is this market?**
   - (A) "Anywhere town": keep as is. *Recommended for now:* costs nothing and doesn't affect the biggest risk.
   - (B) A Malaysian pasar: RM prices, local touches; the fruit artwork could change later; much bigger art scope.
   - (C) A deliberate mix ("a market where the whole world shops"): keep the names, lean into variety.
2. **Rename Panda Lim** (and review Kumar the Genie)? Recommended: yes, a cosmetic change in a later small package.
3. **Can you find 3–5 people who've never played** for the playtest card above?

## 9 · Decisions

| ID | Question | Recommendation | Status |
|---|---|---|---|
| DEC-014 | Use the Concept and Theme Companion for Fruit Market's creative work, alongside the development workflow | Yes; one shared record system | Proposed |
| DEC-015 | Adopt the three design pillars (section 3)? | Yes | Awaiting approval |

| ID | Question | Recommendation | Status |
|---|---|---|---|
| DEC-016 | First creative test = the reveal-comprehension playtest (section 7) | Yes; no code | Awaiting approval |

No conflict with approved decisions: release 1.1.1 (DEC-013) is unaffected, and nothing here changes rules, saves or multiplayer.

# REP-011 · Setting chosen; panda rename planned (PLAN-006)

| | |
|---|---|
| Game | Fruit Market |
| Report date | 2026-10-26 22:16 (Asia/Kuala_Lumpur) |
| Owner input | Answers to REP-010's three questions |
| Type | Creative decisions and planning. **No game changes were made.** |
| Approval status | DEC-017 and DEC-018 **approved**; DEC-019 (PLAN-006) **awaiting approval**; DEC-015 and DEC-016 still awaiting approval |

## 1 · Your answers, recorded

| Question | Your answer | Recorded as |
|---|---|---|
| Where is this market? | "C: a deliberate 'the whole world shops here' mix" | DEC-017, approved |
| Rename Panda Lim? | "sure, but dont run away from the general theme coherence" | DEC-018, approved |
| 3–5 first-time testers? | "yes" | Noted. This confirms testers are available; it isn't approval of the playtest itself (DEC-016) |

## 2 · What option C means in practice

Written into `docs/workflow/CREATIVE-BIBLE.md` (short, one page):

- **A sunny harbour market where the whole world comes to shop.** The seaside colours you already have become the harbour; a port that trades with everywhere explains the mix of fruit.
- **Customers** keep their current mix of names and roles, freely combined, never tying a culture to a cliché.
- **Money** stays plain market dollars ($), the same for every visitor.
- **No game changes are needed for C.** It turns what's already there into a deliberate choice and sets the rules for future content.

## 3 · The rename, keeping the cast coherent

The rivals are animal traders with friendly nicknames: Foxy Rosa (fox), Big Benny (bear), Uncle Toad, Rocky (raccoon), Pinky (pig). The panda's lines are calm and sleepy ("Bamboo-zled you!", "very zen", "Nap time, I guess"). The new name should fit that pattern and keep the lines working.

| Option | Why |
|---|---|
| **Patches** (recommended) | Describes the panda (black-and-white patches), like Pinky and Rocky; no real-world link |
| Bamboo Bea | Keeps the bamboo joke; alliterative like Big Benny |

| Option | Why |
|---|---|
| Sleepy Sid | Matches the napping lines |

Kumar the Genie, reviewed: keep. Genie is a fantasy role like Ninja, Wizard or Astronaut; the name and role don't form a national cliché.

**Checked in the code (read-only):** the panda's reaction lines are found by its picture (🐼), not its name. So saved games and games in progress keep working, and they just show "Panda Lim" until they end. The panda is also one of the three daily-challenge rivals; names don't affect the score check, and PLAN-006 tests that.

## 4 · PLAN-006 · Rename the panda rival (cosmetic)

**Scope**

- Change the display name in the rival list ( `src/engine/data.ts` ) to your chosen name. Picture, lines, difficulty and play stay the same.
- Update the daily-challenge sentence in `RULES.md` .
- Keep "Panda Lim" reserved, so no player can take the old rival name.
- Add tests: an old save naming "Panda Lim" still gets the panda's lines; daily-challenge verification gives the same score; both names are refused as player names.

**Excluded:** other renames, art, lines, rules, AI or balance; renaming inside saved or in-progress games.

**Acceptance**

- N1. New solo games, daily challenges and online rooms show the new name.
- N2. An old save loads, still says "Panda Lim", and the panda's lines still appear.
- N3. The daily-challenge score check is unchanged for the same moves.
- N4. Both names are refused as player names.
- N5. `verify:full` is green in CI; no rule, save-format or server-logic changes.

**When:** built on a separate branch now and released as **1.1.2 after 1.1.1 is live**, so the 1.1.1 candidate you're testing doesn't change.

**Cost:** RM0. **Risk:** very low (one name plus tests). **Rollback:** revert the commit.

## 5 · Decisions

| ID | Decision | Status |
|---|---|---|
| DEC-017 | Setting: a market where the whole world shops | Approved (owner, 2026-10-26) |
| DEC-018 | Rename Panda Lim, keeping the cast coherent | Approved (owner, 2026-10-26) |
| DEC-019 | Rename scope = PLAN-006, shipped in 1.1.2 | Awaiting approval |
| DEC-015 | Three design pillars | Awaiting approval (from REP-010) |
| DEC-016 | Reveal-comprehension playtest with your 3–5 testers | Awaiting approval (from REP-010) |

# REP-012 · PLAN-006 done: the panda rival is now Patches

| | |
|---|---|
| Game | Fruit Market |
| Report date | 2026-10-26 22:48 (Asia/Kuala_Lumpur) |
| Package | PLAN-006, approved by the owner 2026-10-26: "Approve PLAN-006 only, including its listed tests and documentation, name: Patches" |
| Build | Branch `workflow/plan-006 @ f14245a` (not on `main` yet) |
| Live game | **Unchanged** (1.1.0). `main` is still the 1.1.1 candidate |
| Delivery status | Verified (N1–N5 PASS); ships in 1.1.2, after 1.1.1 |

## What changed

- The panda trader is now called **Patches** in new solo games, daily challenges and online rooms. Same picture (🐼), same calm, sleepy lines ("Bamboo-zled you!"), same difficulty and play.
- "Panda Lim" stays blocked as a player name, so nobody can pose as the old rival.
- The daily-challenge sentence in `RULES.md` names Patches.
- Five new tests (`tests/rename.test.ts`). Game code change: one line, plus the blocked-names list.

## Acceptance results

| # | Criterion | Result | Evidence |
|---|---|---|---|
| N1 | New games show the new name | PASS | Rival list and daily lineup contain Patches, not Panda Lim; online rooms draw from the same list |
| N2 | Old saves load, keep "Panda Lim", still get the panda's lines | PASS | Test saves and reloads such a game unchanged; the panda's undercut line is still chosen |
| N3 | Daily-challenge score check unchanged | PASS | The same daily played with either name ends with identical cash and sales for every shop |
| N4 | Both names refused for players | PASS | "Patches", "Panda Lim" and "panda lim" are reserved |
| N5 | Full CI green; no rule, save-format or server-logic change | PASS | CI run 15 on `f14245a` green (68 unit tests plus the full browser suite) |

## How it reaches players

`main` holds the 1.1.1 candidate, and Deploy publishes whatever `main` holds, so this waits on its branch. After 1.1.1 is live: I merge it, you say "PREPARE RELEASE" for 1.1.2, then you press Deploy. Its player note: "The panda trader is now called Patches."

## Risks and costs

- RM0, very low risk.
- **Observation (not part of this package):** the CI machines' browser download took 11–16 minutes in the last 4 runs (it used to take about 20 seconds). The 35-minute limit absorbs it, but caching the download would cut each CI run by about 10 minutes. That would be a small separate package if you want it.

# REP-013 · Approvals recorded: publish 1.1.1, pillars, first playtest

| | |
|---|---|
| Game | Fruit Market |
| Report date | 2026-10-26 22:58 (Asia/Kuala_Lumpur) |
| Owner input | "yes to 1 and 2" (reply to REP-012) |
| Type | Decisions recorded. **No game changes.** Nothing has been published yet |

## 1 · What you approved

| ID | Decision | Status |
|---|---|---|
| DEC-013 | Publish 1.1.1 (reconnect fix + 4-player clarity) | **Approved**. Not yet deployed: you press the button |
| DEC-015 | The three design pillars | **Approved**. Now marked approved in the creative bible |
| DEC-016 | First playtest with 3–5 first-time players | **Approved**. Ready to run with the kit below |

**One gap, recorded honestly:** you didn't say whether you tried the preview yourself. It's recorded as NOT RUN. It's your call; a 5-minute look at https://fruit-market-staging.murshidi-hashim.workers.dev before deploying is still worth it (Settings should show version 1.1.1).

## 2 · Publish 1.1.1 now (about 3 minutes, plus about 15 minutes of automatic checks)

1. Open GitHub → **Fruit-Market** → **Actions** → **Deploy LIVE game**.
2. Tap **Run workflow**, type `PUBLISH`, then tap **Run workflow**.
3. Wait for the green tick. It checks the code, publishes, tests the live site and tags `v1.1.1`.
4. Tell me "deployed". I'll check the live version, mark 1.1.1 published, then bring in the Patches rename and prepare 1.1.2.

`main` is unchanged since the candidate (`83de99b`). These records are on a separate branch until you've deployed.

## 3 · Playtest kit (DEC-016)

**Goal:** find out whether first-time players understand *why* they won or lost a sale.

**Before:** use the live game after 1.1.1 is published (or the preview). Each tester uses their own phone. Don't explain the rules. You only watch and take notes.

**Say only this:** "This is a game about running a fruit stall. Please think aloud while you play. I can't help, but I'll ask a few questions at the end."

**Tasks:** New game → Quick (8 days) → 1 computer shop. Play to the end.

**Note for each tester** (copy this per person):

```
Tester:            Age range:        Plays phone games? (often / sometimes / rarely)
Minutes to first price set:
After day 1–3 sales, in their words: why did they sell or not sell?
Did they mention "cheapest sells first" or fresher fruit? (yes / no)
Did they read the event card? (yes / no / unsure)
Where did they hesitate or get stuck?
Finished the game? (yes / no)   Started another game without being asked? (yes / no)
"What were you trying to do?"
"Why did you lose that customer?"
"Would you play again?"
Anything that made them laugh or frown:
```

**Decision rule (provisional, you set it): continue** if at least 3 of 4 testers can explain a lost sale by day 3; **revise the reveal** (the chalkboard idea in REP-010) if 2 or fewer can.

**Send me:** the notes, as typed text or photos. I'll record what you *saw* separately from what we *think* it means, and propose the smallest next step.

## 4 · Records

DEC-013, DEC-015 and DEC-016 approved, with your words, in the register. Still pending from earlier: DEC-001, DEC-004, DEC-005, PLAN-003 (outline).

# REP-014 · UI and UX polish: what feels unfinished, and a plan for the whole game

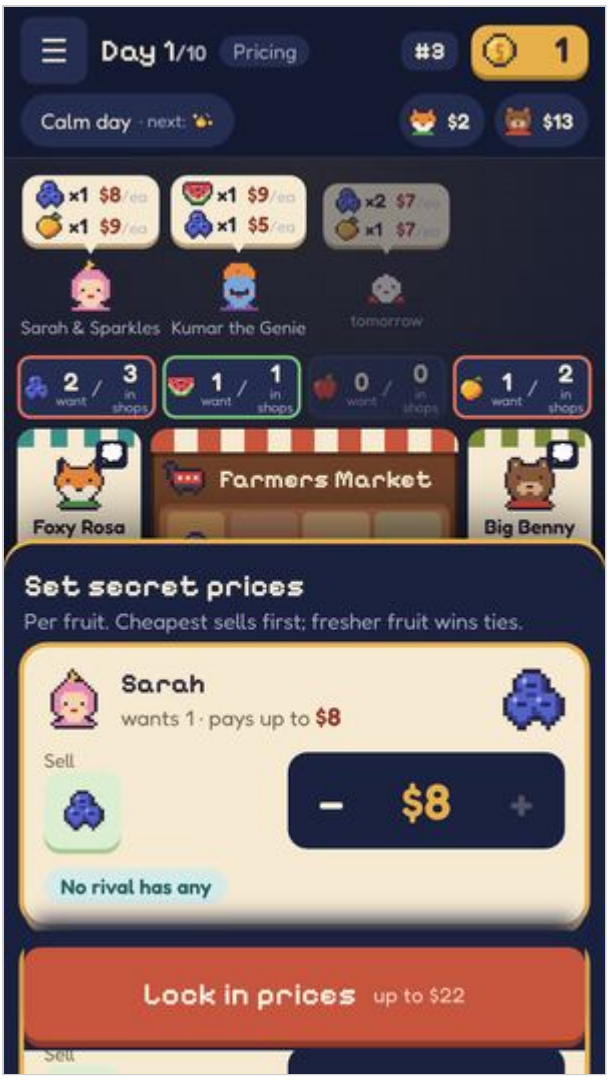
| | |
|---|---|
| Game | Fruit Market |
| Report date | 2026-10-26 23:06 (Asia/Kuala_Lumpur) |
| Owner input | "the look and feel of the game doesnt feel polished" and "the whole UI ... propose changes on the whole UI UX" |
| Build reviewed | 1.1.1 candidate ( `main @ 83de99b` ), phone sizes 360×640 and 390×844 |
| Type | Review and proposal. **No game changes were made.** |
| Approval status | DEC-020, PLAN-007 and PLAN-008 **awaiting approval** |

## 1 · What I looked at

Every main screen at two common phone sizes: home, trait pick, buying, pricing, sales reveal, evening upgrades, results, trophies, online lobby and new game. The same checks the browser tests use: whether anything is cut off, overlaps or wastes space.


Day 1/10
Buying
#3
1
Calm day · next:
$10
$13
×1 $8
×1 $9
Sarah & Sparkles
walk-in
1+ / 3 want / in shops
? / 1 want / in shops
? / 0 want / in shops
1+ / 0 want / in shops
Foxy Rosa
Haggler
10
Farmers Market
Big Benny
Charmer
13
spoils
fresh
You
YOU
Big Basket
4/8
4/5
1
Your move
Tap a fruit in the market · 4/5 bought
Done buying


Day 1/10
Sales
+$8
#1
23
Calm day · next:
$11
$13
×2 $7
×1 $7
Sarah & Sparkles
Kumar the Genie
tomorrow
Foxy Rosa
Haggler
11
SHOWDOWN 1/3
cheapest sells first
Sarah & Sparkles
×1
pays up to $8 each
"Lovely!"
You
offers 1
SOLD
$8
No rivals! 1 blueberry sold at full price.
Big Benny
Charmer
13
You
YOU
Big Basket
1/8
23
Free Fruit
Skip
Selling... tap to speed up







## 2 · Why it doesn't feel polished: four causes

### A. Things that look broken (defects, easy to fix)

1. **Sales reveal:** on shorter phones the explanation is cut off at the bottom ("...sold at full price" disappears behind the bar), and the card covers your own stall.
2. **The "+$8" money pop** flies off the top of the screen.
3. **Grammar:** "**You offers** 1" on the reveal card.
4. **Pricing panel:** the "Lock in prices" button floats over the next customer's card, which shows underneath it.
5. **Evening upgrades:** the Build buttons sit at different heights; card text is long and uneven.
6. **Trait picker:** the title wraps awkwardly and the panel covers your stall.
7. **Mixed icon styles:** emoji ( 🃏 favour count, ⏩ Skip, 💭, 📊, 🔒, 🧑‍🤝‍🧑/🤖) beside the pixel art.
8. **Placeholders that look unfinished:** the grey "walk-in" and "tomorrow" ghosts and the dashed "?" bubble.

9. **Trophies:** achievements have no icons, and locked and unlocked look the same (all grey text).

**B. Too much on screen at once (layout)**

- Before the table you see: the top bar, a second row (event, rival cash), customers, and a 4-box demand strip ("1+ want / 3 in shops"). That's four bands of small information.
- **Numbers shown twice:** your cash (top right and your stall) and rivals' cash (top chips and their stalls).
- **Rival stalls** take about 40% of the width to show mostly "0" rows, which squeezes the farmers market and makes its price labels ($1 $2 $3) tiny.
- **Your stall** is a big grid of empty pastel boxes with tiny "−$2" labels. Hard to read at a glance.

**C. No single focus per moment** Every phase shows the whole table at the same size. When you're pricing, the customers matter most; when you're buying, the market does. Right now nothing steps forward, so your eye has nowhere to land.

**D. The visual system isn't consistent**

- Two fonts are used without a clear rule: the pixel font appears in titles, numbers and small labels alike.
- Cards come in several styles: cream paper, navy chips, outlined boxes.
- Colours carry meaning (red outline, green outline, teal), but it isn't explained or applied consistently.
- Feedback and motion are uneven: some actions animate, others just switch.




## 3 · Proposal: three phases, smallest risk first

### Phase 1 · Fix what looks broken (PLAN-007, build now, low risk)

All nine items in **A**, presentation only:

- the reveal card fits every phone and stops covering your stall
- the money pop stays on screen
- the grammar and other copy are corrected
- the pricing button sits in a solid footer
- upgrade cards are aligned and their text shortened
- the trait picker fits
- the remaining emoji become pixel icons (backlog B-33)
- the walk-in and tomorrow placeholders get proper art
- trophies get icons and a clear locked look

Ships with the Patches rename as **1.1.2**.

### Phase 2 · Calm the table (PLAN-008 mockups first, then a build plan)

- **One compact top bar:** menu · day · phase · today's event icon · your cash and rank. Rivals' cash appears only on their stalls.
- **Slim rival stalls:** face, name, cash, and only the fruit they actually hold. This frees width so the farmers market and its prices get bigger.
- **Plain-language demand:** replace "1+ want / 3 in shops" with icons and words on the customer bubbles, for example "2 shops selling: price war".
- **Your stall:** show your fruit as items with "nights left", not a grid of empty boxes.
- **One focus per phase:** buying enlarges the market; pricing brings customers forward and dims the rest; the reveal gets the full stage; the evening shows only the upgrades.

### Phase 3 · Make it feel finished (after Phase 2 is approved)

- **A small design system:** one type scale (pixel font for titles and big numbers only), one card style, spacing on a fixed grid, and colour roles (green good, red risk, gold money, teal info).
- **Motion and feedback:** phase transitions, a press effect on every button, and the **chalkboard reveal** from REP-010 (price boards flip). The reveal waits for the playtest result, because it's pillar 2.
- **Sound paired with the key moments** (backlog B-34), plus the harbour-market identity (DEC-017) on the home screen: a harbour backdrop, crates, chalkboards.

## 4 · Order of work (recommended)

1. **Now:** publish 1.1.1 and run the first playtest (DEC-016) on the *current* look. That gives a baseline, so we can show whether the new UI really makes the game clearer.
2. **Now, in parallel:** build Phase 1 (PLAN-007). Low risk; goes out as 1.1.2.
3. **Then:** PLAN-008, interactive mockups of the buy, price, reveal and evening screens plus home, as a page you can open on your phone, with two options where there's a real choice (for example, the rival stalls). **No game code.** You pick.
4. **Then:** a build plan for the chosen design (Phases 2 and 3), released as **1.2.0**, followed by a second playtest compared with the baseline.

## 5 · PLAN-007 acceptance (Phase 1)

- P1. Every item in **A** has before-and-after screenshots at 360×640 and 390×844.
- P2. Nothing is cut off or overlapping, and nothing scrolls sideways, on those sizes (automated check on each phase screen).
- P3. The visual-diff tool shows changes only on the screens being fixed.
- P4. The speed budget still passes (no long tasks over 200 ms; page size within budget).
- P5. `verify:full` is green; no rule, save or server changes.

**PLAN-008 acceptance (mockups):** an interactive page you can open on your phone, covering 5 screens, with clear options where a choice is needed, and every idea mapped to a problem above. No game code.

## 6 · Costs and risks

- RM0. No new tools or services.
- **Risk:** Phase 2 changes how the table is laid out, which could confuse returning players. Mitigation: mockups first, then the second playtest.
- **Risk:** polish can expand forever. Mitigation: each phase has a fixed list and its own approval.

## 7 · Questions for you (max 3)

1. **What bothers you most?**

   - (a) cluttered screens
   - (b) things that look broken
   - (c) it lacks animation or "feel"
   - (d) the art style itself

   (This sets the Phase 2–3 priorities. If it's (d), that's a bigger art decision.)

2. **Is there a game whose look you'd call "polished"?** A name or a screenshot helps me aim the mockups.

3. **Approve PLAN-007 (Phase 1 fixes) and PLAN-008 (mockups)?**

## 8 · Decisions

| ID | Question | Recommendation | Status |
|---|---|---|---|
| DEC-020 | How to polish the UI and UX | Three phases; fix defects first; mockups before any layout change; a playtest before and after | Awaiting approval |
| PLAN-007 | Phase 1: fix what looks broken | Build now; ship in 1.1.2 | Awaiting approval |
| PLAN-008 | Interactive mockups for Phases 2–3 | Design only, no game code | Awaiting approval |

# REP-015 · Art direction: what "It Takes Two" can teach a 2D phone game

| | |
|---|---|
| Game | Fruit Market |
| Report date | 2026-10-26 23:18 (Asia/Kuala_Lumpur) |
| Owner input | Bothers most: (a) clutter, (c) lack of animation and feel, (d) the art style itself. Polished reference: *It Takes Two* |
| Type | Creative direction options and a revised plan. **No game changes.** |
| Approval status | DEC-021 and the revised PLAN-008 **awaiting approval**; PLAN-007 still awaiting approval |

## 1 · What this changes

REP-014 started with fixing broken-looking details. Your answer puts the priority elsewhere: **the art style, the feel, and the clutter**. That's a bigger and more important decision, so it should be designed and tested before anything is built.

## 2 · What makes *It Takes Two* feel polished (the qualities we can borrow)

*It Takes Two* (Hazelight, 2021) is a 3D console game, so we can't copy its look or technology, and we must not reuse any of its art or characters. Its polish comes from qualities that translate to 2D:

| Quality | In *It Takes Two* | What it could mean for Fruit Market |
|---|---|---|
| **Toy-like, touchable materials** | The world is shrunk to toy size: wood, felt, clay, fabric | Stalls of painted wood, fruit like shiny clay or wooden toys, cloth awnings, chalkboard price tags |
| **Chunky, rounded shapes** | Soft silhouettes, big heads, readable at a glance | Rounder cards and buttons, bigger and simpler characters, fewer thin lines |
| **Warm light and depth** | Soft shadows, glow, layers of foreground and background | Gentle drop shadows, a warm late-afternoon harbour backdrop, depth between the table and the cards |
| **Everything reacts** | Squash and stretch, bounces, particles on every action | Fruit hops into your basket, coins bounce, buttons squish, the price reveal *flips* |
| **Expressive characters** | Faces and poses show feeling all the time | Customers smile, frown or shrug at the reveal; rivals react with a pose, not just a text line |
| **One thing to look at** | The camera always frames the important moment | Each phase zooms into what matters (market, customers, reveal) and dims the rest |

## 3 · Three art directions (choose one)

| | **A · Polished pixel** | **B · Toy-box tabletop (recommended)** | **C · 3D toy diorama** |
|---|---|---|---|
| Idea | Keep pixel art; make it consistent and animated | A smooth 2D illustrated style: toy-like, rounded, warm, with soft shadows and textures | Real 3D: a miniature market you look down on |

| | A · Polished pixel | B · Toy-box tabletop (recommended) | C · 3D toy diorama |
|---|---|---|---|
| Fixes your (d) art style? | **Weakly**: same look, just cleaner | **Yes**: a clearly new, warmer, toy-like look | Yes, and closest to the reference |
| Fixes (c) feel | Partly | Yes: built around motion from the start | Yes |
| Phone speed | Excellent | Good (vector art; we keep the speed budget) | Risky on cheap phones; heavier download |
| Scope | Small | Medium-large: redraw fruits, ~30 characters, rivals, ~40 icons, UI | Very large: a new rendering engine and 3D models |
| Who makes the art | Claude, in code (as now) | Claude, in code as vector art (consistent, revisable), **or** a hired illustrator later | 3D modelling is beyond what Claude can do well; it needs an artist (paid) |
| Cost | RM0 | RM0 if Claude draws it; an illustrator would be a separate paid decision | Paid artist plus large engineering |
| Risk | Doesn't satisfy (d) | Art quality by Claude is good but not AAA; judge it on the style test before committing | Highest cost, slowest; could stall the game |

**Recommendation: B, proven by a small style test first.** It addresses all three of your concerns, stays fast on phones, costs nothing if I draw it, and can be judged before any real work. If the test looks too plain, the next step would be to brief a paid illustrator. That would be a separate, priced decision under your RM500/month ceiling.

## 4 · Revised PLAN-008 · Style test plus layout mockups (design only, no game code)

1. **Style test:** the *same* buying screen drawn three ways (A, B, C as a flat render), plus one fruit at its four freshness stages, one customer and one rival in direction B. It's a page you open on your phone.
2. **Layout mockups in the chosen direction** for buy, price, reveal, evening and home, with:
   - the clutter fixes from REP-014 (one top bar, slim rival stalls, demand in plain words)
   - one focus per phase
   - motion notes (what moves, when) and a short animated example of the reveal flip
3. **Your choice:** you pick the direction and the layout options. Only then do I propose the build plan(s).

**Acceptance:** M1, a phone-viewable page with the style test (3 directions on one screen) and the 5 mockup screens. M2, every idea is linked to a problem in REP-014 or a quality in section 2. M3, no game code changed. All art is original; reference games are inspiration only.

## 5 · Where PLAN-007 (the 9 broken-looking fixes) fits now

You didn't pick (b), and a new art style would redraw most of those screens anyway. Two options:

- **Keep it small (recommended):** do only the fixes that survive any art change. That's the grammar ("You offers"), the money pop going off-screen, and the sales text being cut off. Ship them in 1.1.2 with Patches; park the rest until the new style.
- Or drop PLAN-007 entirely and fold everything into the new design.

## 6 · Order of work

1. Publish 1.1.1 (your Deploy) and run the first playtest on the current look (baseline).
2. PLAN-008 style test, then you choose a direction.
3. Layout mockups in that direction, then you choose.
4. A build plan for the new look and feel (likely several packages), released as **1.2.0**, then a second playtest.

## 7 · Questions (max 3)

1. **Approve the revised PLAN-008** (style test plus mockups, no game code)?
2. **PLAN-007:** keep it small (3 fixes in 1.1.2) or drop it?
3. **Is a paid illustrator possible later,** if the B style test isn't good enough? A yes or no now helps me plan; any spending would still need its own quote and approval.

## 8 · Decisions

| ID | Question | Recommendation | Status |
|---|---|---|---|
| DEC-021 | Art direction for the polish | Explore A/B/C in a style test; recommend B (toy-box tabletop) | Awaiting approval |
| PLAN-008 (revised) | Style test plus layout mockups, design only | Approve | Awaiting approval |
| PLAN-007 | Broken-looking fixes | Reduce to 3 fixes that survive any art change | Awaiting your choice |

# REP-016 · Toy-box mockups ready for your review (PLAN-008)

| | |
|---|---|
| Game | Fruit Market |
| Report date | 2026-10-26 23:57 (Asia/Kuala_Lumpur) |
| Package | PLAN-008 (design only), approved: "go with plan b" |
| Mockups | https://claude.ai/artifact/YQUpnHeNNWkAUqFhKRz9VB (private to you) |
| Type | Design mockups. **No game code changed.** The live game is unchanged |
| Delivery status | Implemented; waiting for your review and choices |

## What's on the page (phone-sized, tap to try)

| Screen | What to notice | Problem it answers |
|---|---|---|
| **Style sample** | Fruit at four freshness stages (shiny, then spotted, then droopy); Grandpa Tan and Patches as rounded toy figures; wood, chalkboard, coin and a pressable button | Art style (d); "freshness you can see" (pillar 1) |
| **Home** | Harbour backdrop with bobbing boats, a market sign "Where the whole world shops", big pressable buttons, "2 to 4 players online" | Harbour identity (DEC-017); clutter (a) |
| **Buying** | One top bar; customers with plain-language demand ("3 shops sell apples"); slim rival chips with only what they hold; big crates (tap to buy: coins drop, the basket count bounces, fruit hops onto your stall); fruit shows "nights left" | Clutter (a): no repeated cash, no "1+ want / 3 in shops"; feel (c) |
| **Pricing** | One customer in focus, the table dimmed; chalkboard price with big – and +; a hint about rivals in words; a solid "Lock in prices" footer | No focus (REP-014 C); floating button defect |
| **Sales reveal** | Ring the bell: boards flip highest to lowest, the cheapest wins, coins fly to the winner, " too high" on yours, plus a plain explanation and the rival's quip | Pillar 2 and the reveal-comprehension test; feel (c) |
| **Evening** | Dusk sky, three equal upgrade crates with aligned buttons, a "BUILT" stamp, a favour card, "Lights out" | Misaligned upgrade buttons; feel (c) |

## Limits of the mockups

- They're **mockups, not the game**: the numbers are examples, and only the parts needed to show the idea are interactive.
- Characters other than Grandpa Tan, Patches, Foxy Rosa and Cowboy John are not drawn yet; the build would redraw all of them in this style.
- I didn't add sound; that comes with the build.
- The style is original. *It Takes Two* informed the qualities only.

## What I need from you

1. **Overall:** does this look and feel like the direction you want? (yes / closer to … / no)

2. **Anything you love or dislike** on a specific screen (a comment on the page works too).
3. If yes, I'll propose the **build plan** to turn the real game into this style. It's several packages: art and the shared style first, then each screen. It would be released as 1.2.0, after the baseline playtest.

## Acceptance (PLAN-008)

| # | Check | Result | Evidence |
|---|---|---|---|
| M1 | Phone-viewable interactive page with the style sample and 5 screens | PASS | Published at the link above: 6 phone screens plus a shared fruit part. Your review is the real check |
| M2 | Every idea linked to a REP-014 problem or REP-015 quality | PASS | Table above |
| M3 | No game code changed | PASS | No changes under `src/` or `server/` |

# REP-017 · Mockups revised: the cunning edition (PLAN-008, DEC-022)

| | |
|---|---|
| Game | Fruit Market |
| Report date | 2026-10-27 00:12 (Asia/Kuala_Lumpur) |
| Package | PLAN-008 (design only); direction DEC-021, tone DEC-022 |
| Owner feedback | "yes but align more with the cunning aesthetic given the undercut mechanism of the game" |
| Mockups | https://claude.ai/artifact/YQUpnHeNNWkAUqFhKRz9VB (same link, version 9, private to you) |
| Type | Design mockups. **No game code changed.** The live game is unchanged |
| Delivery status | Implemented; waiting for your review |

## The idea in one line

The market is sunny by day, but **pricing is a secret game**. When something is secret, the screen turns **plum and gold**, rivals look sly, and the reveal plays like a showdown. Cheeky, never mean.

## What changed, screen by screen

| Screen | Cunning touches |
|---|---|
| **Style sample** | Plum header; Foxy Rosa as the sly mascot (raised brow, wink, smirk: "Psst... I'll go a dollar lower."); a wax-sealed price board; the dotted "Psst..." whisper bubble; palette row: market colours by day, plum and gold for secrets |
| **Home** | New tagline "Undercut. Outfox. Sell it fresh." (kept "Where the whole world shops"); Foxy Rosa peeks out from behind the sign every few seconds; online buttons in plum; "Outfox a stranger" |
| **Buying** | A whisper: "Psst... Foxy Rosa is sitting on 2 apples. Who blinks first?"; sly rival faces (Foxy Rosa narrowed eyes, Patches half-lidded) |
| **Pricing** | Plum top bar ("top secret"); "Seal your prices"; "Only you see this" over your chalkboard; the rival hint is now a whisper from Foxy Rosa; a "Rivals are scheming..." row with their sealed boards and shifty eyes; footer "Seal my prices" with a wax seal |
| **Sales reveal** | Dark stage with a curtain and spotlight; the winning board glows gold; headline "Outfoxed by $1!" with Foxy Rosa's quip and face |
| **Evening** | "Plotting time"; the favour card is "Up your sleeve"; a whisper: "the rivals are plotting tonight too". Your layout edit (favour card lower) is kept |

## Fit with the approved pillars

- **Pillar 2, outprice rivals in secret:** now the strongest visual idea.
- **Pillar 3, a cheerful town market:** kept. Rivals tease and wink; nobody is cruel; daytime stays sunny. If playtesters read the rivals as mean, we soften them (DEC-022 review trigger).
- **Privacy rule:** whispers only repeat what the game already shows (who holds what). They never reveal a rival's secret price or choice.

## Limits

- Mockups, not the game; numbers are examples. Characters other than Foxy Rosa, Patches, Grandpa Tan and Cowboy John are not drawn yet. No sound yet.
- Plum screens need a contrast check in the build (text sizes and colours were chosen for it, not measured).

## Acceptance (PLAN-008, added check)

| # | Check | Result | Evidence |
|---|---|---|---|
| M4 | Owner feedback applied; owner's own page edit kept | PASS | Version 9 published; Evening spacer kept |
| M3 | No game code changed | PASS | Changes only in docs/workflow |

## What I need from you

1. Is this the right amount of cunning? (more / about right / less)
2. If about right, I'll propose the **build plan** for 1.2.0 (art and shared style first, then each screen).

# REP-018 · Mockups rebuilt: the cosy pixel edition (PLAN-008, DEC-023)

| | |
|---|---|
| Game | Fruit Market |
| Report date | 2026-10-27 00:17 (Asia/Kuala_Lumpur) |
| Package | PLAN-008 (design only); look decision DEC-023 (awaiting your confirmation) |
| Owner feedback | "its still not enough, maybe balance the artstyle back to pixel art, the ui/ux here looks more polished but lacks the cosiness of how the previous board was set" · "no i meant the version before you tweaked here, that felt cosy" |
| Mockups | https://claude.ai/artifact/YQUpnHeNNWkAUqFhKRz9VB (same link, version 10, private to you) |
| Type | Design mockups. **No game code changed.** The live game is unchanged |
| Delivery status | Implemented; waiting for your review |

## The idea in one line

**Pixel heart, smooth frame.** The first toy-box mockups (the cosy ones) are back, but the characters, fruit and icons are now the game's own pixel art, and headings use the game's pixel font.

## What changed from the cunning edition

- **Cosy restored:** the first toy-box layout and warm colours (cream, teal, wood, terracotta). Your Evening layout edit is kept. The dark plum screens, curtain and spotlight are gone.
- **Pixel art:** the fruit at four freshness stages, Grandpa Tan, Cowboy John, Foxy Rosa, Patches, you, the upgrades, bell, coin and favour card all use the game's own sprites (built from `src/ui/pixel.tsx`). Foxy Rosa and Patches get sly half-closed eyes.
- **Light sly touch, in warm colours:** dotted "Psst..." whispers from Foxy Rosa (buying, pricing), "Seal your prices" and "only you see this", "Outfoxed by $1!" on the reveal, "...and every price is a secret" on the home sign, with Foxy Rosa peeking out from behind it.

## Why this is cheaper to build

The full vector redraw (DEC-021) is no longer needed. The build reuses and tidies the existing sprites; most of the work is the new frame and layout.

## Limits

- Mockups, not the game; numbers are examples. No sound yet.
- The sly eyes for Foxy Rosa and Patches are mockup variants; the build would add them as sprite variants.

## Acceptance (PLAN-008, added check)

| # | Check | Result | Evidence |
|---|---|---|---|
| M5 | Cosy pixel revision; your page edit kept | PASS | Version 10 published; Evening spacer kept |
| M3 | No game code changed | PASS | Changes only in docs/workflow |

## What I need from you

1. Does this feel cosy again, and is the pixel balance right? (yes / more pixel / less pixel)
2. If yes, I'll record DEC-023 as approved and propose the 1.2.0 build plan.

# REP-019 · Mockups rebuilt as a board game: one shared market table (PLAN-008, DEC-024)

| | |
|---|---|
| Game | Fruit Market |
| Report date | 2026-10-27 00:25 (Asia/Kuala_Lumpur) |
| Package | PLAN-008 (design only); layout decision DEC-024 (awaiting your confirmation) |
| Owner feedback | "revamp the current UI/UX to match the cosiness of the previous UI/UX board layout" · "the one currently feels very menu based, instead of board game" |
| Mockups | https://claude.ai/artifact/YQUpnHeNNWkAUqFhKRz9VB (same link, version 12, private to you) |
| Type | Design mockups. **No game code changed.** The live game is unchanged |
| Delivery status | Implemented; waiting for your review |

## The idea in one line

**You sit at the market table.** Every game screen is the same board, like the current game, now with the cleaner, pixel-art, warm look.

## The table

- **Street (top):** customers queue as pixel figures with a small tag ("apple ×2 · up to $10") and one plain hint ("3 shops sell apples").
- **Middle:** Foxy Rosa's stall, the Farmers market, Patches' stall, each under a striped awning. Rival stalls show only what matters: cash and what they hold.
- **Bottom:** your stall, with your fruit and how many nights each has left.

## Each phase on the same table

| Phase | What happens on the board |
|---|---|
| Buying | Tap crates in the market; fruit hops onto your stall. Foxy Rosa whispers. |
| Pricing | Tap a customer on the street to serve them. Rival stalls show sealed boards ("sealed..."). Your stall becomes your price board, which only you see. "Seal my prices". |
| Reveal | Ring the bell: the boards flip **on the stalls** (Patches $10, you $9, Foxy Rosa $8). Foxy Rosa's stall glows, coins fly to it, Grandpa Tan walks over, then "Outfoxed by $1!". |
| Evening | Night falls over the same table: lanterns on the rival stalls ("plotting..."), the market covered, and the upgrades at your stall. Favour card kept low, as in your earlier edit. |

## Limits

- Mockups, not the game; numbers are examples. Home and the style sample are unchanged apart from a subtitle.

- NOT RUN: I did not view the page myself; the four screens' scripts were checked to load.

## Acceptance (PLAN-008, added check)

| # | Check | Result | Evidence |
|---|---|---|---|
| M6 | Board-game layout on one shared table | PASS | Version 12 published |
| M3 | No game code changed | PASS | Changes only in docs/workflow |

## What I need from you

1. Does this feel like a cosy board game now? (yes / closer to … )
2. If yes, I'll record DEC-023 and DEC-024 as approved and propose the 1.2.0 build plan.

# REP-020 · Look approved; market made bigger (PLAN-008, DEC-023, DEC-024)

| | |
|---|---|
| Game | Fruit Market |
| Report date | 2026-10-27 01:21 (Asia/Kuala_Lumpur) |
| Package | PLAN-008 (design only) |
| Owner feedback | "going back to your mockup, i like it now, but tweak the layout so that the fruit market is more visible (now it feels to small/ narrow)" |
| Decisions | DEC-023 (cosy pixel look) and DEC-024 (one shared table) approved |
| Mockups | https://claude.ai/artifact/YQUpnHeNNWkAUqFhKRz9VB (version 13, private to you) |
| Type | Design mockups. **No game code changed** |

## What changed

- Rival stalls are slimmer (80 to 64 px wide); the Farmers market grows from about 206 to 234 px.
- Crates are taller (58 to 74 px) with bigger pixel fruit (28 to 42 px) and bigger prices.
- The market title is larger; the street strip is a little shorter to make room.
- The same slimmer stalls are used on pricing, reveal and evening, so every screen matches.

## Validation

| Check | Result | Evidence |
|---|---|---|
| Market wider, crates and fruit bigger | PASS | Version 13 published |
| Screen scripts load | PASS | All four parse |
| Viewed on a phone by me | NOT RUN | Please check the Buying screen |
| No game code changed | PASS | Changes only in docs/workflow |

## Next

Propose the 1.2.0 build plan for this look (separate packages, each needing your approval).

# REP-021 · Vertical decay board and the delivery lorry (PLAN-008, DEC-025)

| | |
|---|---|
| Game | Fruit Market |
| Report date | 2026-10-27 01:28 (Asia/Kuala_Lumpur) |
| Package | PLAN-008 (design only) |
| Decision | DEC-025, approved by your instruction |
| Mockups | https://claude.ai/artifact/YQUpnHeNNWkAUqFhKRz9VB (version 14, Buying screen, private to you) |
| Type | Design mockups. **No game code or rules changed** |

## What's on the Buying screen now

- **Market as a decay tracker, vertical:** four columns left to right (blueberry, watermelon, apple, mango). New and ripening fruit at the top, spoiling at the bottom, with a colour bar (green, gold, brown). Every slot shows its price; empty slots stay as dashed outlines, like today's game. The tracks match the real rules: blueberry 3 slots, watermelon 5 + 1 unripe, apple 7, mango 4 + 2 unripe.
- **Your stall uses the same tracker** as four small strips; fruit you buy pops into its slot.
- **The lorry:** when buying starts it drives in along the street, Grandpa Tan and Cowboy John shuffle left to make room, and the crates drop one by one onto the board. Press **Done buying** and the lorry drives off, the customers spread back out, and the button changes to "Lorry's gone · set prices →". A small **↻ replay lorry** button lets you watch it again.

## Validation

| Check | Result | Evidence |
|---|---|---|
| Slots and prices follow the game's tracks | PASS | Matches src/engine/data.ts |
| Buying moves fruit to the right slot on your stall | PASS | Logic test of the screen |
| Lorry leaves on Done buying | PASS | Logic test of the screen |
| Viewed on a phone by me | NOT RUN | Please check the animation |
| No game code changed | PASS | Changes only in docs/workflow |

## Notes for the build

- The lorry must be quick (about 2 seconds) and skippable with a tap, so it never slows play.
- The other screens keep the shared table; only Buying shows the market.

# REP-022 · Build plan for the cosy table look (PLAN-009 to PLAN-011)

| | |
|---|---|
| Game | Fruit Market |
| Report date | 2026-10-27 01:58 (Asia/Kuala_Lumpur) |
| Owner request | "ok now do this changes" (read as: build the mockup look into the game) |
| Type | Planning only. **No game code changed** |
| Approval status | PLAN-009 awaiting approval; PLAN-010 and PLAN-011 are outlines |

## Good news first

The current game already has rival seats, a rival pop-up, a market and the want / in-shops strip. So the build mostly **restyles and rearranges** what exists; rules, saves and the server stay as they are.

## Three packages, one at a time

| Plan | What you get | Size |
|---|---|---|
| **PLAN-009** | Rivals seated across the top; tap a name to see their freshness chips; full-width market board (fruit columns, top-aligned, colour-coded, key, want / in-shops on each column); your stall as freshness chips; smaller customers | Largest; first |
| PLAN-010 | The delivery lorry: drives in, customers step aside, crates land; leaves before selling; skippable | Small |
| PLAN-011 | Pricing, reveal and evening on the table (sealed boards, flips in the seats, lanterns); folds in PLAN-007's text fixes | Medium |

Release as **1.2.0** once all three are done and checked on the preview.

## Checks for PLAN-009 (all must PASS)

verify and verify:full; board slots and prices tested against the rules; column chips tested against today's numbers; screenshots with 2, 3 and 4 players at small phone, normal phone and desktop with no clipping; secret-price leak check stays 0; old saves load.

## Risks and costs

- RM0. No new dependencies.
- **Small phones (360×640):** the 7-row board is tall. I'll make the rows shorter there; if it still doesn't fit, the table scrolls rather than squashing.
- **Release order:** `main` is holding the 1.1.1 candidate until you press Deploy. I'll build on the work branch, and nothing reaches the preview until 1.1.1 is live.
- **Lorry art (PLAN-010):** made with Gemini. Google doesn't claim ownership of it, but AI-made art may not be protected by copyright, so others could copy it. The alternative is a pixel lorry

drawn in the game's own sprite style, which would match the other art.

## Decisions needed

1. Approve PLAN-009?
2. Lorry: your Gemini picture, or a pixel lorry in the game's sprite style?
3. PLAN-007 (old small fixes): fold its text fixes into PLAN-011 and close it?

# REP-023 · PLAN-009 built: seated rivals and the new market board

| | |
|---|---|
| Game | Fruit Market |
| Report date | 2026-10-27 02:19 (Asia/Kuala_Lumpur) |
| Package | PLAN-009, approved ("yes" to REP-022) |
| Branch | `workflow/plan-009` , commit b2b495a. **Not on the preview or the live game yet** |
| Delivery status | Verified |

## What changed in the game

- **Rivals sit across the table** in one row: portrait, name, cash and what they hold. Tap a seat and their stall drops down as **freshness chips**, the same as your own.
- **The market is the centre:** full width, one column per fruit, newest at the top, every slot coloured by freshness, with a colour key. Each column starts with its **want / in shops** chip.
- **Your stall** shows chips such as "×4 ripens in 2" or "3 nights"; a red "!" marks fruit on its last day.
- Tutorial and how-to text updated for the new board. Rules, saves, server and online play are unchanged.

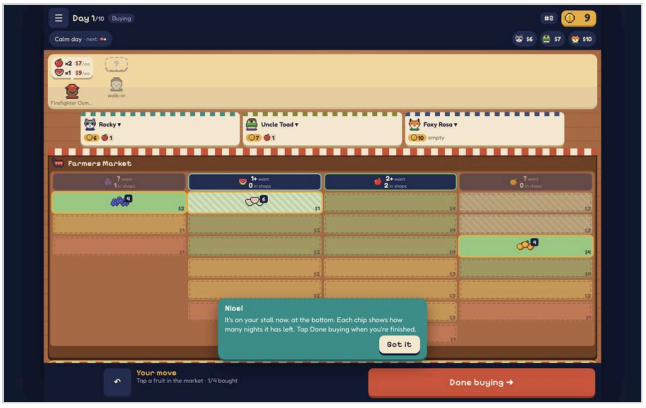
## Screenshots


Day 1/10
Buying
#3
8
Calm day · next:
$10
$10
$7
×1 $10/ea
×1 $7/ea
Princess Siti
walk-in
Big Benny ▾
10 empty
Rocky ▾
10 empty
Patches ▾
7 1
Farmers Market
? want 1 in shops
1+ want 1 in shops
? want 0 in shops
1+ want 0 in shops
Nice!
It's on your stall now, at the bottom. Each chip shows how many nights it has left. Tap Done buying when you're finished.
Got it
Your move
Tap a fruit in the market · 1/4 bought
Done buying


Day 1/10
Buying
#3
8
Calm day · next:
$10
$10
$7
×1 $10/ea
×1 $7/ea
Princess Siti
walk-in
Big Benny ▴
10 empty
Rocky ▾
10 empty
Patches ▾
7 1
Big Benny's stall
Local Favourite
10
Nothing on the shelf yet.
Nice!
It's on your stall now, at the bottom. Each chip shows how many nights it has left. Tap Done buying when you're finished.
Got it
Your move
Tap a fruit in the market · 1/4 bought
Done buying


Day 1/10
Buying
#3
2
Calm day · next:
$16
$10
Princess Siti
walk-in
Patches ▾
16 empty
Uncle Toad ▾
10 empty
Farmers Market
? want 0 in shops
1+ want 0 in shops
? want 0 in shops
1+ want 0 in shops
unripe
fresh
okay
not so
You
YOU
Local Favourite
4/8
4/4
2
×4 ripens in 2
Your move
Tap a fruit in the market · 4/4 bought
Done buying


Day 1/10
Buying
#4
8
Calm day · next:
$10
$10
$10
Dr Priya
walk-in
Uncle Toa...
10 empty
Rocky ▾
10 empty
Pinky ▾
10 empty
Farmers Market
1+ want 1 in shops
? want 0 in shops
2+ want 0 in shops
? want 0 in shops
Nice!
It's on your stall now, at the bottom. Each chip shows how many nights it has left. Tap Done buying when you're finished.
Got it
Your move
Tap a fruit in the market · 1/4 bought
Done buying



## Validation

| # | Check | Result |
|---|---|---|
| B1 | Quick checks (lint, types, 73 unit tests, build) | PASS |
| B2 | Full browser tests (solo, speed budget, online with a leaver, quick match, four players, reconnect) | PASS |
| B3 | Board slots and prices match the rules (30 games) | PASS |
| B4 | Want / in-shops numbers unchanged (30 games) | PASS |
| B5 | 2, 3 and 4 players on small phone, phone and desktop: no sideways scroll or errors | PASS |
| B6 | Rival drop-down shows only public stock; secret-price leak check 0 | PASS |
| B7 | Rules, saves, server untouched | PASS |

## Notes

- On small phones the table scrolls to reach your stall (as planned). Long names shorten ("Uncle Toa…") with three rivals on the smallest phones.
- Pricing, reveal and evening still use today's panels; that is PLAN-011. The lorry is PLAN-010.

## Next

1. Press **Deploy LIVE game** for 1.1.1 and tell me "deployed". Then I can merge this to `main`, which puts it on the **preview** for you to try.
2. Lorry art choice for PLAN-010: your Gemini picture, or a pixel lorry in the game's style.

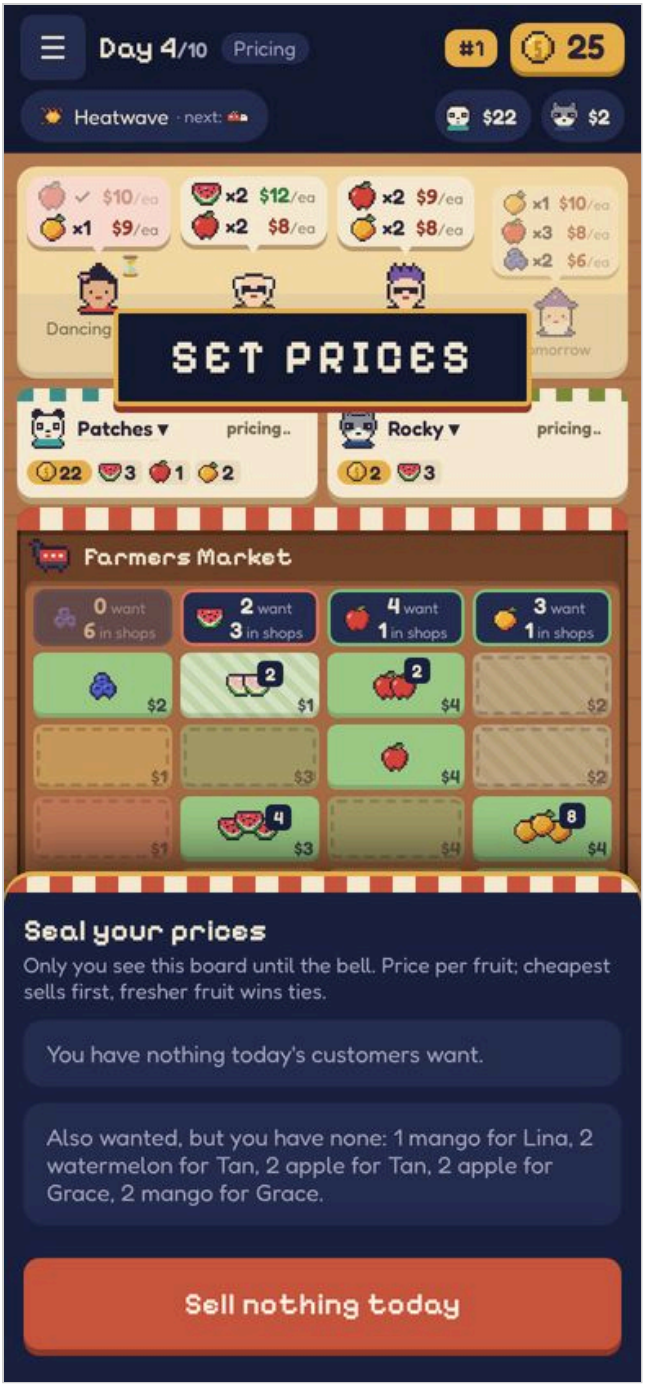
# REP-024 · PLAN-011 built: pricing, reveal and evening on the table

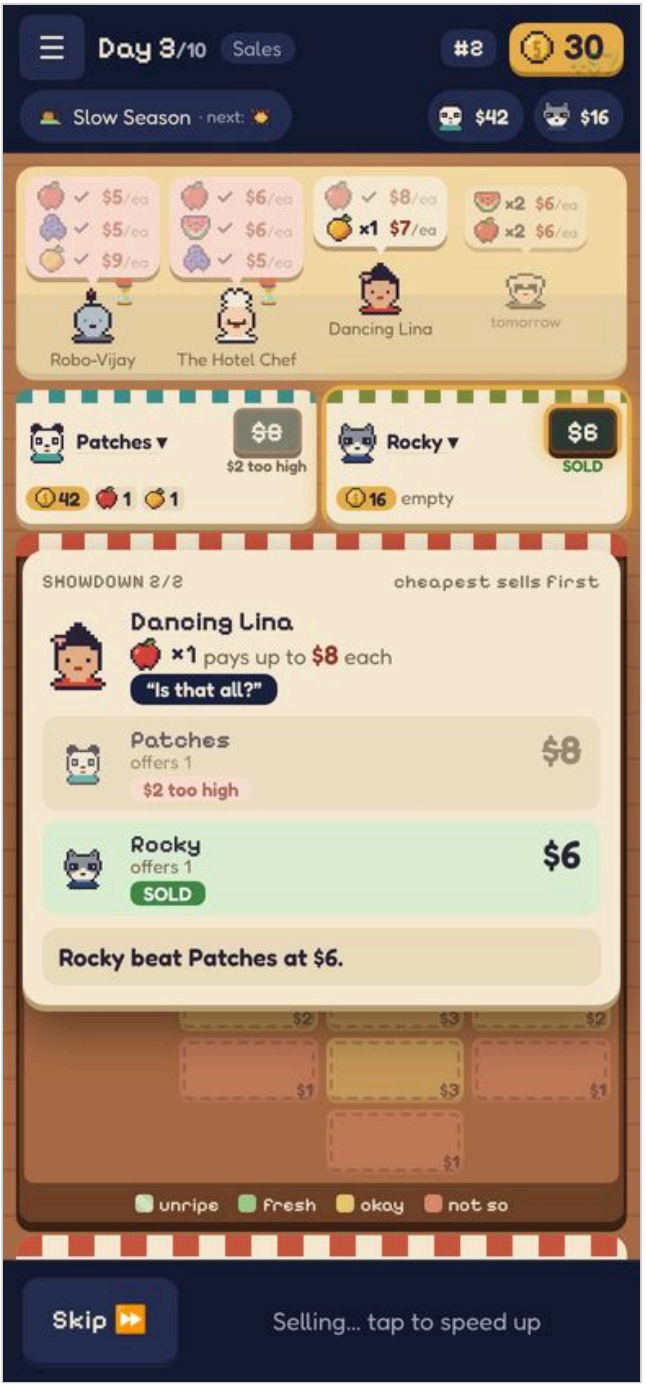
| | |
|---|---|
| Game | Fruit Market |
| Report date | 2026-10-27 02:51 (Asia/Kuala_Lumpur) |
| Package | PLAN-011, approved ("use the gemini lorry, approve PLAN-011") |
| Branch | `workflow/plan-009` , commit 4408cd8. **Not on the preview or the live game yet** |
| Delivery status | Verified |

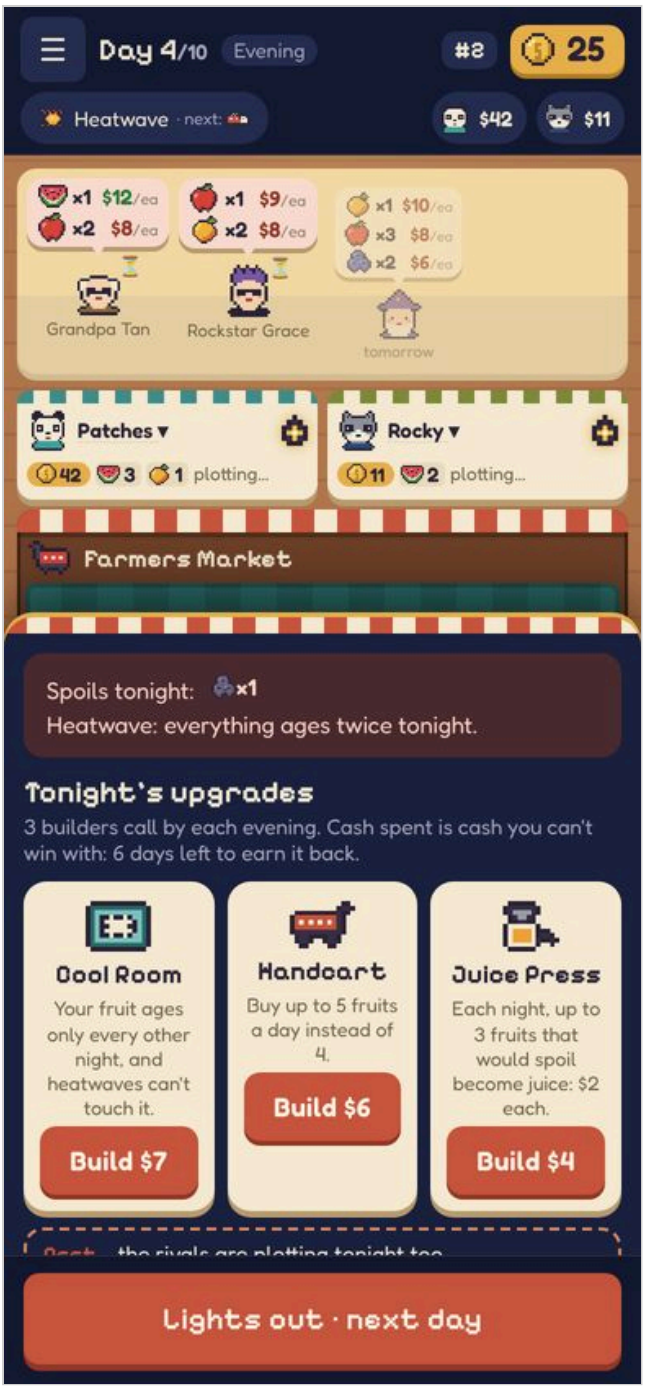
## What changed in the game

- **Pricing:** each rival's seat shows "pricing...", then a sealed board once they're done. Nobody sees a price. Your panel says **Seal your prices**, with a "Psst..." read of the competition, and the button says **Seal my prices**.
- **The reveal happens on the table:** the boards flip open on each seat and on your stall, highest price first. The winner's seat glows, and tags like **SOLD** or **$1 too high** sit under the boards. If a rival undercuts you, the card says **Outfoxed!**
- **Evening:** lanterns and "plotting..." on the rival seats, and the market crates are covered for the night. Favour cards sit under **Up your sleeve**, with a "Psst... the rivals are plotting tonight too."
- **Old small fixes (PLAN-007, now closed):** "You offer" instead of "You offers", money pop-ups stay on screen, and the showdown card scrolls instead of being cut off.

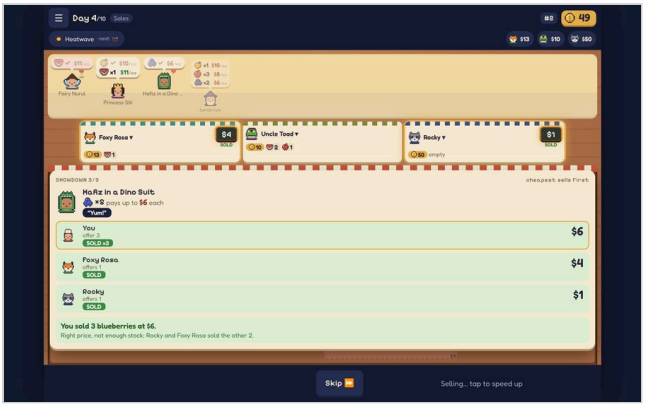
## Screenshots


Day 4/10
Pricing
#1
25
Heatwave · next:
$22
$2
SET PRICES
Patches ▾
pricing..
Rocky ▾
pricing..
Farmers Market
Seal your prices
Only you see this board until the bell. Price per fruit; cheapest sells first, fresher fruit wins ties.
You have nothing today's customers want.
Also wanted, but you have none: 1 mango for Lina, 2 watermelon for Tan, 2 apple for Tan, 2 apple for Grace, 2 mango for Grace.
Sell nothing today


Day 3/10
Sales
#2
30
Slow Season · next:
$42
$16
Robo-Vijay
The Hotel Chef
Dancing Lina
tomorrow
Patches ▾
$8
$2 too high
Rocky ▾
$6
SOLD
empty
SHOWDOWN 2/2
cheapest sells first
Dancing Lina
×1 pays up to $8 each
"Is that all?"
Patches
offers 1
$2 too high
$8
Rocky
offers 1
SOLD
$6
Rocky beat Patches at $6.
unripe
fresh
okay
not so
Skip
Selling... tap to speed up


Day 4/10
Evening
#8
25
Heatwave · next:
$42
$11
Grandpa Tan
Rockstar Grace
tomorrow
Patches ▾
Rocky ▾
plotting...
plotting...
Farmers Market
Spoils tonight: ×1
Heatwave: everything ages twice tonight.
Tonight's upgrades
3 builders call by each evening. Cash spent is cash you can't win with: 6 days left to earn it back.
Cool Room
Your fruit ages only every other night, and heatwaves can't touch it.
Build $7
Handcart
Buy up to 5 fruits a day instead of 4.
Build $6
Juice Press
Each night, up to 3 fruits that would spoil become juice: $2 each.
Build $4
Lights out · next day


Day 4/10
Sales
+$9
#8
35
Minimum Price · next
$48
$10
$31
Rockstar Grace
Dancing Lina
Hafiz in a Dino ...
Hui Min th
Patches ▾
Foxy Rosa...
Uncle Toa...
empty
SOLD
$1 too high
SHOWDOWN 3/3
cheapest sells first
Hui Min the Elf
×1 pays up to $10 each
"A bargain!"
Foxy Rosa
offers 1
$1 too high
Patches
offers 1
SOLD
$9
Skip
Selling... tap to speed up



## Validation

| # | Check | Result |
|---|---|---|
| T1 | Full browser tests incl. secret-price leak check 0; screenshots at 3 sizes, no sideways scroll | PASS |
| T2 | Quick checks (73 tests, lint, types, build) | PASS |
| T3 | Rules, saves, server and online protocol untouched | PASS |
| T4 | PLAN-007 fixes | PASS |

## Not done

- The customer walking over to the winner (from the mockup) is not built. Fruit and coins already fly between the stall and the customer. It can follow later if you want it.

## Next

1. **Deploy 1.1.1** (Actions → Deploy LIVE game → `PUBLISH`) and tell me "deployed". Then I merge the new look to `main`, and it appears on the **preview** for you to play.
2. Approve **PLAN-010** (the lorry with your Gemini picture). It's the last piece before 1.2.0.

# REP-025 · Balance and pacing audit: how Fruit Market is won, lost and paced

| | |
|---|---|
| Game | Fruit Market (rules version 2, `main` at 15f358a) |
| Report date | 2026-10-27 03:36 (Asia/Kuala_Lumpur) |
| Type | Audit only. **No code changed.** Planning and documentation work, not a game change |
| Method | Read the rules engine line by line ( `src/engine/game.ts` , `data.ts` , `ai.ts` , `daily.ts` , `rng.ts` ), the mode set-up ( `src/ui/App.tsx` , `src/ui/progress.ts` , `server/room.ts` , `server/meta.ts` , `src/net/protocol.ts` ) and the scoring screens; then measured with computer-only simulations ( `npm run balance` at 300 games per test, plus 4 extra read-only probes, 300–400 games each) |
| Approval status | Not applicable (findings and proposals; any change needs a plan and your approval) |

**Caution on the numbers.** Every measured figure comes from computer players playing each other. People price, buy and take risks differently, so treat win rates as the shape of the game, not as human results. Human results need the playtest (DEC-016) and the telemetry proposed in section 6.

---

## A. How the game is won and lost (overview)

- **You win by having the most cash when the last day's sales end.** Nothing else counts: unsold fruit, upgrades and favour cards are worth $0 at the end ( `ranking()` in `src/engine/game.ts` ).
- **Nobody is ever knocked out.** There is no bankruptcy, no health, no timer and no early finish. Every game lasts exactly its length: 8, 10 or 14 days. "Losing" just means not finishing first.
- **Ties:** most cash wins; if level, most fruit sold; if still level, the **seat order decides silently** (seat 0, which is you in solo, daily and league games). The results screen then wrongly says it was "decided on fruit sold".
- **Where money comes from:** buying fruit cheaply at the farmers market and selling it to customers through **secret, sealed prices** where the **cheapest offer sells first**. Freshness limits the top price you may ask and breaks price ties. Everything else (upgrades, traits, favour cards, events) moves the margins.
- **What really decides games** (measured): pricing skill (Hard beats Normal 84% of the time head to head), buying only what customers want, the Signboard and Cool Room upgrades, and the event deck. Turn order is a deliberate catch-up: the poorest shop buys first.
- **Long-term goals outside a single game:** the 6-stage Market League (win to advance, unlimited retries), the daily challenge (score = final cash on a shared leaderboard), and 12 achievements.

---

## B. Win / loss condition table

| # | Outcome | Exact check (code) | When it's evaluated | Notes and problems |
|---|---|---|---|---|
| 1 | **Game ends** | `finishReveal()`: `if (s.day >= s.config.days) phase = 'over'` | After the final day's sales reveal | No evening on the last day: no upgrades, no ageing, no night fines |
| 2 | **Win (1st place)** | `ranking()`: sort by `cash` desc, then `stats.sold` desc | Phase `over` | Third tie-break is the array's original order (stable sort) = seat order. **Hidden outcome** |
| 3 | 2nd–4th place ("loss") | Same ranking | Phase `over` | No other penalty |
| 4 | Dead heat message | `Results.tsx`: `margin === 0` → "A dead heat, decided on fruit sold!" | Results screen | **Wrong text** when sold counts are also equal (then seat order decided). Measured: 0.33% of 3–4-player games |
| 5 | No defeat / elimination | none exists | — | Cash can't go below 0 (buys need cash; the inspector fine is capped at your cash) |
| 6 | "Broke" state (soft loss) | Not in code; arises when cash < $1 and no ripe stock | Any morning | You can still play but can't buy (every fruit costs ≥ $1). Only favour cards (Free Cash, Free Fruit) can rescue you. Measured: 0.3–0.6% of computer players in 3-player games, 2.2% in 4-player Hard games, typically from day 4–6 |
| 7 | Online leaver | Server: grace period, then a computer plays the seat (`aiActFor`) | Live | The match always finishes; the winner is still decided by cash |
| 8 | Local quit | Leaving a local game | Any time | No result; only `quit_day` telemetry |
| 9 | Daily challenge score | `server/meta.ts`: board sorted by `cash`, then `sold`, then submit time; the server replays your moves to verify | On submit | Your "win" against the 3 rivals is separate from your leaderboard rank |
| 10 | League stage cleared | `recordLeague()`: `ranking(g)[0] === me` and you're on that stage | Results screen | Lose = replay the same stage (new shuffle). Clearing stage 6 unlocks the League Champion achievement and the Royal colour |
| 11 | Achievements (optional) | `checkAchievements()` in `src/ui/progress.ts` | Results screen | `rich` ($200) reached by 0–1.8% of computer players and effectively impossible in 4-player Hard games (best seen $185). `comeback` needs being last on day 5 |

**Order inside a day** (it matters for every outcome above): morning (event revealed; delivery added) → buying in turn order → customers arrive → everyone seals prices → sales resolved customer by customer (yesterday's waiting customers first, then today's) → evening upgrades (not on the last day) → night: inspector fine, then everything ages, then the next day starts.

### Conflicting or unreachable outcomes found

- On the **final day**, Health Inspector and Heatwave's extra ageing do nothing (they act at night, and there is no last night). Priority can't be played (no evening). These cards still appear as the final day's event.
- The **seat-order tie-break** favours seat 0 (you in solo, daily and league games, the room host online).
- `rich` ($200) is practically unreachable in 4-player games.

---

## C. Balance values (grouped by system, with code references)

Legend for "Type": **C** = constant in `src/engine/data.ts` (one place, easy to tune); **H** = hard-coded number inside logic; **F** = formula; **T** = also written in player-facing text (must be edited too). ⚠ marks a value that needs edits in several places. ★ marks the highest-impact controls.

### C1. Run length, start and end

| Value | Current | Changes | Affects | Where | Type |
|---|---|---|---|---|---|
| Game length ⚠ | 8 / 10 / 14 days (default 10); league 8 or 10; daily 10 | Chosen at set-up | Everything; final cash roughly scales with days (3-player Hard: $61 / $81 / $127) | `App.tsx` `setDays`, options list; `server/room.ts` allowed list; `src/net/protocol.ts` `RoomConfig`, `DEFAULT_CONFIG`; `progress.ts` `LEAGUE`; `daily.ts` | H ×5, T (`RULES.md`, `GAME-BRIEF.md`) |
| Starting cash ⚠ | $10 | Nest Egg makes it $16 | Day 1–2 buying power, upgrade timing | `START_CASH` (data.ts); `applyTrait()` `START_CASH + 6` | C, H, T (`RULES.md`, `HowTo.tsx` "$10", trait text "$16 instead of $10") |
| Winner rule ★ | most cash → most sold → seat order | Never | The goal | `ranking()` game.ts | H |
| Final-day stock value | $0 | Never | End-of-game buying | implicit | — |

### C2. Supply: the farmers market

| Value | Current | Changes | Affects | Where | Type |
|---|---|---|---|---|---|
| Delivery size ★ | each card fruit: `max(1, round(base × scale))`; fruits not on the card: base 2 | Daily | How much fruit exists; price pressure | `startDay()`; `FARMER_CARDS` (20 cards, 3 fruits each, 4–6 per fruit) | C + F |
| Supply scale ★ | `buyCapacity × (1 + 0.5 ×` | Rises over the game; | ≈100% of total buy capacity | `supplyScale()` game.ts; 17.15 = `FARMER_AVG_QTY` | F with H 1, 0.5 |

| Value | Current | Changes | Affects | Where | Type |
|---|---|---|---|---|---|
| | `progress) ÷ 17.15` | rises when anyone builds a Handcart or has Big Basket | on day 1 → 150% on the last day | | |
| Unripe deliveries | 9 of 20 farmer cards; mango (2 nights) or watermelon (1 night); only if "Unripe fruit" is on | Per card | The best margins in the game (unripe mango $2 → sells as fresh) | `FARMER_CARDS[].unripe`, `FRUIT.*.unripeSlots` | C |
| Market fruit ages | 1 step a night (+1 Heatwave) | Nightly | Cheap old fruit available every day | `nightfall()` | H |
| Buy cost by freshness | blueberry 2/1/1 (fresh/okay/not so), unripe 1; watermelon 3/2/1, 1; apple 4/3/1, 1; mango 4/2/1, unripe 2; floor $1 | Events ±$1 | Margins | `FRUIT.*.cost`, `marketCost()` | C, T (`RULES.md` table) |
| Buy limit ⚠ | 4 a day; 5 with Handcart; +1 Big Basket | Upgrade or trait | Throughput, **and everyone's supply and demand** (see finding E7) | `BASE_BUY`, `CART_BUY`, `buyLimit()` | C, T (upgrade text, `RULES.md`, `HowTo.tsx`) |
| Shelf size ⚠ | 8; 14 with Bigger Shelf | Upgrade | Rarely binding | `BASE_SHELF`, `BIG_SHELF` | C, T |

## C3. Demand: customers

| Value | Current | Changes | Affects | Where | Type |
|---|---|---|---|---|---|
| Demand target ★ | `buyCapacity × (0.40 + 0.45 × progress)` units/day, split over (1 + walk-ins) customers | Daily | How much can be sold at all; price wars | `demandScale()` game.ts | F with H 0.4, 0.45 |
| Card size | 24 customers, 3–6 fruit each (avg 4.375), 1–3 fruit types | Deck reshuffles when empty | Order sizes | `RAW_CUSTOMERS`; `CARD_AVG_QTY` | C |
| Order rounding | `max(1, round(qty × scale))` | Per draw | Small early games get more demand than the target (every want is at least 1) | `drawCustomer()` | H |

| Value | Current | Changes | Affects | Where | Type |
|---|---|---|---|---|---|
| Walk-ins ⚠ | 1 a day; 2 from the halfway day (day 6 of 10, day 5 of 8, day 8 of 14); events ±1 | At halfway | More, smaller orders → more showdowns in the second half | `baseWalkIns()` (threshold 0.5) | H |
| Forecast customer | 1, known a day early | Daily | Planning | `customersArrive()` | H |
| Waiting customers | Unserved new customers wait 1 more day and are served first; then leave (no penalty) | Daily | Carry-over demand | `finishReveal()`, `resolve()` | H |
| Customer top price ★ | card `max` + fruit shift (apple −1, mango +2), + event; weighted averages: blueberry $6.5, watermelon $7.4, apple $7.7, mango $9.1 | Events | Revenue ceiling | `RAW_CUSTOMERS[].max`, `PRICE_SHIFT`, `effectiveMax()` | C |
| Event extra customers | Influencer, Gym Club, Hotel Chef | With their event | One-day demand spike | `EVENT_CARDS[].extra` | C (not price-shifted, unlike normal cards) |

## C4. Fruit life and selling

| Value | Current | Changes | Affects | Where | Type |
|---|---|---|---|---|---|
| Life tracks ★ | blueberry 3 slots (not so, okay, fresh); watermelon 5 (+1 unripe); apple 7; mango 4 (+2 unripe) | Fruit moves one slot a night | Spoilage risk | `FRUIT.*.ripe`, `unripeSlots` | C, T |
| Freshness discount ★ | blueberry −4/−2/0; watermelon −3/−1/−1/0/0; apple −2/−1/−1/−1/0/0/0; mango −3/−1/0/0 | With age | Top price allowed per fruit | `FRUIT.*.drop`, `capAt()` | C, T |
| Price floor | $1; Minimum Price event: watermelon 5, apple 5, mango 6, blueberry 4 | Event | Price wars | `minPrice()` | H, C |
| Who sells first | lowest price → fresher fruit → tie edge (Favoured Trade card, or Haggler v richer shop) → buying order (poorest first) | Every sale | All income | `resolve()` sort | H |
| Customers pay | your sealed price (not their top price) | — | Undercutting | `resolve()` | H |
| Signboard tip ⚠ | +$1 per fruit sold | Upgrade | Income | `resolve()`; display repeated in `animator.tsx` | H ×2, T |

## C5. Turn order and catch-up

| Value | Current | Where | Type |
|---|---|---|---|
| Buying order ★ | Day 1 random; then **poorest first** (ties round the table); Priority card jumps the queue | `startDay()` | H |
| Tie-break by buying order | Poorer shop wins full price-and-freshness ties | `resolve()` | H |

## C6. Events (13 cards, one a day from day 2, tomorrow's shown a day early; reshuffled when used)

| Event | Effect | Measured income vs same-day average (2p / 4p) |
|---|---|---|
| Festival | +$3 all prices, +1 walk-in | +11.1 / +8.1 |
| Superfood | +$3 blueberry, Gym Club wants 4 at $8 | +10.6 / +4.5 |
| Mango Craze | +$3 mango, Influencer wants 3 at $11 | +10.4 / +4.2 (largest gap between shops: 25–35) |
| Hotel Order | Hotel Chef orders 7 | +6.6 / +1.0 |
| Bad Harvest | delivery ×0.5, +$2 prices | +1.7 / +2.6 |
| Heatwave | +$4 watermelon, everything ages 2 steps tonight | −1.1 / −0.1 |
| Minimum Price | price floors | −2.3 / +1.1 |
| Good Harvest | +3 each fruit, costs −$1 | −3.6 / −2.0 |
| Health Inspector | $2 fine per not-so fruit at night (capped by cash) | −3.9 / −2.0 |
| Price Hike | costs +$1 | −4.0 / −2.3 |
| Lorry Breakdown | no delivery | −6.0 / −2.9 |
| Rainy Day | no walk-in, costs −$1 | −8.0 / −4.8 |
| Slow Season | −$2 all prices, no walk-in | **−12.0** / −7.3 |

Where: `EVENT_CARDS` (data.ts, C), applied in `startDay()`, `marketCost()`, `priceMod()`, `customersArrive()`, `nightfall()`; text repeated in `RULES.md`.

## C7. Upgrades (each evening each shop is offered 3 it hasn't built; it may build any of those it can afford)

| Upgrade | Cost | Effect (code) | Measured: AI that never builds it (3p, control 37% win) | Payback estimate |
|---|---|---|---|---|
| Signboard ★ | $8 | +$1 per fruit sold | **26.3%** (biggest loss) | ~2.5 days (2p, 3.2 sales/day) to ~3.2 days (4p, 2.5/day) |

| Upgrade | Cost | Effect (code) | Measured: AI that never builds it (3p, control 37% win) | Payback estimate |
|---|---|---|---|---|
| Cool Room | $7 | ages only on even-numbered nights ( `s.day % 2 === 1` skips); immune to Heatwave | 32.3% | Indirect: keeps fruit fresh (higher top price, wins freshness ties) |
| Bigger Shelf | $3 | 8 → 14 shelf | 29.7% | Only when hoarding |
| Handcart | $6 | 4 → 5 buys a day | 32.0% | Low: shops are limited by customers, not by buying |
| Shop Bell | $4 | 2 favour cards now; hand 3 → 4 | 34.3% | ~2 cards × ~$2–3 value |
| Juice Press | $4 | up to 3 spoiled fruit a night → $2 each | **37.3% (no loss at all)** | Never for careful play: computers spoil 0.1–0.7 fruit per game |

Rushing any single upgrade on the first evening loses (13–26% win). Building all six loses badly (–$87 versus rivals). No upgrades at all: 21% (vs 37% for the computer's normal choices). Where: `UPGRADES` (C, T), effects in `buyLimit`, `shelfSize`, `handLimit`, `resolve`, `ageShop`, `buyUpgrade` (H). The Juice numbers (3, $2) are hard-coded in `ageShop()`.

### C8. Merchant traits (pick 1 of 3 on day 1; computers take a random one)

| Trait | Effect | Where | Measured win rate (3p, fair 33.3%) |
|---|---|---|---|
| Bargain Hunter | first fruit each day $1 cheaper (min $1) | `buyCost()` | **42.0%** (+$7.7 vs rivals) |
| Charmer | 3 favour cards at start; hand +1 | `applyTrait()`, `handLimit()` | 39.0% |
| Nest Egg | start with $16 | `applyTrait()` | 38.0% |
| Local Favourite | +$3 per customer you complete alone (no daily limit) | `resolve()` | 35.3% |
| Haggler | wins tied prices against richer shops | `tieEdge()` | 30.7% |
| Big Basket | +1 buy a day | `buyLimit()` | **29.0%** |

### C9. Favour cards (12-card deck; earned when you alone complete a customer; at most 1 per shop per day; hand limit 3)

Free Cash +$3 · Free Fruit (any market fruit, free, ignores the buy limit) · Priority (buy first tomorrow) · Good Riddance (sell all ripe stock back for $1 each; it returns to the market) · Favoured Trade (win tied prices today) · Fridge (all your ripe fruit one step fresher). Where: `FAVOURS`, `FAVOUR_DECK` (C), `playFavour()` (H: the $3, the $1), `resolve()` (earning rule).

### C10. Computer difficulty

| Knob | Easy | Normal | Hard | Where |
|---|---|---|---|---|
| Buy if value − cost ≥ | 0.3 (+ random ±2) | 0.8 | 0.3 | `pickPurchase()` |
| Pricing | rival's cap minus 0–1 (random) | best response, then −0 or −1 | best response, then −1; reads tomorrow's event | `planBids()`, `HARD` constant |
| Keeps unsold value | — | 30% | 10% | `planBids()` |
| Upgrades | skips 60% of nights | list order with day limits, keeps ≥ $6 | same | `pickUpgrade()` |
| Measured head to head (2p) | — | beats Easy 72% | beats Normal 84%, Easy 91% | `npm run balance` |

### C11. Modes

| Mode | Set-up | Where |
|---|---|---|
| Daily | You vs Normal, **Hard**, Normal; 10 days; same decks for everyone | `daily.ts` |
| League | 1: 1 Easy (8 days) · 2: 2 Easy (8) · 3: 1 Normal · 4: 2 Normal · 5: Hard + Normal · 6: 3 Hard (all 10 days) | `LEAGUE` in `progress.ts` |
| Online | 2–4 people, optional computers, 8/10/14 days, toggles | `server/room.ts` |

**Versioning rule:** any change to C1–C10 that alters results needs `RULES_VERSION` (`src/net/protocol.ts`) raised. That **discards games saved in progress** (`store.ts`), keeps old and new apps out of each other's online rooms, and separates daily scores. League and mode set-ups (C11) can change without it.

---

## D. Progression and pacing map

**Units:** money in $, time in game days. Computer-only measurements, 10-day games unless stated.

### Early game: days 1–3 ("set-up")

- Day 1 is thin: one forecast customer plus one walk-in, orders scaled to 40% of buying capacity. Average income per shop on day 1: **$6.3 (2p), $4.9 (4p)**, against $10 starting cash. Most shops spend nearly all their cash on day 1.
- Supply is about **2.5× new demand on day 1** (formula: supply 100% vs demand 40% of capacity). Unbought market fruit ages and becomes cheap, so not-so-fresh fruit at $1 is almost always available.
- First upgrades: computers buy the Signboard on day ~2.6, Handcart ~2.5–3.1, Cool Room ~2.8, Bell ~2.8–3.4. The AI only builds when it keeps ≥ $6 afterwards, so the $8 Signboard needs $14 in hand.
- Traits make the biggest difference here: Nest Egg's +$6 is about one full day's income at this point.

### Mid game: days 4–6

- Income climbs: 2p **22–27*aday* * *,4*p* * *13–16 a day**.
- Walk-ins double at the halfway day (day 6 of 10), so there are more but smaller orders, more showdowns and longer reveals.
- Juice Press is usually built here (day ~5) and pays nothing back for careful players.
- **Who's ahead on day 5 usually wins:** 66% (2p), 54% (3p), 47% (4p), against fair chances of 50/33/25%.

### Late game: days 7–10

- Income plateaus: 2p **28–30*aday* * *,4*p* * *16–19 a day** (demand reaches 85% of capacity; supply 150%).
- Upgrades stop (the computer needs 3–6 days left for each). Last day: no evening; unsold fruit is worth $0.
- Computers still buy a full basket (~4) on the last day to sell the same day.

### Whole run

| Measure | 2 players | 3 players | 4 players | Source |
|---|---|---|---|---|
| Average final cash (Hard mirror) | $131 | $81 | $56 | balance, probes |
| Final cash range (10th–90th percentile) | $92–170 | $49–117 | $24–91 | probes |
| Median winning margin | $27 | $17 | $15 | balance |
| Close games (≤ $3) | 8% | 10% | 12% | balance |
| Blowouts (≥ $20) | 63% | 45% | 41% | balance |
| Lead changes per game | 2.7 | 3.6 | 4.2 | balance |
| Comeback from last on day 5 | 33% | 19% | 9% | balance |
| Sales at ≤ $2 ("price war") | 6% | — | 16% | balance |
| Sales near the customer's top price | 51% | — | 20% | balance |
| Fruit spoiled per shop per game | 0.1 | — | 0.7 | balance |

**Profit per fruit bought** (3p): mango $2.59 > blueberry $2.21 > watermelon $1.85 > apple $1.58. In 4-player games every fruit falls to $1.1–1.8.

**Real time per run: not measured.** From the animation code, each showdown takes about 2.3–3.2 seconds (0.55× with "Fast"); people's thinking time dominates and isn't recorded. Assuming 30–90 seconds per day, a 10-day game takes about 5–15 minutes. Telemetry should measure this (section 6).

**League difficulty curve** (computer standing in for you, 300 games per stage):

| Stage | Easy-level player | Normal-level player | Hard-level player |
|---|---|---|---|
| 1 Village Square (1 Easy, 8 days) | 47% | 65% | 87% |
| 2 Harbour Steps (2 Easy, 8 days) | 27% | 67% | 84% |
| 3 Old Town (1 Normal) | 29% | 54% | 84% |
| 4 Spice Lane (2 Normal) | 7% | 38% | 80% |

| Stage | Easy-level player | Normal-level player | Hard-level player |
|---|---|---|---|
| 5 Grand Bazaar (Hard + Normal) | 5% | **11%** (~9 tries) | 54% |
| 6 Royal Market (3 Hard, 4 players) | 5% | **5%** (~20 tries) | 28% |
| Daily challenge (Normal, Hard, Normal) | 2% | **8%** | 41% |

**Downtime, spikes and snowballing**

- Downtime: day 1 (little to sell) and long showdown sequences in the late game.
- **Difficulty spike:** league stages 4 → 5 → 6 for an average player (38% → 11% → 5%).
- **Snowball:** the day-5 leader wins well above fair, and blowouts are common. The poorest-first buying rule does help: the old rotating order had 62% blowouts in 3-player games against 45% now. But being poor enough to buy first on 4 or more days still means winning only 15.5% of the time.
- **Unrecoverable:** a shop with less than $1 and no ripe fruit can't buy again without a lucky favour card.
- **Player count changes the game more than any setting:** 4-player games earn less than half what 2-player games earn.

---

## E. Prioritised findings and proposed adjustments

**Confirmed** = shown by the code or by measurement. **Hypothesis** = needs testing or human playtest.

| # | Finding | Evidence | Player impact | Where to change | Proposal and side effects |
|---|---|---|---|---|---|
| E1 | **League cliff at stages 5–6** (confirmed) | Normal-level player: 38% → 11% → 5% | Average players stall for 10–20 tries; the "League Champion" goal feels out of reach | `LEAGUE` in `src/ui/progress.ts` (no rules change, no `RULES_VERSION` bump) | Add a step between (e.g. Hard + Normal + Normal), or make stage 6 three players. Test until each step loses no more than ~10–15 points for a Normal-level player. Side effect: easier champion achievement |
| E2 | **4-player economy is squeezed** (confirmed) | Final cash $56 vs $131; 16% of sales at ≤ $2; only 20% near top price | More price wars and fewer good days; `rich` achievement out of reach; Royal Market feels grim | `demandScale()` and `supplyScale()` coefficients in `game.ts`; could depend on the number of shops | Hypothesis: lower supply growth (the 0.5) or raise demand (0.40/0.45) for 3–4 shops. Tune **together**: the gap between supply and demand drives prices. Side effect: changes daily scores (always 4 players), so needs `RULES_VERSION` |

| # | Finding | Evidence | Player impact | Where to change | Proposal and side effects |
|---|---|---|---|---|---|
| E3 | **Snowballing and blowouts** (confirmed; how much is too much is a hypothesis) | Day-5 leader wins 54% (3p); blowouts 41–63%; close games 8–12% | Many games are decided by mid-game | `startDay()` order, `resolve()` tie rules, the event deck | Options to test one at a time: tie advantage for the poorest beyond full ties; smaller late-game event swings. Side effect: skill matters less |
| E4 | **Juice Press is a dead choice** (confirmed) | AI without it wins as often (37.3% vs 37%); spoilage < 1 fruit per game | Wastes $4 and an evening for new players who pick it | `UPGRADES.juicer` and `ageShop()` (3 and $2 hard-coded) | Rework so it helps careful players too, e.g. also juices unsold fruit on the last day, or more per fruit. Needs testing so it doesn't become a farm (I tested "buy $1 dying fruit to juice": it **loses**, 17% win) |
| E5 | **Trait spread 29–42%** (confirmed at 300 games; ±3 points noise) | Bargain Hunter 42%, Big Basket 29%, Haggler 30.7% | Trait pick quietly decides some games | `TRAITS` text + `buyCost()`, `buyLimit()`, `tieEdge()` | Haggler almost never fires because freshness breaks ties first (code order). Make it apply before freshness, or give it a small extra. Big Basket is weak because buying capacity isn't the limit: try +1 shelf space or a $1-off second fruit. Trim Bargain Hunter only if playtests agree. Test each trait **independently** |
| E6 | **Handcart and Big Basket feed everyone** (confirmed in code) | `buyCapacity()` adds every shop's limit into demand and supply | Your Handcart also gives rivals more customers and fruit | `buyCapacity()`, `demandScale()`, `supplyScale()` | Decide if intended (the comment says it keeps the market tight). If not, base the scales on the shop count instead. Side effect: Handcart becomes stronger |
| E7 | **Hidden seat tie-break and wrong text** (confirmed) | `ranking()` stable sort; Results text | 0.3% of games; seat 0 wins dead heats | `ranking()`, `Results.tsx` | Add an explicit third rule (e.g. fewer fruit spoiled, or a shared win) and correct the text. Low effort |
| E8 | **Final-day events that do nothing** (confirmed) | Inspector and Heatwave ageing act at night; no last night | A "dramatic" final event that changes nothing | `startDay()` event draw | Don't draw night-only events for the last day, or apply them at the final sales. Low effort |
| E9 | **No way back from broke** | 0.3–2.2% of computer players; | A player can sit out the rest | `canBuy()`, `startDay()` | Hypothesis: a tiny safety net (e.g. $1 |

| # | Finding | Evidence | Player impact | Where to change | Proposal and side effects |
|---|---|---|---|---|---|
| | (confirmed, rare) | humans may be higher | of a game | | morning credit when cash and ripe stock are both 0). Measure humans first (E-telemetry) |
| E10 | **Slow Season and Rainy Day are big shared losses** (confirmed size; fun is a hypothesis) | −12/−8 a day in 2p | Can feel like a wasted day | `EVENT_CARDS` | Soften (e.g. Slow Season −1*instead of* −2) if playtesters dislike them |
| E11 | **Daily challenge is hard for an average player** (confirmed for the computer stand-in) | Normal-level wins 8%; Hard rival in the line-up | Daily "wins" are rare, but leaderboard rank still rewards | `daily.ts` rival levels | Optional: Normal, Normal, Hard on alternate days. Changing it changes daily seeds' outcomes; do it at a date boundary |
| E12 | **Rounding inflates small orders** (confirmed, minor) | `max(1, round(qty × scale))` at scale 0.46–0.7 | 2-player early days have slightly more demand than intended | `drawCustomer()` | Leave unless tuning E2 |
| E13 | **Numbers duplicated in text** (confirmed) | Section C ⚠ items | A balance change can leave the rules, how-to or upgrade text wrong | `RULES.md`, `HowTo.tsx`, `UPGRADES` / `TRAITS` text, `animator.tsx` tip | When tuning, update together; better: build those texts from the constants |
| — | Checked, **not** a problem | "Juice farming" (17% win, loses); "buy only the cheapest old fruit" (0% win); last-day buying (sold the same day, warned "Unsold = $0") | — | — | No change |

## F. Missing information and playtest questions

### Not available from the code or simulations

- How people actually price (do they undercut by $1 like the Hard computer, or price high?).
- Real play time per day and per game, and where people stop (only `quit_day` exists).
- Human win rates per difficulty, trait and mode; how often humans go broke.

- Your target experience: how often should an average player win against Normal? Is a 4-player game meant to feel tighter? Should the League be finishable by most players?

**Playtest questions (fit the DEC-016 kit)**

1. After the first game: did you understand why you won or lost a sale? (freshness and ties)
2. Did any day feel pointless (day 1, Slow Season, a broke stall)?
3. Which upgrade did you build first and why? Did the Juice Press feel useful?
4. Which trait did you pick, and did it feel strong?
5. Did the game feel decided before the end? From which day?
6. How long did it take, and was that the right length?

---

## 6. Tuning and measurement plan

**Order of testing (expected impact first)**

1. **League steps (E1)**: no rules change, quick to simulate and ship.
2. **Player-count economy (E2)**: supply and demand coefficients, tuned **together** (their ratio matters), checked in 2, 3 and 4 players.
3. **Dead or weak choices (E4, E5, E6)**: Juice Press, Haggler, Big Basket, each **independently**, one change per test run.
4. **Snowballing (E3)** after the economy, because price wars and margins change it.
5. **Small fixes (E7, E8)**: tie-break and final-day events. Safe; ship with the next rules change.
6. **Difficulty and daily line-up (E11)** last, once the economy is settled.

**Tune together:** demand 0.40/0.45 with supply 1.0/0.5; customer top prices with fruit costs (margin); Signboard tip with upgrade cost. **Tune alone:** each trait, each upgrade, each event, the league line-up.

**Telemetry to add** (anonymous counters, in line with `docs/ANALYTICS.md`; adding events changes data handling, so it needs its own approved plan):

- Game result by mode, number of players, length and rival levels (win/lose, final rank, winning margin bucket).
- Trait picked × win; upgrades built (which, which day) × win.
- Fruit spoiled and unsold at the end (buckets); fruit sold; share of sales at the customer's top price.
- Rank on day 5 × final rank (snowball and comeback).
- Game duration in minutes (bucket) and minutes per day.
- Times a player was broke (cash < $1, no ripe stock).
- League: attempts per stage and clears; daily: rank against the three rivals.

**Small test matrix** | Axis | Values | |---|---| | Players | 2, 3, 4 | | Rivals | all Easy, all Normal, mixed, all Hard | | Length | 8, 10, 14 days | | Strategy for the test seat | computer default; no upgrades; Signboard first; each trait fixed; cautious buyer (only forecast demand) | | Toggles | events on/off; unripe on/off |

Run each cell 300–400 games with `npm run balance` (plus the probes used here), compare against this report's baseline, then confirm the top 2–3 changes with 3–5 human playtesters before any release.

---

## Appendix: probes used for this report (read-only, not committed)

- `curves` : supply and demand per day from `demandScale()` / `supplyScale()` for 2–4 players.
- `probes` : juice farming; broke shops; ties; last-day buying; final cash spread.
- `ladder` : league stage and daily win rates for Easy-, Normal- and Hard-level stand-ins (300 games each).
- `bottom` : "buy only the oldest, cheapest fruit" strategy (400 games).

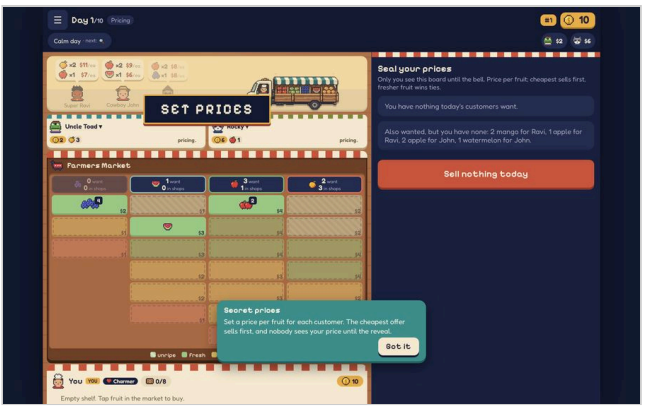
# REP-026 · PLAN-010 built: the farmer's lorry

| | |
|---|---|
| Game | Fruit Market |
| Report date | 2026-10-27 07:54 (Asia/Kuala_Lumpur) |
| Package | PLAN-010, approved ("approve") |
| Build | commit bc2e760, merged to `main` for the preview |
| Delivery status | Verified |

## What changed

- Each morning your Gemini lorry drives into the street and parks; the customers step aside.
- The delivered fruit flies from the lorry into its own slot on the market board.
- When buying ends and pricing starts, the lorry drives off. On "Lorry Breakdown" days it doesn't come.
- It's quick (about 1 second to arrive) and never blocks play. Tapping speeds the unloading; reduced-motion turns the driving off.




## Validation

| Check | Result |
|---|---|
| Quick checks (73 tests, lint, types, build) | PASS |
| Full browser tests incl. speed budget and secret-price leak check 0 | PASS |
| Phone and desktop screenshots, no sideways scroll or errors | PASS |
| Rules, saves, online play untouched | PASS |

## Next

All of the new look (PLAN-009, PLAN-010, PLAN-011) and the Patches rename are on the preview. When you're happy, say **PREPARE RELEASE** and I'll set up 1.2.0 for you to publish.

# REP-027 · Balance redesign: fruit identity, strategy depth and pacing (proposal, no rule change)

| | |
|---|---|
| Game | Fruit Market (rules version 2, `main` at 66bc299) |
| Report date | 2026-10-27 09:04 (Asia/Kuala_Lumpur) |
| Type | Analysis, experiments and a patch plan. **Production rules unchanged.** All variants ran in a separate lab copy of the engine |
| Builds on | REP-025 (balance and pacing audit) |
| Proposed plan | PLAN-012 "Market rebalance: tighter town, fair Handcart, better apples" (awaiting approval) |
| Approval status | Awaiting approval (PLAN-012). This report approves nothing |

**How to read the numbers.** Three kinds of statement appear below, and they are labelled:

- **[Code]**: a fact read from the source code.
- **[Sim]**: a measured result from computer players. The ± figures are 95% intervals. With 600 games per cell, a win rate is good to about ±4 points.
- **[Hyp]**: a hypothesis about how people will feel. Only the playtest (section 7) can confirm it.

Computer players are not people. Win rates show the shape of the game, not human results.

---

## 1. Diagnosis: what the strategic game is today

**The core loop (code).** Each day you buy up to 4 fruits (5 with a Handcart), then set sealed prices per customer. The cheapest offer sells first, and freshness caps your price and breaks ties. The goal is the most cash at the end.

**What decides games (sim, current rules, 3 players, 600 games per plan on held-out seeds):**

1. **Two plans dominate.**
   - **Undercut**: always price $1 under the expected rival. It wins 43% (fair share is 33%).
   - **Salvage**: dump old stock cheaply and keep buying fresh. It wins 51%.
   - In 4 players it is worse: 45% and 54% (fair share 25%).
   - **Premium pricing** (hold out for top price) wins only 13% in 3 players and 6% in 4 players.
   - In short, cutting the price is almost always right. That is the opposite of an interesting sealed-bid game.
2. **The market is loose early, so price wars pay.**
   - Customer demand starts at 40% of what the shops can buy, while farmers bring 100%.
   - Shops are therefore fighting over too few customers with too much fruit.
   - 11% of 3-player sales and 16% of 4-player sales happen at $2 or less ("price-war share").

3. **The Handcart (and Big Basket) grows the market for everyone.** [Code] Demand and deliveries are sized from the sum of every shop's buy limit (`buyCapacity()` in `src/engine/game.ts`). So when one shop buys a Handcart, the town gets more customers and more fruit for all shops. That rewards whoever floods the market and blurs what the upgrade is for.
4. **Apples are a trap.**
   - Apples cost $4 fresh but are priced $1 below other fruit (`PRICE_SHIFT` in `data.ts`).
   - [Sim] Net profit per apple bought is $1.64, the lowest of the four fruits (mango is $2.58).
   - 25% of okay-age apples are still unsold at the end.
   - The "safe, long-keeping" identity does not pay, because almost nothing is held long enough for keeping to matter.
5. **Fruit is barely held, so decay rarely bites.**
   - [Sim] The average fruit sells 0.6 days after purchase, and 87–93% of fresh fruit sells the same or next day.
   - The only real "hold" decision is green mango (held 2.6 days, net $3.33 each) and green watermelon.
   - Spoilage is 1.2 fruits per shop per game in 3 players.
6. **Early leaders snowball.**
   - [Sim] The mid-game leader wins 57% in 3 players (fair is 33%).
   - Only 15% of 3-player games and 8% of 4-player games are won by whoever was last at mid-game.
   - 45% of games are blowouts.
7. **4 players is squeezed.** [Sim] Average final cash is $130 in 2 players, $80 in 3 and $56 in 4. Four-player shops barely grow, and price wars are most common there.

**Summary.** The tension the game wants is "freshness and timing against price". Today it collapses into "price lowest, buy fresh, repeat". Fruit identities exist on paper, but only mango's shows up in play.

---

## 2. Fruit identity and decay

### 2a. Before and after (exact values)

Decay is unchanged in the recommendation. The reason is in 2c.

| Fruit | Track, oldest → freshest (price drop $) | Cost: fresh / okay / not so / unripe (now → proposed) | Price shift on cards | Unripe nights | Intended identity |
|---|---|---|---|---|---|
| 🫐 Blueberry | −4 · −2 · 0 (unchanged) | 2/1/1/– (unchanged) | 0 | none | Cheap, fast, sell today |
| 🍉 Watermelon | −3 · −1 · −1 · 0 · 0 (unchanged) | 3/2/1/1 (unchanged) | 0 | 1 | Steady middle; small ripening bet |
| 🍎 Apple | −2 · −1 · −1 · −1 · 0 · 0 · 0 (unchanged) | **4/3/1/1 → 3/2/1/1** | −1 (unchanged) | none | The keeper: safe margin any day, sells late |

| Fruit | Track, oldest → freshest (price drop $) | Cost: fresh / okay / not so / unripe (now → proposed) | Price shift on cards | Unripe nights | Intended identity |
|---|---|---|---|---|---|
| 🥭 Mango | −3 · −1 · 0 · 0 (unchanged) | 4/2/1/2 (unchanged) | +2 | 2 | Premium; buy green and wait |

**Measured fruit economics (sim, 3 players, held-out seeds; "net" = sale price minus cost, per fruit bought):**

| Fruit | Net now | Net with package | Share of fruit bought, now → package | Note |
|---|---|---|---|---|
| Blueberry | $2.14 | $3.43 | 26% → 24% | Still the quick-profit fruit |
| Watermelon | $1.90 | $2.55 | 22% → 24% | Green watermelon nets $3.57 |
| Apple | **$1.64 (last)** | **$2.69** | 22% → 26% | Moves from trap to solid choice |
| Mango | $2.58 | $4.26 | 30% → 25% | Green mango nets $4.07, held 2.9 days |

All fruits gain under the package because the tighter market raises prices (section 3). What matters is the order: apple goes from clearly worst to level with watermelon.

## 2b. Specific investigations

| Question | Test | Result | Verdict |
|---|---|---|---|
| Is apple too expensive? | Apple cost 3/2/1 (A1); apple price +$1 (A2); both (A3) | A1 and A3 both lift apple to mid-table. A3 adds cash and widens the Hard-vs-Normal gap slightly. A1 is the smaller change | **A1 recommended** |
| Should apples keep value even longer? | Apple drop −1 · −1 · 0… | No measurable change to any plan (all within ±3 points) | Rejected: no effect |
| Is blueberry too punishing? | Blueberry drop −3 · −1 · 0 | No measurable change | Rejected: no effect |
| Is green mango too strong? | Green mango $2 → $3 | Premium rises (27%) but Reserve becomes too strong (45%). The Hard-vs-Normal gap shrinks to 57% (3 players), so skill matters less | Optional experiment, not recommended now |
| Should unripe fruit stop arriving when it can't ripen before the end? | Unripe cutoff | No measurable change | Rejected |
| Does unsold fruit at the end need a salvage value? | $1 per fruit at the end; juice at the end | No measurable change | Rejected |

## 2c. Why decay is not the lever

[Sim] Shops sell almost everything within a day, so changing decay curves moves nothing. Decay only matters if holding is a real choice. Holding becomes a real choice when the market is tight and

has gaps: some days your fruit is scarce and worth keeping, other days it is everywhere.

The package creates that through market size, not through faster rot. That honours "do not simply increase decay everywhere or force a spoilage quota". Spoilage actually falls (1.2 → 0.3 per shop) because shops buy more selectively.

---

## 3. A deeper strategy layer

### 3a. The plans tested

Each plan is the Hard computer with one strategic lean layered on top (lab "styles").

| Plan | Lean |
|---|---|
| Undercut | Always price $1 under the best rival guess |
| Salvage | Dump old stock cheaply, restock fresh |
| Premium | Keep prices near the top; accept unsold fruit |
| Reserve | Keep a cash cushion; buy only clear profits |
| Ripen | Prefer green mango and watermelon |
| Adaptive | Learn each rival's discount from past reveals |
| Turnover | Buy the cheapest stock and sell it fast |
| Event | Lean hard into event cards (festival, heatwave and so on) |

### 3b. Matchup table: one plan against two (or three) standard Hard computers

Held-out seeds 100001–100600 (600 games per cell, about ±4 points). Fair share is 33% in 3 players and 25% in 4 players.

| Plan | 3p now | 3p package | 4p now | 4p package |
|---|---|---|---|---|
| Undercut | **43** | 25 | **45** | 21 |
| Salvage | **51** | 32 | **54** | 29 |
| Premium | 13 | 25 | 6 | 16 |
| Reserve | 30 | 36 | – | – |
| Ripen | 27 | 31 | 26 | 23 |
| Adaptive (reads rivals) | 36 | 34 | – | – |
| Turnover | 18 | 21 | – | – |
| Event | 32 | 30 | – | – |
| **Spread (best – worst)** | **38 points** | **15 points** | **48 points** | **13 points** |

**Four different plans at one table (4 players, fair share 25%):**

| | Turnover | Reserve | Ripen | Event |
|---|---|---|---|---|
| Now | 17 | 19 | 29 | 34 |
| Package | 16 | 33 | 32 | 20 |

**Reading this.** [Sim] Under the package no plan is a must-pick, and several different plans sit within a few points of fair. Turnover stays weakest. [Hyp] That is acceptable, because it is the "no thinking" plan. Reading rivals (Adaptive) stays strong, which keeps the sealed-price duel rewarding.

### 3c. Sealed-price analysis

- [Code] The cheapest offer sells first. Ties go to the fresher fruit, then Favoured Trade or Haggler, then buying order.
- [Sim] Price-war share (sales at $2 or less) falls from 11.4% to 7.0% in 3 players and from 16.3% to 7.6% in 4 players.
- [Sim] Sales within $1 of the customer's top price rise from 30% to 55%.
- **Meaning.** With fewer spare fruits around, the question changes from "how low must I go?" to "will they dare to go high, and is my fruit fresh enough to win a tie?". [Hyp] That is the more interesting guess.
- **Adaptive still wins 34%.** Reading rivals still beats autopilot.

### 3d. Handcart and Big Basket: the shared-market effect

- [Code] Today, any shop's Handcart or Big Basket raises everyone's demand and supply.
- **Proposed:** size the town from the base 4 fruits per shop, so upgrades help only their owner.
- [Sim] This one change does most of the work against undercutting (P1 test: Undercut 32 → 24, Reserve 32 → 39, comeback from last 23% → 28%).
- [Sim] Handcart is still bought in 99% of games, so it keeps its value.
- **Trade-off:** average cash falls about 12%. The demand start is raised (0.55 → 0.60) to give some back.

### 3e. Traits (supporting evidence; changes optional)

2,000 games per player count, current rules (development seeds) → package without the 0.60 demand start (held-out seeds).

| Trait | 3p now | 3p package | 4p now | 4p package |
|---|---|---|---|---|
| Haggler | 28.6 | 28.6 | 24.0 | 22.9 |
| Big Basket | 29.6 | 33.3 | 22.4 | 23.2 |
| Bargain Hunter | 38.4 | 36.8 | 29.9 | 29.5 |
| Charmer | 36.1 | 35.3 | 26.1 | 27.4 |
| Local Favourite | 30.0 | 31.5 | 18.4 | 20.5 |
| Nest Egg | 37.4 | 34.6 | 29.2 | 26.3 |
| Spread | 10 points | 8 points | 11.5 points | 9 points |

**Optional trait tweak** (Haggler wins ties before freshness; Local Favourite tip $3 → $4; Nest Egg $16 → $15):

- 3 players: spread 6 points (30.5–36.2).
- 4 players: 21.4–29.4.
- A small gain, kept as an optional experiment.
- Bargain Hunter needing a $3+ fruit and Big Basket also adding 3 shelf spaces: no measurable change.

---

## 4. Pacing

### 4a. Standard game (10 days), held-out seeds

| Measure | 3p now | 3p package | 4p now | 4p package |
|---|---|---|---|---|
| Average final cash | $80 | $105 | $56 | $97 |
| Day 1 income | $5.9 | $8.8 | $5.0 | $9.2 |
| Blowouts (winner far ahead) | 45.5% | 40.0% | 42.5% | 30.0% |
| Mid-game leader wins | 56.8% | 38.7% | 47.8% | 34.8% |
| Last at mid-game but wins | 15.3% | 28.8% | 8.2% | 18.3% |
| Lead changes per game | 3.6 | 4.3 | 4.3 | 4.7 |
| Close games | 10.8% | 15.2% | 14.5% | 15.8% |

In 2 players, final cash goes from $130 to $146. A 4-player game is no longer the poor relation ($97 against $56); the 15% extra 4-player demand is what fixes this.

### 4b. Quick (8 days) and Long (14 days), 3 players

"Now" uses development seeds; "package" uses held-out seeds.

| Measure | 8d now | 8d package | 14d now | 14d package |
|---|---|---|---|---|
| Undercut | 45 | 29 | 40 | 18 |
| Salvage | 52 | 39 | 43 | 23 |
| Premium | 10.5 | 21 | 16.5 | 35 |
| Reserve | – | 34 | – | 41 |
| 4-plan table T/R/Ri/E | 22/13/26/38 | 20/30/24/26 | 15/25/36/24 | **10/45/35/10** |
| Comeback from last | 19% | 28% | 17% | 30% |

- **Quick (8 days) becomes the most balanced length.**
- **Long (14 days) swings towards patient plans** (Reserve and Ripen); Turnover and Event fall to 10%.
- [Hyp] Patience winning a long game may feel right, but it is a watch item. Long-mode tuning is listed as an optional follow-up rather than mixed into this package.

### 4c. Provisional success criteria (computer-only, 3 and 4 players, held-out seeds)

| # | Criterion | Target | Package result |
|---|---|---|---|
| S1 | Against standard Hard computers, no single plan wins more than 1.25 × fair share (42% in 3p, 31% in 4p) | ≤ 42% / ≤ 31% | 36% / 29%: **PASS** |
| S2 | At least 4 of 8 plans within ±7 points of fair (3p) | ≥ 4 | 5: **PASS** |
| S3 | Hard beats Normal head to head (skill still matters) | ≥ 65% | 66%: **PASS (narrow)** |
| S4 | Comeback from last at mid-game | ≥ 20% (3p), ≥ 15% (4p) | 28.8% / 18.3%: **PASS** |
| S5 | 4-player cash at least 60% of 2-player cash | ≥ 60% | 66%: **PASS** |
| S6 | Price-war share | ≤ 10% | 7.0% / 7.6%: **PASS** |
| S7 | Long mode: no plan above 42% in the 4-plan table | ≤ 42% | 45% (Reserve): **FAIL**, watch item |
| S8 | Human enjoyment: clearer choices, "I could have won" feeling | Playtest | **NOT RUN** (needs section 7) |

---

## 5. Test method and evidence

- **Engine copy with switches.**
  - The lab is a copy of the rules engine with switches for each proposed change.
  - With every switch off it reproduces production exactly (checked on 450 games, identical results).
  - The production code was not touched.
- **Paired seeds and seat rotation.** Every variant replays the same seeds. Seats rotate each game, so every plan plays every seat equally often.
- **Many opponent types.**
  - The plans in 3a, plus Normal and Easy computers.
  - Crude single-rule bots, which all lose under both rule sets (0–3% wins), so they are not reported.
- **Sweeps.**
  - Eight runs (a baseline, a plan check and six sweeps): 813 test cells and 384,200 simulated games.
  - Round 1: demand × supply grid.
  - Rounds 2–3: market candidates, apple and mango costs, delivery forecast, end-of-game salvage.
  - Round 4: traits, Handcart basis, broke credit, earlier walk-ins, 4-player demand, Slow Season, unripe cutoff.
  - Round 5: decay variants, trait sample of 2,000 games, 8 and 14 days.
  - Round 6: final candidates on held-out seeds.
- **Validation.**

    - Seeds 1–600 were used to develop the package.
    - Seeds 100001–100600 (and 100001–102000 for traits) were kept back for the final check.
    - The current-rules numbers on held-out seeds match the development numbers within the error margin (Undercut 46 vs 43, Salvage 54 vs 51), so the results are not seed-specific.

- **Market League check** (300 games per stage, a computer standing in for you):

| Stand-in skill | Village | Harbour | Old Town | Spice Lane | Grand Bazaar | Royal Market | Daily win |
|---|---|---|---|---|---|---|---|
| Normal, now | 65 | 67 | 54 | 38 | 11 | 5 | 8 |
| Normal, package | 54 | 41 | 49 | 36 | 19 | 14 | 15 |
| Hard, now | 87 | 84 | 84 | 80 | 54 | 28 | 41 |
| Hard, package | 75 | 58 | 65 | 62 | 45 | 24 | 33 |

  The late-stage cliff softens, but early stages get harder because Easy and Normal computers do relatively better in a tight market. **Risk:** new players may find Harbour Steps harder. The league fix already suggested in REP-025 should follow this package, and the playtest checks it.

- **Rejected or no measurable effect:**
    - credit for broke shops;
    - earlier second walk-in;
    - Slow Season at −$1;
    - Bargain Hunter minimum cost;
    - Big Basket shelf bonus;
    - unripe cutoff;
    - end-of-game salvage or juice;
    - apple and blueberry decay changes.

- **Optional, not in the package:**
    - **Showing tomorrow's farmer delivery.** Undercut 28%, Turnover 14%, Hard-vs-Normal gap up to 72% in 2 players. It is an information feature with UI cost.
    - **Green mango at $3.**

- **Coverage gaps (NOT RUN):**
    - human play;
    - online rooms under new rules (needs verify:full after the patch);
    - Juice Press and Cool Room rework (Juice Press stays near-useless: 0.24 juiced per game);
    - favour-card balance;
    - every event card individually;
    - 2-player plan matchups (only mirror and skill tests);
    - Easy-level opponents in the plan tests.

- **Reproducibility.**
    - The lab, experiment lists and raw results are archived as `docs/workflow/reports/img/REP-027-lab.tar.gz`, with a README inside.

---

## 6. Recommendation: one cohesive package (PLAN-012)

### 6a. Essential changes

For each change: what decision it improves, why it improves that decision, and what it could break.

| # | Change | Decision improved | Why | What could break |
|---|---|---|---|---|
| E1 | **Tighter town.** Demand starts at 60% of base capacity instead of 40% (growth unchanged at +45%). Deliveries are 80% + 20% growth instead of 100% + 50% | "How low do I price?" becomes "how high dare I go?" | Fewer spare fruits, so cutting is no longer automatic; premium and reserve become real plans | Higher cash totals make the achievement thresholds easier; new players see fewer bargains at the farmers market |
| E2 | **Town size uses the base 4 fruits per shop.** Handcart and Big Basket no longer grow the market for everyone | "Should I buy a Handcart?" is about my own capacity, not flooding the town | Removes a hidden subsidy to market-flooding play; the main reason undercutting fell | Cash about 12% lower than E1 alone (offset by E1's 0.60 start); Handcart is slightly less exciting |
| E3 | **4-player towns have 15% more demand** | Whether to fight or share in big games | 4-player shops were starved ($56 average) | If over-tuned, 4 players becomes easier than 3 (now $97 against $105: fine) |
| E4 | **Apple cost 4/3/1 → 3/2/1** | "Which fruit fills my last slot?" | Apple becomes a real safe choice instead of a trap ($1.64 → $2.69 net per fruit) | Apple is now equal to watermelon on cost; identity rests on keeping a week, which the text must say |
| E5 | **RULES_VERSION 2 → 3** and compatibility work | – | Required by the rules-change policy | Old saves and old clients are cut off (by design, with clear messages) |

### 6b. Optional experiments (not in PLAN-012)

1. Trait tweak (Haggler before freshness, Favourite $4, Nest Egg $15).
2. Green mango $3.
3. Show tomorrow's delivery.
4. Long-mode tuning (demand growth for 14 days).
5. Juice Press and Cool Room rework.
6. League stage rebalance (follows REP-025).

### 6c. Exact patch plan

| File | Change |
|---|---|
| `src/engine/data.ts` | Add `export const MARKET = { demandStart: 0.6, demandGrowth: 0.45, supplyStart: 0.8, supplyGrowth: 0.2, fourPlayerDemand: 1.15 }`. Apple `cost: { fresh: 3, okay: 2, notso: 1, unripe: 1 }`. Apple `theme`: "Keeps a week and barely loses value. Never the best price, never a loss." Update the header comment (apple no longer "costs the most") |
| `src/engine/game.ts` (lines 179–209) | New `marketCapacity(s) = s.players.length * BASE_BUY` used by `demandScale` and `supplyScale` instead of `buyCapacity`. `demandScale`: `marketCapacity(s) * (MARKET.demandStart + MARKET.demandGrowth * progress) * (n === 4 ? MARKET.fourPlayerDemand : 1)`. `supplyScale`: `marketCapacity(s) * (MARKET.supplyStart + MARKET.supplyGrowth * progress) / FARMER_AVG_QTY`. Remove the unused `buyCapacity` if nothing else uses it. Fix the stale comment at line 179 ("3 fruit each a day") |
| `src/engine/ai.ts` | No code change. The AI reads `demandScale`, `FRUIT` costs and prices at run time (lab parity confirmed). Re-measure Easy, Normal and Hard gaps after the patch |
| `src/net/protocol.ts` | `RULES_VERSION = 3`. Old clients get the existing "please update" message from `server/room.ts` (line 203) and `server/meta.ts` (line 70) |
| `src/ui/store.ts` | No code change: line 66 already discards a game in progress with "from an older version of the rules". Release notes say so |
| `server/worker.ts` (lines 107 and 110) | Daily board key `daily:\${date}` → `daily:\${date}:r\${RULES_VERSION}`, so release-day scores under old and new rules never share a board. Replays already use server rules (`verifyDaily`) |
| `src/engine/daily.ts` | No change (same seeds; the outcomes simply follow the new rules) |
| `RULES.md` | Fruit table (apple $3/$2/$1, new character line). "The town grows": orders and deliveries are sized for 4 fruits per shop, whatever upgrades or traits you have; 4-player towns have 15% more customers. Handcart row: "Buy up to 5 fruits a day instead of 4 (the town doesn't grow with it)." Fix the existing wrong price-cap example (it says okay apples lose $2 and not-so lose $4; the real figures are $1 and $2). Balance notes refreshed |
| `src/ui/HowTo.tsx`, `src/ui/tutorial.ts` | No apple cost or demand numbers found; re-read after the patch. Fruit lines come from `FRUIT.theme` |
| `src/engine/data.ts` `UPGRADES.cart.text` / `TRAITS.basket.text` | Add "(the town doesn't grow with it)" to the Handcart; Big Basket unchanged |
| `src/ui/progress.ts` | Keep "Fruit Tycoon" ($200) and "Market Day" ($40/day). They become easier; earned badges stay. Owner choice listed in section 8 |
| `tests/` | New tests: demand and supply are the same with or without a Handcart; demand rises 15% at 4 players; apple cost values. Update any snapshots |
| Balance records | Re-run `npm run balance` and the plan checks on the patched engine; attach to the PLAN-012 verification report |

**Release:** this would ship as a minor version (1.2.0 or 1.3.0) with player notes, for example "Markets are livelier: customers come sooner, and big undercuts pay less. Apples are cheaper. Games in progress from the old version can't be resumed."

## 7. Human playtest plan

- **Who.** 4–6 people. Mix newcomers and people who have played 3+ games. At least one session with 4 players on phones (pass-and-play or online).
- **Format.**
  - Each tester plays current rules (live game) and the package (a preview copy built from a branch), in alternating order.
  - Each set is 2 games: one Standard and one Quick or Long.
  - Testers are not told which version is which.
- **Questions after each game** (1–5 scale, plus one sentence):
  1. Did you have more than one sensible way to play?
  2. Did prices feel like a guessing duel or a race to the bottom?
  3. Did each fruit feel different? Which one did you avoid, and why?
  4. Did you feel you could still win in the last third?
  5. Was the game the right length?
  6. Was anything confusing?
- **Watch for:**
  - the price testers set relative to the top price;
  - whether anyone buys a Handcart and says why;
  - whether apples get bought;
  - league stage 2 difficulty for newcomers.
- **Pass mark.**
  - The package scores at least as well on questions 1, 2 and 4 for most testers.
  - No tester finds it more confusing.
  - Any FAIL goes back to the lab.
- **No telemetry.** Notes are written by hand. No new data collection, as instructed.

---

## 8. Decisions for the owner

1. **Approve PLAN-012** (E1–E5 as above), or ask for changes.
2. Achievement thresholds: keep $200 and $40 (recommended, simplest), or raise them in the same plan.
3. Run the human playtest before or after building the preview copy (recommended: build the package on a branch preview, then playtest, then publish).

# REP-028 · PLAN-012 built: market rebalance, optional tweaks and the cosy home screen

| | |
|---|---|
| Game | Fruit Market, rules version 3 (branch `workflow/plan-012`) |
| Report date | 2026-10-27 10:35 (Asia/Kuala_Lumpur) |
| Type | Implementation and verification of PLAN-012 (approved: "yes approvrd together with the optional changes and the new layout from the mockup") |
| Builds on | REP-027 (balance proposal) |
| Approval status | Approved (PLAN-012). Publishing the live game still needs its own approval |

**How to read the numbers.**

- **[Code]** marks a fact about the code.
- **[Sim]** marks a computer-player result: 600 games per test, about ±4 points, on seeds 100001–100600, which were never used for tuning.
- **[Hyp]** marks a guess about people. Only the playtest can confirm it.

---

## 1. What was built

| Item | Change | Where |
|---|---|---|
| E1 | Customers start at 60% of the town's buying power (was 40%), growing +45%. Farmers bring 80% + 20% growth (was 100% + 50%) | `MARKET` in `src/engine/data.ts`; `demandScale`, `supplyScale` in `src/engine/game.ts` |
| E2 | The town is sized for 4 fruits per shop. Handcart and Big Basket help only their owner | `marketCapacity()` in `game.ts`; Handcart text |
| E3 | 4-shop towns have 15% more customers | `MARKET.fourPlayerDemand` |
| E4 | Apples cost $3 / $2 / $1 (were $4 / $3 / $1) | `FRUIT.apple` |
| E5 | `RULES_VERSION` 3; daily boards keyed by rules version (`daily:<date>:r3`), so release-day scores never mix | `src/net/protocol.ts`, `dailyBoardKey()` in `server/meta.ts`, `server/worker.ts` |
| O1 | Haggler wins ties against richer shops before freshness is compared. Local Favourite tip $4 (was $3). Nest Egg starts with $15 (was $16) | `hagglerEdge()` in `game.ts`; `TRAITS`, `NEST_EGG_BONUS`, `LOCAL_FAVOURITE_TIP` |
| O2 | Green mango costs $3 (was $2) | `FRUIT.mango` |
| O3 | **Tomorrow's lorry**: tomorrow's farmer card is drawn a day early and shown to everyone as plenty or few per fruit. Computer players use it to hold scarce fruit | `nextFarmer` in `game.ts`; `NextLorry` in `src/ui/table.tsx` (market and morning card); `src/engine/ai.ts` |
| O5 | **Juice Press**: each older fruit sold (okay or not so) earns $1 extra as juice; the nightly juicing of spoiling fruit stays. The sale pop-up includes it | `JUICE_TIP`; `resolve()` in `game.ts`; `src/ui/game/animator.tsx` |

| Item | Change | Where |
|---|---|---|
| O5 | **Cool Room**: unchanged (see section 3) | – |
| O4 | **Long (14-day) games**: unchanged (see section 3) | – |
| O6 | **Market League** stages retuned (section 2d) | `LEAGUE` in `src/ui/progress.ts` |
| L1 | **Home screen in the cosy pixel layout** from the mockup: harbour scene with bobbing boats, bunting, the "Fruit Market" sign with Foxy Rosa peeking out, fruit on the pier, then New game, Play a friend / Quick match, Daily challenge and Market League cards, and How to play / Trophies / Settings. Motion stops when reduced motion is on | `src/ui/App.tsx` , `src/ui/styles.css` |
| Text | Rules page (fruit table, the town grows, tomorrow's lorry, ties, Handcart, Juice Press, traits, balance notes). The wrong apple price-cap example is fixed. How-to mentions the lorry. Sale explanations name the Haggler rule | `RULES.md` , `src/ui/HowTo.tsx` , `src/engine/story.ts` |

**Saves and online play.**

- A game in progress from the old version shows the existing "older version of the rules" notice and is cleared ( `src/ui/store.ts` , unchanged).
- Old app versions are refused by online rooms and the daily board with the existing "please update" message.

---

## 2. Evidence

### 2a. The game code plays exactly like the tested package

[Sim] Production engine against the lab with the package switched on: **450 games** (2–4 players, 8/10/14 days, mixed levels). **0 differences in final cash.**

### 2b. Strategy matchups, 3 players (fair share 33%)

One plan against two Hard computers.

| Plan | Before (rules 2) | After (rules 3) |
|---|---|---|
| Undercut | 43 | 27 |
| Dump old stock (Salvage) | 51 | 36 |
| Premium prices | 13 | 28 |
| Reserve (keep cash) | 30 | 36 |
| Ripen green fruit | 27 | 34 |
| Read rivals (Adaptive) | 36 | 37 |
| Turnover | 18 | 23 |
| Events | 32 | 29 |
| **Spread** | **38 points** | **14 points** |

In 4 players (fair share 25%): Undercut 45 → 20, Salvage 54 → 28, Premium 6 → 20.

Four plans at one 4-player table:

| | Turnover | Reserve | Ripen | Events |
|---|---|---|---|---|
| Before | 17 | 19 | 29 | 34 |
| After | 16 | 32 | 33 | 20 |

### 2c. Pacing and the success criteria from REP-027

| # | Criterion | Before | After | Result |
|---|---|---|---|---|
| S1 | No plan above 42% (3p) / 31% (4p) | 51% / 54% | 37% / 28% | **PASS** |
| S2 | At least 4 of 8 plans within ±7 of fair (3p) | 3 | 7 | **PASS** |
| S3 | Hard beats Normal in 2 players at least 65% of the time | 80% | **60%** | **FAIL** (see 3) |
| S4 | Comeback from last at half-time, 3p at least 20% / 4p at least 15% | 15.3% / 8.2% | 25.8% / 19.2% | **PASS** |
| S5 | 4-player cash at least 60% of 2-player cash | 43% ($56 / $130) | 67% ($100 / $150) | **PASS** |
| S6 | Price-war sales at most 10% | 11.4% / 16.3% | 5.8% / 6.7% | **PASS** |
| S7 | Long mode: no plan above 42% at a 4-plan table | – | 42% (Reserve); Reserve alone wins 50% against Hard | **FAIL** (see 3) |
| S8 | Human enjoyment | – | – | **NOT RUN** (playtest) |

Other measures, 3 players:

- Blowouts: 45.5% → 36.3%.
- Mid-game leader wins: 56.8% → 43.5%.
- Lead changes per game: 3.6 → 4.3.
- Average cash: 2 players $130 → $150, 3 players $80 → $108, 4 players $56 → $100.

**Quick (8-day) games** are the most even length. At the 4-plan table the plans score 23 / 28 / 25 / 24. Salvage reaches 40% against Hard computers, still under the 42% limit.

### 2d. Traits, upgrades and the league

**Traits** (2,000 games per player count):

- 3 players: from 29.9% (Charmer) to 37.7% (Bargain Hunter), an 8-point spread (was 10).
- 4 players: from 23.3% to 28.5%, a 5-point spread (was 11.5).

**Upgrades.** A Hard computer building only one upgrade, 3 players. Building nothing wins 10.8%.

| Upgrade | Win % |
|---|---|
| Signboard | 20.7 |
| Cool Room | 15.8 |
| Handcart | 14.7 |
| Shop Bell | 13.5 |
| Juice Press | 12.8 (was below nothing: 9.7) |
| Bigger Shelf | 12.2 |

Every upgrade is now worth more than none.

**Market League** (300 games per stage, a computer standing in for you):

| Stand-in | Village | Harbour | Old Town | Spice Lane | Grand Bazaar | Royal Market |
|---|---|---|---|---|---|---|
| Normal, rules 2 and old league | 65 | 67 | 54 | 38 | 11 | 5 |
| Normal, rules 3 and new league | 64 | 64 | 52 | 41 | 33 | 20 |
| Hard, rules 3 and new league | 76 | 76 | 64 | 58 | 48 | 35 |

New stages:

1. Village Square: 1 Easy rival, 8 days.
2. Harbour Steps: 1 Easy rival, 8 days.
3. Old Town: 1 Normal rival.
4. Spice Lane: 2 Easy rivals.
5. Grand Bazaar: 2 Normal rivals.
6. Royal Market: 1 Hard and 2 Normal rivals.

Stages 3–6 are 10 days. The old final stage (three Hard rivals) was a wall.

### 2e. Checks

| Check | Result | Evidence |
|---|---|---|
| `npm run verify` (records, lint, types, 81 unit tests, build) | PASS | exit 0 |
| `npm run verify:full` (browser tests, online leak check, reconnect, quick match, four players) | PASS | exit 0; see section 5 |
| Home screen at 390×844, 360×640 and 1280×800 | PASS | no sideways scroll, no page errors; screenshots checked |
| Tomorrow's lorry visible on the morning card and under the market | PASS | screenshot at 390×844 |

---

## 3. What did not work, and what I did instead

- **Long games (O4).** I tried 10 different levers in the lab:
    - demand growth, demand start, supply start, supply growth;

    - when the second walk-in arrives;
    - the Cool Room price;
    - and others.

  None stopped patient plans (Reserve and Ripen) winning most 14-day games; Reserve stays at 39–52%. Long mode therefore keeps the standard rules. [Hyp] Patience winning a long game may feel right to players. The playtest should check it.

- **Cool Room (O5).**

    - A cheaper Cool Room ($4–5) adds little.
    - A Cool Room that stops ageing completely makes the Reserve plan too strong (45–49%).
    - Under the new market the current Cool Room is already worth buying (15.8%, against 10.8% for nothing), so it stays unchanged.

- **Juice Press (O5).** Four juice designs were tested. Juicing any leftover fruit was worse, because it turns good fruit into juice. The +$1 "juice tip" on older fruit was the only design that helped without boosting any plan.

- **Skill gap (S3).** Hard beats Normal 60% of the time (target 65%). Most of the drop comes from green mango at $3: without it the figure is about 70%, but Ripen and Turnover weaken.

    - Mango is on the preview as approved.
    - Changing it back to $2 is a one-line change. Decide after playtesting.
    - [Hyp] A smaller gap makes the game friendlier to new players and less punishing against Hard.

---

## 4. Risks

- Fewer spare fruits on Day 1: the market looks sparser. [Hyp] This may feel tight to new players.
- The $200 "Fruit Tycoon" and $40 "Market Day" achievements become easier (thresholds kept, as recommended).
- League progress is kept by stage number, so a player midway through keeps their stage under the new line-up.

## 5. Browser tests (`npm run verify:full`, exit 0)

- **Solo game:** results screen reached, no page errors.
- **Performance budget:** BUDGET OK.
- **Online match:**
    - 40 frames checked during pricing, 0 leaks of secret prices;
    - a refresh rejoined on Day 2;
    - both phones showed the same final cash;
    - rematch started.
- **Abandoned match:** a computer took over and the other player finished alone.

- **Quick match:** both players reached the game.
- **Four players:** lobby counts OK, a 5th person refused, all 4 in the game.
- **Reconnect stress:** 5 cycles, 0 failed (median 255 ms after a reload, 538 ms after a drop).
- **Old apps:** turned away by online rooms ( `tests/room.test.ts` ) and by the daily board ( `tests/meta.test.ts` ).

Lab files (engine copy, sweeps 7–11, parity check against the game code) are archived in `docs/workflow/reports/img/REP-028-lab.tar.gz` .

# REP-029 · PLAN-013: full-screen pricing and event sheets, clearer rivals while pricing

| | |
|---|---|
| Game | Fruit Market, rules version 3 (branch `workflow/plan-013`) |
| Report date | 2026-10-27 12:25 (Asia/Kuala_Lumpur) |
| Type | UI only. No rule, balance, server or protocol change |
| Owner request | "edit the UI/UX so that 1. secret price and events pops up to fill the screen (instead of just the bottom half of the screen 2. rework the rivals indicator during secret price, now its not clear still" |
| Approval status | Approved (PLAN-013). Publishing needs its own approval |

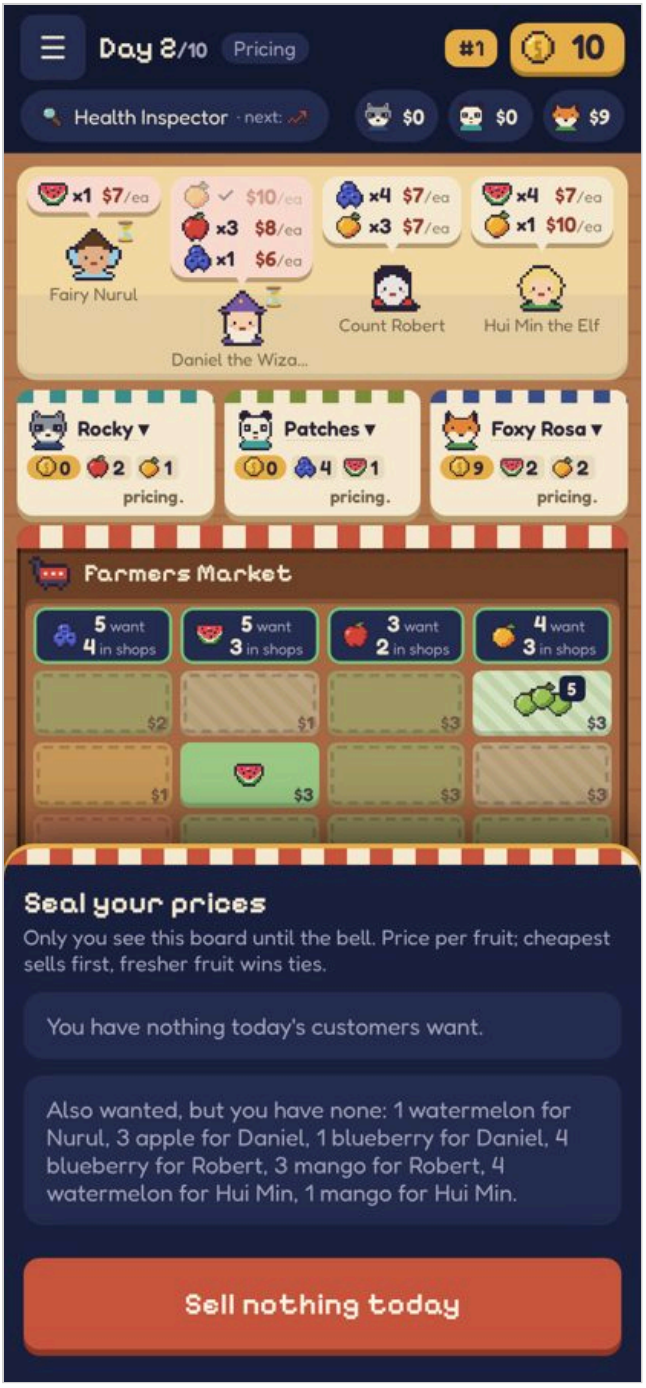
## What changed

1. **Pricing fills the screen.** On phones the "Seal your prices" sheet now covers everything below the top bar. It was the bottom half before. Desktop keeps its side panel.
2. **Events fill the screen.**
   - The morning card (day, standing, today's event, tomorrow's event, tomorrow's lorry) opens as a full-screen sheet. "Open the market" stays at the bottom.
   - Tapping the event chip in the top bar opens the same full-screen view, with "Back to the market".
3. **"Your rivals" strip at the top of the pricing sheet.** Each rival gets one card showing:
   - their portrait and name;
   - a clear status: **sealed** (green, wax seal), **deciding** (amber, a person still choosing) or **next in line** (amber, a computer, which always prices after the people at the table);
   - the fruit they hold that today's customers want, or **sits out** if they have none.

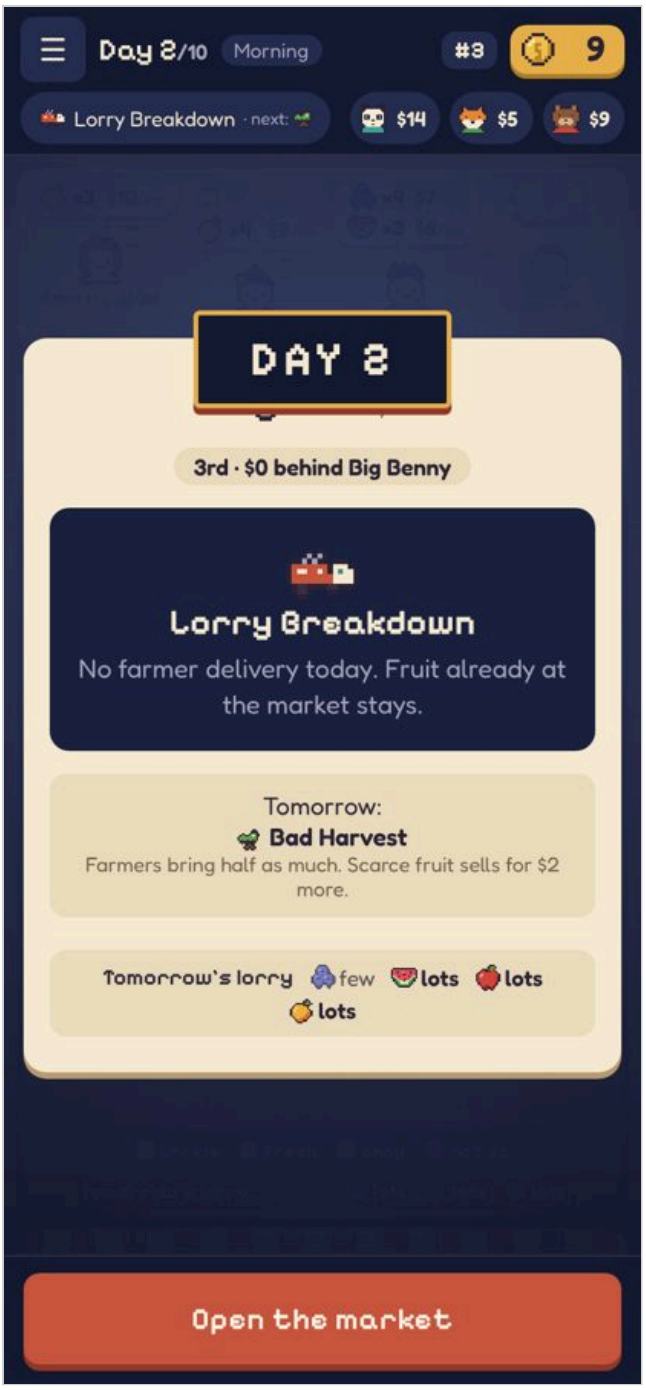
   A summary line reads, for example, "1 of 3 sealed · nobody sees a price until the bell".
4. **Seat badges** use the same words and colours: a larger amber "deciding" or "next in line" pill, and a "sealed" label under the wax seal.

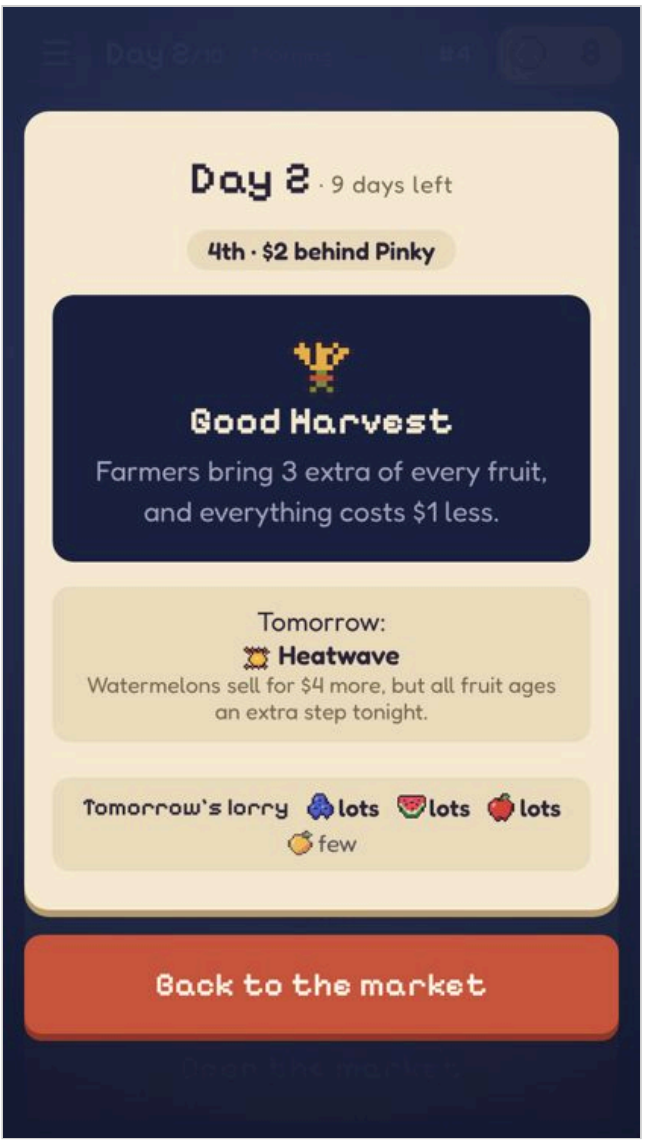
Only public facts are shown: whether a shop has sealed, and the stock that is already visible on its stall. No price is shown before the bell.

## Before and after (390×844)


Day 2/10
Pricing
#1
10
Health Inspector · next:
$0
$0
$9
×1 $7/ea
Fairy Nurul
$10/ea
×3 $8/ea
×1 $6/ea
Daniel the Wiza...
×4 $7/ea
×3 $7/ea
Count Robert
×4 $7/ea
×1 $10/ea
Hui Min the Elf
Rocky ▾
0 2 1
pricing.
Patches ▾
0 4 1
pricing.
Foxy Rosa ▾
9 2 2
pricing.
Farmers Market
5 want 4 in shops
5 want 3 in shops
3 want 2 in shops
4 want 3 in shops
$2
$1
$3
5
$3
$1
$3
$3
$3
Seal your prices
Only you see this board until the bell. Price per fruit; cheapest sells first, fresher fruit wins ties.
You have nothing today's customers want.
Also wanted, but you have none: 1 watermelon for Nurul, 3 apple for Daniel, 1 blueberry for Daniel, 4 blueberry for Robert, 3 mango for Robert, 4 watermelon for Hui Min, 1 mango for Hui Min.
Sell nothing today


Day 2/10
Pricing
#1
6
Heatwave · next:
$0
$0
$0
Seal your prices
Only you see this board until the bell. Price per fruit; cheapest sells first, fresher fruit wins ties.
Your rivals
0 of 3 sealed · nobody sees a price until the bell
Patches
next in line
2 2 2
Pinky
next in line
2 1
Uncle Toad
next in line
1 2
Vijay
wants 3 · pays up to $7
Sell
$4
Psst... 5 rival fruits chasing 4
≤$7 ≤$5 ≤$7
Chris
wants 1 · pays up to $7
Sell
$7
Psst... 5 rival fruits chasing 4
≤$7 ≤$5 ≤$7
Also wanted, but you have none: 4 mango for Lina, 4
Seal my prices up to $12




## Checks

| Check | Result | Evidence |
|---|---|---|
| `npm run verify` | PASS | exit 0: records, lint, types, 81 unit tests, build |
| `npm run verify:full` | PASS | exit 0: solo game, BUDGET OK, online 40 pricing frames with 0 price leaks, computer takeover, quick match, four players, reconnect 5 cycles 0 failed |
| Screens at 390×844, 360×640, 1280×800 | PASS | no sideways scroll, no page errors |

## Code

- `src/ui/Game.tsx` : the morning card moves into a full-screen sheet, the pricing tray gets the full-screen class, and seat boards record computer rivals.
- `src/ui/panel.tsx` : new `RivalWatch` strip.
- `src/ui/game/panels/Popovers.tsx` : full-screen event view.
- `src/ui/table.tsx` : seat badges.
- `src/ui/styles.css` : PLAN-013 block.

# REP-030 · Rules re-analysis for a tabletop-ready market game (proposal, no rule change)

| | |
|---|---|
| Game | Fruit Market, rules version 3 ( `main` at 416f381) |
| Report date | 2026-10-27 12:59 (Asia/Kuala_Lumpur) |
| Type | Analysis, lab experiments and a proposal. **Game code unchanged** |
| Owner request | Re-analyse fruit pricing (lorry and customer), decay, events, traits, power-ups and supply/demand control, against three guidelines: (1) playable on a physical table, (2) a real market feel with inventory, forecasting, strategy and undercutting, balanced all round, (3) each fruit's playstyle in line with its payout |
| Proposed plan | PLAN-014 "Tabletop edition rules" (awaiting approval) |

**Labels. [Code]** = how the game works now. **[Sim]** = computer-player result: 600 games per test on seeds never used for tuning, about ±4 points. **[Hyp]** = a guess about people, for the playtest.

---

## 1. Verdict against the three guidelines (rules version 3)

| Guideline | Today | Main gap |
|---|---|---|
| 1. Playable on a table | **Partly** | Deliveries and order sizes come from formulas (player count × game progress, rounded); 4-player ×1.15; up to ~10 secret prices per player per day; sliding every fruit one step each night is slow by hand |
| 2. Market feel | **Partly** | Undercutting and forecasting work; prices never react to shortage or glut; fruit is sold within half a day (85–90% bought fresh, 0.3 spoiled per shop per game), so inventory and decay barely matter; shared-loss events (Slow Season, Rainy Day) are luck with no decision |
| 3. Fruit playstyle = payout | **No** | Blueberries earn the best return on cost of all four ($1.72 → $5.61) though they are meant to be the cheap volume fruit; apples (safest) earn more than watermelons |

---

## 2. How the rules work now [Code]

### 2a. Fruit: lorry price, customer price, decay

| Fruit | Farmers-market cost fresh / okay / not so / green | Customer top price (24 cards, after shift) | Price drop by age, oldest → freshest | Life on the shelf |
|---|---|---|---|---|
| Blueberry | 2 / 1 / 1 / – | 5–8 (shift 0) | −4 · −2 · 0 | 3 nights |
| Watermelon | 3 / 2 / 1 / 1 | 6–9 (0) | −3 · −1 · −1 · 0 · 0 | 5 nights (+1 green) |

| Fruit | Farmers-market cost fresh / okay / not so / green | Customer top price (24 cards, after shift) | Price drop by age, oldest → freshest | Life on the shelf |
|---|---|---|---|---|
| Apple | 3 / 2 / 1 / – | 6–10 (−1) | −2 · −1 · −1 · −1 · 0 · 0 · 0 | 7 nights |
| Mango | 4 / 2 / 1 / 3 | 7–11 (+2) | −3 · −1 · 0 · 0 | 4 nights (+2 green) |

- The farmers market cost is fixed per freshness stage; events change it ±1.
- The customer's price is the card value plus the fruit shift plus the event, minus the age drop.
- Fruit at the farmers market ages too, and every night all fruit moves one step.

### 2b. Supply and demand control (the tabletop problem)

- **Deliveries:** one farmer card a day lists three fruits (4–6 each); the fourth fruit comes as 2. Each number is then multiplied by a formula: 4 per shop × (0.8 + 0.2 × game progress) ÷ 17, rounded. Tomorrow's card is shown a day early.
- **Customers:** one expected customer, known a day early, plus 1 walk-in (2 from halfway), drawn face down. Every order size is multiplied by a formula: 4 per shop × (0.6 + 0.45 × progress), × 1.15 with 4 players, ÷ cards, rounded.
- **Selling:** every shop sets a secret price and quantity for every customer and fruit. Then:
  - cheapest sells first;
  - ties go to a Haggler facing a richer shop, then fresher fruit, then Favoured Trade, then turn order.

### 2c. What events, traits and upgrades do to outcomes

**Events** (from REP-025, income vs a normal day, 2p / 4p):

- Big gains: Festival +11 / +8, Superfood +11 / +5, Mango Craze +10 / +4 (the largest gap between shops).
- Neutral: Heatwave, Minimum Price.
- Losses: Slow Season **−12** / −7, Rainy Day −8 / −5, Lorry Breakdown −6 / −3.
- Most events hit every shop equally. They reward planning only when they are announced a day early, which they are.

**Traits** [Sim, rules 3]:

- 3 players: 30–37%, a 7-point spread (fair 33%).
- 4 players: 23–29%.

Close enough.

**Upgrades** [Sim]: building only one upgrade, against 10.8% for building none:

| Upgrade | Win % |
|---|---|
| Signboard | 20.7 |
| Cool Room | 15.8 |
| Handcart | 14.7 |

| Upgrade | Win % |
|---|---|
| Shop Bell | 13.5 |
| Juice Press | 12.8 |
| Bigger Shelf | 12.2 |

All are worth something; the Signboard is the clear best.

**Favour cards:** small one-off boosts. Fine on a table.

---

## 3. Proposed "Tabletop edition" (rules version 4)

### 3a. Core changes (tested)

| # | Change | Replaces | Physical form | What decision it improves |
|---|---|---|---|---|
| T1 | **Printed deliveries.** Each farmer card shows 3-player amounts: its three fruits at 2–4 (today's number −2), the fourth fruit at 1. With 2 players, one less of each listed fruit (minimum 1); with 4 players, one more of every fruit | The supply formula | Numbers on the cards plus one line in the rules | None directly; makes the game runnable by hand with almost the same flow |
| T2 | **Printed orders and a walk-in rule.** Customer cards are used as printed. Walk-ins: players − 2 in the first half of the game, players − 1 in the second half | The demand formula and ×1.15 | The deck and one line in the rules | Same as T1 |
| T3 | **One posted price per fruit.** Each shop secretly sets one price per fruit on a 4-slider price board behind its screen, and puts the fruit it will sell in its "for sale" crate. On reveal, customers (waiting ones first) buy from the cheapest shop whose fruit is fresh enough for the price | ~10 secret prices per customer and fruit | A price board and a screen | The core duel becomes "what will they post for mangoes today?", which reads like a real market stall and is quick on a table |
| T4 | **Market pulse.** Each fruit has a marker on a track from −$1 to +$2. After sales: if any customer still wants that fruit, the marker goes up $1 (shortage). If at least players + 1 ripe ones are left at the farmers market, it goes down $1 (glut). Otherwise it drifts one step back to 0. Customers pay the marker on top of their card price | Fixed prices | One cube per fruit on a small track | Prices respond to supply and demand. Spotting tomorrow's shortage pays, and flooding a fruit is punished |
| F1 | **Fruit payouts lined up with risk.** Blueberry customers pay $1 less; apple customers $1 less again (apple shift −2) | Card shifts | Printed on the customer cards | See 3c |

### 3b. Results [Sim, 3 players unless noted]

| Measure | Rules 3 now | T1+T2 | + T3 + T4 | + F1 (proposed) |
|---|---|---|---|---|
| Undercut | 27 | 22 | 29 | 28 |

| Measure | Rules 3 now | T1+T2 | + T3 + T4 | + F1 (proposed) |
|---|---|---|---|---|
| Dump old stock | 36 | 32 | 32 | 28 |
| Premium prices | 28 | 30 | 31 | 33 |
| Reserve | 36 | 38 | 33 | 34 |
| Ripen | 34 | 35 | 39 | 38 |
| Read rivals | 37 | 35 | 38 | 35 |
| Turnover | 23 | 24 | 24 | 26 |
| Events | 29 | 30 | 24 | 28 |
| **Spread** | 14 | 16 | 15 | **12** |
| 4 players: Undercut / Salvage / Premium | 20 / 28 / 20 | 20 / 30 / 22 | 21 / 28 / 21 | 23 / 29 / 20 |
| Hard beats Normal (2p) | 60 | 59 | 63 | 62 |
| Comeback from last at half-time (3p / 4p) | 25.8 / 19.2 | 28 / 19 | 32.3 / 19.2 | 30.2 / 18.0 |
| Price-war sales (3p / 4p) | 5.8 / 6.7 | 5.0 / 7.1 | 3.9 / 4.2 | 5.9 / 7.0 |
| Average cash (2p / 3p / 4p) | 150 / 108 / 100 | 137 / 116 / 94 | 146 / 133 / 116 | 135 / 122 / 103 |
| 4-plan table (Turnover / Reserve / Ripen / Events) | 16/32/33/20 | 16/40/26/19 | 18/29/35/18 | 15/29/38/19 |

**Daily flow, formula against printed numbers** (fruit delivered / units customers want, first half → second half):

| Players | Formula (now) | Printed numbers |
|---|---|---|
| 3 | 9.7 → 11.0 / 8.4 → 12.1 | 10.1 → 10.1 / 8.7 → 13.1 |
| 4 | 13.7 → 15.3 / 12.8 → 17.3 | 14.1 → 14.1 / 13.1 → 17.4 |
| 2 | 6.8 → 8.0 / 5.5 → 8.1 | 7.1 → 7.1 / 4.4 → 8.8 |

The printed rules reproduce the market closely.

Traits under the proposal: 31–37% in 3 players.

**Caution.** The computers' posted price is a rough stand-in: they take their lowest planned price for each fruit. People will price more cleverly, so T3 must be playtested.

### 3c. Fruit identities and payout [Sim, proposal]

| Fruit | Target playstyle | Cost (fresh) | Average sale | Net per fruit bought | Risk |
|---|---|---|---|---|---|
| 🥭 Mango | **Speculator:** buy green for $3, wait 2 nights, sell big on Mango Craze or shortage days | $3.61 | $9.15 | **$5.15** | Highest: 2-night wait, spoils fast once ripe, few delivered |

| Fruit | Target playstyle | Cost (fresh) | Average sale | Net per fruit bought | Risk |
|---|---|---|---|---|---|
| 🫐 Blueberry | **Quick flipper:** cheap and plentiful; sell today or lose $2 a night | $1.72 | $5.58 | **$3.51** | High: 3-night life, steepest decay |
| 🍉 Watermelon | **Steady bulk seller:** big orders, slow decay, a 1-night green bet | $2.51 | $6.20 | **$3.15** | Medium |
| 🍎 Apple | **The keeper:** lasts a week, sells any day, the safe stock for a Lorry Breakdown or a shortage | $2.79 | $6.08 | **$3.00** | Lowest |

Now the more a fruit can hurt you, the more it pays: mango > blueberry > watermelon > apple. Under rules 3 apples paid more than watermelons, and blueberries gave the best return on cost.

### 3d. Supporting proposals (not simulated: they need people, or computer players that plan ahead)

1. **The queue: forecasting the crowd.**
    - Tomorrow's walk-ins are drawn tonight and placed face down.
    - The card backs show which fruits each customer wants, but not how many or at what price.
    - With the face-up expected customer and tomorrow's lorry, players can forecast demand: "two walk-ins want apples tomorrow, and the lorry brings few".
    - Physically: printed card backs. Digitally: a row of silhouettes with fruit icons.
2. **Market Days.** Every third day is a busy fair day: +2 walk-ins, and the lorry doesn't come.
    - Shops must stock up the day before, which is where shelf space, decay and apples matter.
    - [Sim] Comebacks rise to 34% and blowouts fall to 31%.
    - But the computers don't plan ahead, and skill matters less in their games (Hard beats Normal 52%).
    - [Hyp] This may be the strongest inventory mechanic, so it should go to a paper playtest before any decision.
3. **Crates instead of sliding.** On a table, fruit goes into a crate stamped with the day it arrived. Its age (today minus that day) gives its freshness from a printed chart. Nothing needs sliding each night, and the digital game already works like this underneath.
4. **Events with decisions.**
    - Keep the events that are announced a day early and reward preparation: Lorry Breakdown, Mango Craze, Superfood, Heatwave, Bad Harvest, Hotel Order.
    - Replace the two pure shared losses (Slow Season, Rainy Day) with single-fruit shocks such as "Blueberry glut: +2 blueberries per player, blueberry customers pay −$1" or "Apple shortage: no apples delivered".
    - Scale Good Harvest by players: +1 of each fruit per 2 players, rounded up.
    - Never draw a night-only event (Inspector, Heatwave ageing) for the final day.
    - Needs a simulation round before numbers are fixed.
5. **Traits and upgrades:** no changes needed for the table; all are printed cards.
    - Juice Press stays the weakest upgrade (12.8%). [Hyp] It gains value with Market Days and stocking ahead.

### 3e. What it would mean for the app

- Rules version 4: old saves cleared; old apps refused online; separate daily boards (all already handled).
- **Pricing screen:** four fruit price sliders plus "how many to put out", instead of one card per customer. It is simpler and fits the full-screen sheet from PLAN-013.
- **Board:** a market pulse marker on each fruit column, and walk-in silhouettes in the street (if 3d-1 is approved).
- **Computer players:** must learn posted pricing properly before release. Today's stand-in is crude.
- **Rules page and how-to:** rewritten. The digital game and a paper version share one rulebook.

---

## 4. Risks

- **Posted prices remove per-customer pricing.** A shop can no longer charge a rich customer more than a poor one. [Hyp] Simpler and more market-like, but less fine control for experts. Hard beats Normal 62%, still under the 65% aim.
- **Market pulse adds one track and a rule after sales.** [Hyp] It is easy on a table, but it is a new idea for players.
- **The computers can't judge** Market Days, the queue or event redesigns, so those depend on playtests.
- **It is a large change:** a new pricing screen, computer pricing work, and a rules rewrite.

## 5. Playtest plan

1. **Paper prototype:** print the farmer and customer cards with the new numbers, use coins and tokens, play 3 evenings with 3–4 people. Test T1–T4, the queue and Market Days. Time each day; target 6 minutes or less on a table.
2. **Digital preview:** after building, the same questions as REP-027 section 7, plus:
    - "Did the posted prices feel like a market?"
    - "Did you ever stock up ahead?"

## 6. Recommendation

| Order | Item | Status |
|---|---|---|
| 1 | Approve **PLAN-014 core: T1 + T2 + T3 + T4 + F1** (tested above), with computer posted-price pricing and the new pricing screen | Awaiting approval |
| 2 | Paper playtest of the queue, Market Days and crates, then decide | Proposal |
| 3 | Event redesign (3d-4) after a simulation round | Proposal |

Lab files (engine copy, sweeps 12–15) are archived in `docs/workflow/reports/img/REP-030-lab.tar.gz`.

# REP-031 · Fruit Market: game concept and mechanics (Tabletop edition, rules version 4)

| | |
|---|---|
| Game | Fruit Market. Design: Murshidi bin Md Hashim |
| Report date | 2026-10-27 13:46 (Asia/Kuala_Lumpur) |
| Type | Final mechanics (worked out and lab-tested) and a concept summary for benchmarking against board games |
| Plan | PLAN-014, approved: "approve all the idea, work out the mechanics, and summarise to me game back highlighting the concept and game mechanics as i want to benchmark against similar genre board games and traditional popular board games" |
| Status | Design final for the build. The app still runs rules version 3 until PLAN-014 is built |

**Labels. [Sim]** = computer-player result: 600 games per test on seeds never used for tuning, about ±4 points. **[Hyp]** = a guess about people, to be checked in the paper playtest.

---

## 1. The game on one page

**Pitch.** You run a fruit stall in a busy harbour market. Each day you buy fruit from the farmers' lorry, guess what the crowd will want and what your rivals will charge, then post your prices in secret. The cheapest stall makes the sale. But fruit goes off every night, so stock you can't sell is money rotting on your shelf. After 8, 10 or 14 days, the richest stall wins.

| | |
|---|---|
| Players | 2–4 (solo against computer shops in the app) |
| Length | Quick 8, Standard 10 or Long 14 days. App: about 15–20 minutes. Table: about 45–60 minutes **[Hyp]** |
| Age / weight | 10+. Light-to-medium strategy **[Hyp]**: simple turns, with depth in reading the market and rivals |
| Win condition | Most cash at the end. Unsold fruit and upgrades are worth nothing. Ties go to the shop that sold more fruit |
| Elimination | None. Everyone plays every day |
| Luck vs skill | Luck from the order of the farmer, customer and event decks, softened by a one-day look ahead. Skill in buying, timing, pricing and reading rivals. [Sim] Hard computer beats Normal 62% |

**The four pillars (your brief).**

| Pillar | How the game delivers it |
|---|---|
| **A real market** | Supply comes off a lorry; demand comes from customers with budgets. One posted price per fruit; the cheapest stall sells first. The market pulse moves prices up after a shortage and down after a glut |
| **Inventory and decay** | Every fruit ages one step a night and sells for less as it ages; the oldest spoils. Shelves are limited. Fruits rot at different speeds |

| Pillar | How the game delivers it |
|---|---|
| **Demand forecasting** | The crowd queues a day ahead: the expected customer is face up, and the walk-ins are face down with their wanted fruits printed on the back. Tomorrow's event and tomorrow's lorry are also shown a day early |
| **Strategy and undercutting** | Sealed simultaneous pricing creates the "do I go $1 lower?" duel. Eight different plans are viable (section 4) |

---

## 2. Components (tabletop)

| Component | Count | Purpose |
|---|---|---|
| Fruit tokens | 4 kinds (about 40 each) | Blueberry, watermelon, apple, mango |
| Market board | 1 | Farmers market (crates by arrival day), customer queue, market pulse track, day track |
| Shop mats | 4 | Shelf (8 spaces), crates by arrival day, "for sale" crate |
| Price boards and screens | 4 | One slider per fruit ($1–$14), hidden until the reveal |
| Farmer cards | 20 | Tomorrow's lorry: 3 fruits in quantity plus 1 of the fourth; some mark one fruit as arriving green |
| Customer cards | 24 | Front: name, wanted fruits, quantity, top price for fresh fruit. Back: fruit icons only |
| Event cards | 13 | One a day from Day 2, each announced a day early |
| Favour cards | 12 (6 kinds × 2) | One-off bonuses earned by serving a whole order alone |
| Upgrade cards | 6 | Evening draft |
| Trait cards | 6 | Pick 1 of 3 at the start |
| Coins, cubes, markers | | Cash; one market pulse cube per fruit; day marker |

**Crates instead of sliding.** Fruit sits in a crate marked with the day it arrived. Its age (today minus that day) gives its freshness from the chart on the mat. Nothing needs moving each night, and the app shows the same thing as a decay track.

---

## 3. A day, step by step

1. **Morning**
   - Reveal today's event (announced yesterday) and turn up tomorrow's.
   - **The lorry arrives:** put today's farmer card's fruit into the farmers market. Green fruit goes to the green crate.
   - Turn up **tomorrow's lorry** card.
2. **Buying**
   - **Poorest shop first** (ties go round the table).

    - Each shop buys up to **4 fruits** (5 with a Handcart), paying the cost for that freshness.
    - Shelf limit 8 (14 with a Bigger Shelf).
3. **The crowd**
    - Customers still waiting from yesterday stand at the front.
    - Today's expected customer and today's walk-ins are turned face up.
    - Draw **tomorrow's queue**: 1 expected customer face up, and the walk-ins face down (backs show their fruits).
4. **Pricing (secret)**
    - Behind your screen, set **one price per fruit** on your price board.
    - Put the fruit you are willing to sell in your "for sale" crate.
    - Everyone reveals at once.
5. **Sales**
    - Customers are served in queue order. For each fruit a customer wants, they buy from the **cheapest stall**.
    - Your price must not exceed what they pay for your fruit's freshness: their top price + market pulse + today's event − the age drop.
    - You hand over your oldest fruit that is fresh enough.
    - **Ties:** a Haggler facing a richer shop, then fresher fruit, then Favoured Trade, then turn order.
    - Paid on the spot.
    - Serve a customer's whole order by yourself to draw a **favour card** (one a day).
    - An unserved new customer waits until tomorrow; a waiting customer then leaves.
6. **Market pulse** (for each fruit)
    - **Shortage:** a customer still wants it, so the cube goes up $1 (to +$2 at most).
    - **Glut:** at least players + 1 ripe ones are left at the farmers market, so the cube goes down $1 (to −$1 at least).
    - Otherwise the cube moves one step back towards 0.
7. **Evening**
    - Upgrade draft: each shop sees 3 upgrades and may build any it can afford.
    - Evening favour cards can be played.
8. **Night**
    - All fruit ages one step, and fruit past its last step spoils.
    - The Juice Press turns up to 3 spoiling fruits into $2 each.
    - The Cool Room ages your fruit only every other night.
    - Night events apply (Health Inspector fines, Heatwave ageing).

**Supply and demand on a table (no formulas).**

- **Lorry:** the card's three fruits arrive at the printed number (2–4); the fourth fruit arrives as 1.
    - 2 players: one less of each listed fruit (minimum 1).
    - 4 players: one more of every fruit.
- **Walk-ins:** players − 2 in the first half of the game, players − 1 in the second half.
    - 2 players: 0 → 1.
    - 3 players: 1 → 2.
    - 4 players: 2 → 3.
- [Sim] These printed rules reproduce the old formula flow almost exactly. In 3 players, about 10 fruits a day are delivered, and customers want about 9 units a day early and 13 late.

**Optional rule: Market Days** (off by default, for the paper playtest).

- Every third day is a fair: +1 walk-in, and **no lorry**.
- Tomorrow's lorry shows "Market Day", so shops must stock up the day before.
- See section 6 for why it is optional.

---

## 4. Fruit: price, decay and playstyle

| | 🫐 Blueberry | 🍉 Watermelon | 🍎 Apple | 🥭 Mango |
|---|---|---|---|---|
| **Playstyle** | Quick flipper | Steady bulk seller | The keeper | Speculator |
| Lorry cost fresh / okay / not so / green | $2 / $1 / $1 / – | $3 / $2 / $1 / $1 | $3 / $2 / $1 / – | $4 / $2 / $1 / $3 |
| Customer top price (fresh) | $4–7 | $6–9 | $5–9 | $7–11 |
| Price drop by age, oldest → freshest | −4 · −2 · 0 | −3 · −1 · −1 · 0 · 0 | −2 · −1 · −1 · −1 · 0 · 0 · 0 | −3 · −1 · 0 · 0 |
| Shelf life | 3 nights | 5 nights (+1 green) | 7 nights | 4 nights (+2 green) |
| Risk | High: steepest decay | Medium | Lowest | Highest: wait to ripen, fast spoil, few delivered |
| [Sim] Net profit per fruit bought | $3.51 | $3.15 | $3.00 | $5.15 |
| Best moment | Sell the day you buy | Big orders, Heatwave | Stock for a Lorry Breakdown or shortage | Buy green before a Mango Craze or a shortage |

The more a fruit can hurt you, the more it pays.

---

## 5. Events, traits, upgrades and favours

**Events** (13 cards; one a day from Day 2, each shown a day early):

| Event | Effect |
|---|---|
| Festival | Customers pay +$3, plus an extra walk-in |
| Blueberry Glut *(new, replaces Slow Season)* | +1 blueberry per player delivered; blueberry customers pay −$1 |
| Heatwave | Watermelons +$4; all fruit ages an extra step tonight |
| Apple Blight *(new, replaces Rainy Day)* | No apples delivered; apple customers pay +$1 |
| Mango Craze | Mangoes +$3; the Influencer wants 3 |
| Superfood Trend | Blueberries +$3; the Gym Club wants 4 |
| Good Harvest *(scaled)* | +1 of every fruit per 2 players (rounded up); lorry prices −$1 |

| Event | Effect |
|---|---|
| Bad Harvest | Half delivery (rounded up); customers pay +$2 |
| Price Hike | Lorry prices +$1 |
| Minimum Price | Price floors per fruit |
| Health Inspector | $2 fine per not-so fruit tonight. Never the final day's event (draw again) |
| Lorry Breakdown | No delivery today |
| Hotel Order | The Hotel Chef orders 3 apples, 2 watermelons and 2 blueberries |

**Traits** (pick 1 of 3 at the start, public):

| Trait | Effect |
|---|---|
| Haggler | Wins ties against richer shops, even over fresher fruit |
| Big Basket | +1 fruit a day |
| Bargain Hunter | First fruit each day $1 cheaper |
| Charmer | Start with 3 favour cards; hold 4 |
| Local Favourite | +$4 per customer you serve completely alone |
| Nest Egg | Start with $15 instead of $10 |

[Sim] 31–38% wins each in 3 players.

**Upgrades** (evening draft, 3 offered, built once):

| Upgrade | Cost | Effect |
|---|---|---|
| Handcart | $6 | Buy 5 a day |
| Signboard | $8 | +$1 per fruit sold |
| Cool Room | $7 | Ages every other night; Heatwave-proof |
| Bigger Shelf | $3 | 14 spaces |
| Juice Press | $4 | +$1 per older fruit sold, and up to 3 spoiling fruits a night become $2 each |
| Shop Bell | $4 | Draw 2 favour cards; hold 4 |

**Favour cards** (2 of each): Free Cash ($3), Free Fruit, Good Riddance (sell ripe stock back at $1 each), Fridge (all fruit one step fresher), Favoured Trade (win ties today), Priority (buy first tomorrow).

---

## 6. Balance evidence (computer players)

| Measure | Rules 3 (app now) | Rules 4 (this design) |
|---|---|---|
| Win share of 8 plans, 3 players (fair 33%) | 23–37% | 25–42% (Ripen at the top) |
| Win share, 4 players (fair 25%) | 20–28% | 18–25% |

| Measure | Rules 3 (app now) | Rules 4 (this design) |
|---|---|---|
| Hard beats Normal (2 players) | 60% | 56–62% |
| Comeback from last at half-time (3p / 4p) | 26% / 19% | 28–32% / 18–19% |
| Price-war sales at $2 or less (3p) | 6% | 5% |
| Quick (8 days): 4-plan table | – | 19 / 30 / 30 / 21 (even) |
| Long (14 days): 4-plan table | Reserve 42% | Ripen 49% (watch item) |
| Traits (3 players) | 30–37% | 31–38% |

**Market Days** (optional rule):

- Every third day with +1 walk-in gives the most even plans of anything tested (24–35%).
- It also gives 38% comebacks and spoilage halves, because shops stock ahead.
- But Hard and Normal computers become even (50%), because the computers don't plan ahead.
- [Hyp] People who plan ahead may make it the most skilful mode. The paper playtest decides.

**Open items for the build:**

- Shop Bell, Juice Press and Cool Room are worth little on their own under posted prices (12.8–14.5%, against 13.5% for no upgrade). They need retuning.
- Long games favour the ripening plan.
- The computer players need proper posted-price logic.

---

## 7. For your benchmark: mechanism profile

Mechanism names follow the usual board-game catalogue terms, so you can compare like for like.

| Mechanism | In Fruit Market | Weight in this game |
|---|---|---|
| Simultaneous action selection / sealed pricing | Secret price board, revealed together | Core |
| Lowest-price-wins competition (a reverse auction) | Cheapest stall serves the customer | Core |
| Dynamic market pricing | Market pulse after shortages and gluts; lorry prices by freshness | Core |
| Perishable inventory / resource decay | Fruit ages nightly, loses value, spoils | Core |
| Demand forecasting / hidden information | Face-down queue with fruit-icon backs; tomorrow's lorry and event shown | Core |
| Open market buying | Shared farmers market, first come, first served | Medium |
| Catch-up turn order | Poorest buys first | Medium |
| Variable player powers | Traits | Light |
| Card drafting | Evening upgrade draft (3 offered) | Light |

| Mechanism | In Fruit Market | Weight in this game |
|---|---|---|
| Event deck with look ahead | One event a day, announced a day early | Medium |
| Contract fulfilment | Serve a whole order alone for a favour | Light |
| Engine building | Upgrades that pay each day | Light |

**Starting points for comparison** (well-known games sharing a mechanism; please verify details in your own benchmark):

| Game | Shared idea | Where Fruit Market differs |
|---|---|---|
| Power Grid | A resource market whose prices rise as supply runs out; the trailing player buys first | Fruit also rots, and selling prices are set by the players in secret |
| Food Chain Magnate | Customers buy from the cheapest restaurant; price-setting competition | Much lighter; demand is forecast from a visible queue rather than built by marketing |
| For Sale / Modern Art | Blind or sealed bidding duels | Bidding is for customers each day, and goods spoil |
| Monopoly (traditional) | Win by cash; rents and luck | No elimination, no property; every day is a fresh price duel; catch-up turn order |
| Catan (traditional) | Variable supply each turn | Supply is shown a day ahead and bought, not rolled; no trading between players |
| Sushi Go / 7 Wonders | Simultaneous play and drafting | Drafting is a side system (upgrades); the core is pricing and stock |

**Possible distinctive hooks** for your benchmark to confirm: perishable stock with a different decay speed per good; one posted price per good settled by a simultaneous reveal; and a self-adjusting price track driven by unmet demand.

---

## 8. Next steps

1. **Paper playtest:**
    - print the cards with these numbers;
    - 3 sessions of 3–4 players, with Market Days on in one;
    - time each day.
2. **Build PLAN-014 in the app** (rules version 4):
    - new pricing screen (four sliders);
    - market pulse on the board;
    - queue silhouettes;
    - printed supply and demand;
    - event changes;
    - Market Days as an option in the game set-up;
    - computer posted pricing;
    - upgrade retune.
3. **Benchmark** using section 7, then feed changes back before the build is finalised.

Lab files (sweeps 16–18) are archived in `docs/workflow/reports/img/REP-031-lab.tar.gz`.

# REP-032 · Tweaks from the benchmark review: a readable price duel, three prototypes and the Market Gossip card

| | |
|---|---|
| Game | Fruit Market (app on rules version 3; design target rules version 4, REP-031) |
| Report date | 2026-10-27 14:34 (Asia/Kuala_Lumpur) |
| Type | Design proposal with lab checks. **Game code unchanged** |
| Input | The owner's benchmark review of REP-031 (Stockpile, Food Chain Magnate, Modern Art, For Sale, Catan, Monopoly) and the request: "suggest tweaks based on the above, dont take current game turn as the final form, feel free to tweak accordingly, introduce one favour cards which allows player to see the demand for the face down card" |
| Proposed plan | PLAN-015 (awaiting approval). It reshapes PLAN-014 before that is built |

**Labels. [Sim]** = computer players: 600 games per test on unseen seeds, about ±4 points. **[Hyp]** = a guess about people, to test at the table.

**Correction to REP-031.** Food Chain Magnate does not simply give customers to the cheapest restaurant. Whether a restaurant can serve the order, price plus distance, and tie-breakers all matter. Thank you for the check.

---

## 1. The direction

The review's main point holds: **make the price duel exciting and easy to read; don't add machinery.** The hook stays:

> **"How much can I charge before my rival, or the rot, gets me?"**

Every tweak below either:

- removes bookkeeping;
- makes the reveal clearer;
- gives a real alternative to "$1 lower"; or
- gives the session an arc.

The new favour card is the **only** new information mechanism, as the review advised.

---

## 2. A new day: five beats instead of eight

| Beat | What happens | Why |
|---|---|---|
| **1. Dawn: the "Tomorrow" strip** | The lorry unloads today's fruit. One strip shows everything known about tomorrow: the lorry card, the event, the expected customer (face up) and the walk-ins (face down, backs show their fruit) | All forecasting in one place, so there are no separate things to check |
| **2. Market** | Buy up to 4 fruits, poorest shop first | Unchanged |
| **3. Open stalls** *(new)* | Each shop puts the fruit it will sell today into its "for sale" crate, **face up**. Quantities are public before anyone prices | Makes the alternative to undercutting visible: "He only has two apples, so I can charge more and serve the rest" |
| **4. The bell** | Prices are set secretly on the price board, then **everyone reveals at once**. This is the centrepiece | One clear dramatic moment a day |
| **5. The rush** | Customers come in queue order; each fruit goes from the cheapest stall, and leftover demand **spills to the next stall**. Every sale is shown as customer → stall. Then night: fruit ages | Players see why they sold or didn't |

**Removed from the core:**

- the market pulse track, now an optional advanced rule;
- evening admin on most days (upgrades are bought only on Market Day evenings, see section 4);
- the Haggler tie rule. Ties become **fresher fruit, then poorer shop**, which uses one catch-up idea, not three.

---

## 3. Prototype A: the core market (readability)

| Tweak | Change | Evidence |
|---|---|---|
| A1 **Uniform freshness** | Every fruit sells at full price when fresh, −$1 when okay, −$2 when not so. Fruits differ only in **how many nights** they stay in each stage. One rule replaces four price-drop tables | [Sim] Plans win 26–38% (was 25–42%); Hard beats Normal 56% (same); comebacks 32% (was 28%) |
| A2 **Street prices (option)** | Each fruit has one printed street price ($5 blueberry, $7 watermelon, $6 apple, $9 mango). Customer cards show only what and how many, plus a "generous" +$2 icon on a few cards | [Sim] Balanced (plans 25–43%), but skill matters a little less (Hard beats Normal 53%). Keep as the beginner option |
| A3 **Catch-up stays "poorest buys first"** | Keep cash order; don't switch to net worth | [Sim] Adversarial test: a shop that buys every upgrade wins 33% (fair), so there's no exploit. Ordering by net worth makes hoarding cash viable (13% → 32%) but makes upgrades optional, and it's harder to count by hand |
| A4 **Simpler ties** | Fresher fruit, then poorer shop | [Hyp] Fewer exceptions to remember |

**Prototype A without traits, favours and upgrades** [Sim]:

- Games are close (17% blowouts).
- **Ripening green fruit dominates** (60% wins), and Hard beats Normal only 52%.
- The layers we're removing currently keep that plan in check. If people find it the same, Prototype B must fix it.

---

## 4. Prototype B: different businesses (upgrades that change decisions)

There are four "stall specialities", one per plan. Each changes *what you do*, not just how much you earn.

| Speciality | Cost | Effect | Decision it changes | Replaces |
|---|---|---|---|---|
| **Cold Room** | $5 | Your fruit ages only every other night; Heatwave-proof | Hold stock for a shortage or Market Day | Cool Room ($7) |
| **Juice Bar** | $4 | +$1 for each okay or not-so fruit sold; up to 3 spoiling fruits a night become $2 each | Buy cheap old fruit at the farmers market on purpose | Juice Press |
| **Barrow** | $6 | Buy 5 a day and hold 12 | Volume and turnover | Handcart + Bigger Shelf |
| **Premium Display** | $8 | Your **fresh** fruit may be priced $1 above a customer's normal limit | Stay expensive, and sell to whoever is left after the cheap stalls run out | Signboard |

- Removed: **Shop Bell** (the Market Gossip card takes over the information role).
- **When:** upgrades are drafted only on the evening after each Market Day (see section 5): three build moments a game, not ten.
- **Traits** stay as starting stall capabilities.
- [Hyp] The upgrades are designed to change decisions, so measure them by whether they change purchasing, holding and pricing, not only by win rate. The computer players need teaching before the lab can rate them.

---

## 5. Prototype C: market rhythm (the arc)

| Day (10-day game) | Rhythm |
|---|---|
| 1–2 | **Opening:** small crowds; learn the rivals |
| **3** | **Market Day:** +1 walk-in, **no lorry**; then the evening upgrade draft |
| 4–5 | Ordinary trading; demand grows |
| **6** | **Market Day**, then the draft |
| 7–8 | Second-half crowds (more walk-ins) |
| **9** | **Market Day**, then the last draft |
| **10** | **Last Call:** customers pay +$1; afterwards the jam maker buys leftover fruit at $1 each |

Market Days are printed on the day track, so everyone knows they're coming. In 8-day games they fall on days 3 and 6; in 14-day games on 3, 6, 9 and 12.

- **Why:**
  - preparation, holding stock for a day with no lorry, becomes a planned, satisfying choice;
  - the jam maker softens the final-day dump without making stock valuable.
- [Sim] Market Days every third day:
  - plans become the most even of anything tested (24–35%);
  - comebacks rise to 38%;
  - spoilage halves.
- **But** Hard and Normal computers become even (50%), because the computers don't plan ahead.
- [Hyp] Following the review: lower spoilage is not automatically better. Test whether preparing creates hard choices, or just an obvious stock-up.
- A $1 jam-maker salvage was tested earlier: no balance change.

---

## 6. The new favour card: **Market Gossip** 👂

> **Market Gossip** *(evening)*. Secretly look at the front of every face-down walk-in card in tomorrow's queue: who they are, what they want, how many, and their top price. Don't show anyone. Everyone sees that you played it.

- **Replaces Free Cash** (2 copies; the deck stays at 12). Free Cash is the only favour that is pure money with no decision.
- **Why it fits the review:**
  - it is the single, contained piece of private information that the review suggested;
  - rivals know you know *something*, and your buying gives clues ("why is she buying so many mangoes?");
  - that adds Stockpile-style reading without forecasts, secret objectives and trading all at once.
- **How it's earned:** like any favour, by serving a customer's whole order by yourself. It rewards the sales that build rivalry.
- **App:**
  - the walk-in silhouettes turn face up **only on your screen**, with a small 👂 badge on your seat for rivals;
  - online, the server must send the card fronts to that player only, the same privacy rule as secret prices, with a leak check added to the browser tests.
- **Needs:** the face-down queue (REP-031). It works whether or not the street-price option is used; with street prices it reveals quantities and "generous" icons.
- [Hyp] Test it on its own, after A–C, so its effect can be told apart (as the review advised).

**Favour deck after the change** (2 of each):

- Market Gossip *(new)*
- Free Fruit
- Good Riddance
- Fridge

- Favoured Trade
- Priority

---

## 7. App feedback: understand why you won or lost

| Feature | Status today | Tweak |
|---|---|---|
| Which customer bought from which stall | Showdown cards per line | One **"rush" view**: customers walk to the stalls they bought from |
| Why a sale was missed | Tags: "$1 too high", "lost tie: fresher won", "Right price, not enough stock" | Keep; add "too old for that price" (freshness) |
| Day's money in, money out and waste | Day income only | **Day ledger:** sales, purchases, spoilage and upgrades, with net for the day |
| Reveal pacing | Fast, skippable | A short pause after the bell before the rush resolves |
| Forecast and rival stock | Lorry strip, rival drop-downs, rivals strip while pricing | All in the one Tomorrow strip, plus the public "for sale" crates |
| Playtest modes | Toggles for events, upgrades, unripe | Add **"Core market" (Prototype A)** and **"Market Days"** toggles in the set-up, so playtests can compare versions |

---

## 8. Test plan (the review's targets)

| Step | Version | Question | Measure |
|---|---|---|---|
| 1 | A (core) on paper and in the app | Is the core fun on its own? | Teach in ~5 minutes; new players finish unaided; players can explain their best and worst decision |
| 2 | B (specialities) | Does specialising improve rivalry and replay? | Did upgrades change what people bought, held or priced? Did a "stay expensive" win happen? |
| 3 | C (rhythm), on the better of A and B | Does preparation improve the arc? | Hard choices before Market Days; the final two days still have real decisions |
| 4 | + Market Gossip | Does one private clue add reading? | Did anyone change their play after seeing a rival use Gossip? |
| Always | Same group repeatedly, plus fresh groups | Repetition and dominant plans; teaching problems | Replay requests; time per game; the reveals people remember |

---

## 9. What changes against the approved PLAN-014

| PLAN-014 item | Proposal |
|---|---|
| Printed supply and demand, posted prices, crates, queue, fruit payouts | **Keep** |
| Market pulse in the core | Move to an **optional advanced rule** |
| Market Days optional | Make the rhythm **standard in Prototype C** (fixed days on the track) |
| Event redesign | Keep |
| Upgrades unchanged (retune) | Replace with the **four specialities**, drafted on Market Day evenings |
| Tie rules | Simplify to fresher, then poorer |
| Free Cash | Replace with **Market Gossip** |

Lab files: `docs/workflow/reports/img/REP-032-lab.tar.gz` .

# REP-033 · Rules version 4 built: open stalls, the bell, Market Days, specialities and Market Gossip

| | |
|---|---|
| Game | Fruit Market (app now on rules version 4, "tabletop edition") |
| Report date | 2026-10-27 15:37 (Asia/Kuala_Lumpur) |
| Type | Build and validation report |
| Plans | PLAN-015 (approved: "approve plan 15") built together with the approved PLAN-014 items |
| Branch | `workflow/plan-015` , merged to `main` for the preview. **Live game unchanged** |

**Labels. [Sim]** = computer players, 300 games per test, about ±5 points. **[Hyp]** = a guess about people.

---

## 1. What was built

**The day (five beats).**

1. **Dawn:** the lorry unloads; the Tomorrow strip shows the next lorry, event, expected customer and the face-down walk-ins (fruit icons on the backs).
2. **Market:** buying, poorest shop first.
3. **Open stalls (new):** each shop secretly fills its for-sale crate; all crates show together.
4. **The bell:** one secret price per fruit, revealed together, with a short pause before the rush.
5. **The rush:** customers buy from the cheapest stall, leftover demand spills to the next; then evening and night.

**Prototype A (core market).**

- Uniform freshness: −$1 okay, −$2 not so, for every fruit.
- Ties: Favoured Trade, then Haggler, then fresher fruit, then poorer shop, then turn order.
- Cash catch-up kept ("poorest buys first").
- Options in the set-up: **Core market** (no traits, favours or specialities), **Street prices** (beginner) and **Market pulse** (advanced).

**Prototype B (four specialities),** drafted on the evenings after days 3, 6, 9 and 12: Cold Room $5, Juice Bar $4, Barrow $6, Premium Display $8. Shop Bell, Signboard, Bigger Shelf and Handcart are gone.

**Prototype C (rhythm).** Market Days on days 3, 6, 9 and 12 (+1 walk-in, no lorry); Last Call on the final day (+$1, then the jam maker buys leftovers at $1). Set-up toggle "Market Days", on by default.

**Market Gossip** replaces Free Cash (2 copies, evening): secretly see tomorrow's walk-ins. The fronts appear only on your screen, with a 👂 badge.

**PLAN-014 items.**

- Printed farmer cards (2–4 per fruit for 3 shops; 2 shops −1, 4 shops +1; missing fruit 1).
- Walk-ins: players − 2, then players − 1 in the second half. Customer cards are no longer scaled.
- Price shifts: blueberry −1, apple −2, mango +2.
- Events: Blueberry Glut and Apple Blight replace Slow Season and Rainy Day; Good Harvest +1 per 2 shops; Bad Harvest rounds up; no Health Inspector on the final day.
- `RULES_VERSION` 4. Older apps are asked to update; daily scores are kept separately per rules version.

**App feedback.**

- Stall screen and price screen: each shows rivals' crates, which customers can pay your price, and a hint.
- Day ledger: sales, bought, other (fines, favours), net, and fruit spoiling tonight; jam maker money on the results.
- Missed-sale reason "too old for $X". Bell pause before the rush. New tutorial tips.
- `RULES.md` and the in-app How to play rewritten.

**One rule detail added for privacy.** The evening is now held whenever anyone has a favour card. Before, it was held only when someone had an *evening* card, which told rivals what that player held. No balance effect.

## 2. Privacy (online)

| Hidden until everyone commits | How it is protected | Check |
|---|---|---|
| Crates during Open stalls | Server keeps them aside; every view shows all crates empty | Unit test + browser test: PASS |
| Prices during the bell | Same | Unit test + browser test: PASS (0 leaks in 94 frames) |
| Face-down walk-ins | Each player's view shows only the fruit, no name, amount or price, unless that player heard the gossip | Unit test (fronts only to the gossiper) + browser test frame check: PASS |
| Rivals' favour cards | Only the number of cards is sent | Existing test: PASS |

## 3. Validation

| Check | Result | Evidence |
|---|---|---|
| `npm run verify` (records, lint, types, 92 unit tests, build) | PASS | Local run, commit after this report |
| `npm run verify:full` (solo game, performance budget, online match with leaver, quick match, 4 players, reconnect) | PASS | Exit 0; "leaks: 0"; "E2E OK" |
| New screens on a phone-sized browser | PASS | Stall, price, Market Day notice, evening, day ledger checked by screenshot; no page errors |

| Check | Result | Evidence |
|---|---|---|
| Paper or preview playtest (PLAN-015 T3) | NOT RUN | Needs the owner |

## 4. Balance of the production build [Sim]

| Measure | Result | Target or earlier figure |
|---|---|---|
| Hard beats Normal (2 players) | 56% | REP-032: 56% (earlier aim 65%) |
| Hard beats Easy (2 players) | 63% | — |
| Normal beats Easy (2 players) | 55% | — |
| Comeback from last at half-time (3 players) | 31% | REP-032: 32% |
| Day-5 leader wins (3 players) | 34% | lower is more open |
| Blowouts, margin $20+ (2 / 3 / 4 players) | 34% / 27% / 13% | watch 2-player games |
| Traits win (3 players, fair 33%) | 30–38% (Big Basket highest) | REP-028: 30–38% |
| Old "first buyer rotates" rule for comparison | comebacks 8%, blowouts 58% | catch-up still essential |

**Specialities (3 players, fair 33%).**

- Rushing **Barrow** wins 29% and **Premium Display** 27%; skipping either costs a lot (23%, 26%).
- Rushing **Cold Room** wins only 13%, **Juice Bar** 17%. The computers buy Juice Bar rarely (0.14 per game).
- Never building anything wins 19%, so specialities are worth having.
- This matches REP-032's warning: Cold Room and Juice Bar change *what you do*, and the computers don't play those plans well yet, so these numbers undersell them. [Hyp] Check in the playtest whether people find them useful.

**Market feel.**

- 90% of computer sales are within $1 of the customer's top price, and deep price wars are rare (0.1% of sales). Spoilage is near zero.
- Supply is tight enough that the computers rarely need to undercut. [Hyp] People may still undercut; watch for it in the playtest. If not, the next lever is +1 supply per listed fruit (needs a new plan).
- Market Days are lean days (income dips on days 3, 6 and 9), as designed.

## 5. Risks and open items

1. **Cold Room and Juice Bar are weak for computers.** Possible retune: Cold Room $4, Juice Bar $3. Not changed without your approval.
2. **Little undercutting among computers.** The hook ("how low dare I go?") needs a human check.
3. **2-player blowouts at 34%.** Watch in playtests.
4. **Hard vs Normal 56%,** below the old 65% aim. The computers are fair practice, not experts.

5. **Online settings:** online rooms use the defaults (Market Days on, no Core, Street or Pulse options). The playtest toggles are for games on one phone.

## 6. Next steps

1. Preview playtest in the order of REP-032 section 8: Core market first, then specialities, then Market Days, then Market Gossip.
2. Decide on the speciality retune (section 5, item 1).
3. PREPARE RELEASE 1.2.0 when you want rules version 4 on the live game (separate approval to publish).